PRELIMINARY INTACT STABILITY BOOKLET

"MARSOL SPLENDOR"

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1 GENERAL PARTICULARS

NAME	MARSOL SPLENDOR
VESSEL TYPE	SUPPLY VESSEL – FIRE FIGHTING SHIP
IMO NUMBER	9944376
CLASS	BV
REGISTER NUMBER	39612J
CALL SIGN	3E3692
FLAG	PANAMA
PORT OF REGISTRY	PANAMA
CLASS NOTATIONS	I ✠HULL ✠MACH Offshore support vessel (supply -HNLS - WELLSTIM, fire-fighting 1 -water spraying) Special service/WELLTEST Vessel Unrestricted navigation ✠ DYNAPOS AM/AT R, ✠ ALM, INWATERSURVEY
BUILDER	GUANGZHOU SHUNHAI SHIPYARDS LTD.
OWNER	LIBRA WORLDWIDE HOLDING CORP
HULL NUMBER	SH163
DATE OF BUILD	27-AUG-2022
LOA	62.800 m
LBP	57.400 m
BREADTH (mld)	16.80 m
DEPTH (mld)	5.300 m
SUMMER DRFAT	4.200 m

2 INTRODUCTION

Our client, **M/s. MARSOL WORLDWIDE LIMITED**, converted the vessel **MARSOL SPLENDOR** into a **Well Stimulation Vessel**. Accordingly, the equipment required for well stimulation operations has been installed onboard.

Subsequent to the modifications, a preliminary lightship weight estimation was carried out (Document No.: DU5864/1021). The report is attached in the Appendix. The lightship particulars used in this booklet have been adopted from the aforementioned lightship weight estimation report.

Lightship Weight	=	1911.97 T
Lightship VCG	=	5.792 m above BL
Lightship LCG	=	0.592 m AFT. of MIDSHIP
Lightship TCG	=	0.243 m port. off CL

Additionally, for the above-mentioned conversion, the following integral tanks have been converted:

NO.1 CEMENT BULK TANK	→	TREATED INJECTION (P)
NO.2 CEMENT BULK TANK	→	TREATED INJECTION (S)
NO.3 CEMENT BULK TANK	→	BRINE TK-4 (P)
NO.4 CEMENT BULK TANK	→	BRINE TK-4 (S)

NO.3 D.W/S.W.B TK(C)(U)	→	BRINE TK-3 (C) (U)
NO.4 D.W/S.W.B TK(C)(U)	→	BRINE TK-2 (C) (U)

LIQUID MUD TK.(C)	→	ACID TANK.1 (C)
LIQUID MUD / RECOVERY OILTK.(C)	→	BRINE TK-1 (C)

NO.4 F.W TK(P)	→	CORROSION INHIBITOR.P + SCALE INHIBITOR.P
NO.4 F.W TK(S)	→	SCALE DISSOLVER.S + SCALE INHIBITOR.S

R.O. PUMP ROOM	→	ACID PUMP ROOM

The following loading conditions are considered in this report:

- CONDITION NO.1: LIGHTSHIP (NOT A SAILING CONDITION)
- CONDITION NO.2: FULLY LOADED DEPARTURE
- CONDITION NO.3: FULLY LOADED RRIVAL
- CONDITION NO.4: BALLAST DEPARTURE (WITHOUT CARGO)
- CONDITION NO.5: BALLAST ARRIVAL (WITHOUT CARGO)

Following data are taken from the existing FINAL INTACT STABILITY BOOKLET (Doc.No: SH6163-C2) Cover page attached in appendix.

- Down flooding points
- FIFI Calculation details

3 CRANE OPERATION NOTES

In this report, the worst-case scenario for crane operations is considered to be the movement to the port side, as the crane is already installed on the port side. This assessment takes into account the full 360-degree operational range of the crane.

- New Crane specifications: -

  Model - KT/OTC2200-0435-1e
  Total operating weight - 46.500 T
  Boom weight - 32.550 T
  Crane pedestal weight - 13.950 T

Weight and COG of crane components in resting position is as follows: -

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
CRANE BOOM	32.550	17.425a	6.000p	11.504

- The following worst crane operations are checked as per the load chart (refer appendix)
  Side lifting to port side
  - Max. Transverse moment – Lifting 3.500t @ 35.00 m radius (35.530 m boom length – Side Lifting to port side)
  - Max. Vertical moment – Lifting 3.500t @ 9.600 m radius (35.530 m boom length - Side Lifting to port side)

- The wind speed should not exceed 20 knots during crane operation.
- The maximum duration of the lift is < 6 hour.
- The ship speed is limited to 0 knots during crane operation.
- Crane operations should be done in conditions where there is no traffic
- Significant wave height as per crane manufacturer

4 INTACT STABILITY CRITERIA

OSV CHEMICAL CODE – CHAPTER 2 , 2.2

IS CODE 2008 INTACT STABILITY CRITERIA.

The following criteria are equivalent to General Intact Stability Criteria – IMO A.167, but they are recommended where a vessel's characteristics render compliance with A.167 impracticable.

1. The area under the righting lever curve (GZ Curve) should not be less than 0.070 metre-radians 'up to an angle of 15° when the maximum righting lever (GZ) occurs at 15° and 0.055 metre-radians up to an angle of 30° when the maximum righting lever (GZ) occurs at 30° or above. Where the maximum righting lever (GZ) occurs at angles of between 15° and 30°, the corresponding area under the righting lever curve should be:

   $0.055 + 0.001\ (30° - \theta max)$ metre-radians
2. The area under GZ Curve between the angles of heel of 30° and 40° or between 30° and Sf, if this angle is less than 40°, should not be less than 0.03 metre-radians.
3. The righting lever GZ should be at least 0.20 metre at an angle of heel equal to or greater than 30°.
4. The maximum righting lever should occur at an angle of heel not less then 15°.
5. The initial metacentric height GMo should not be less than 0.15 metre.

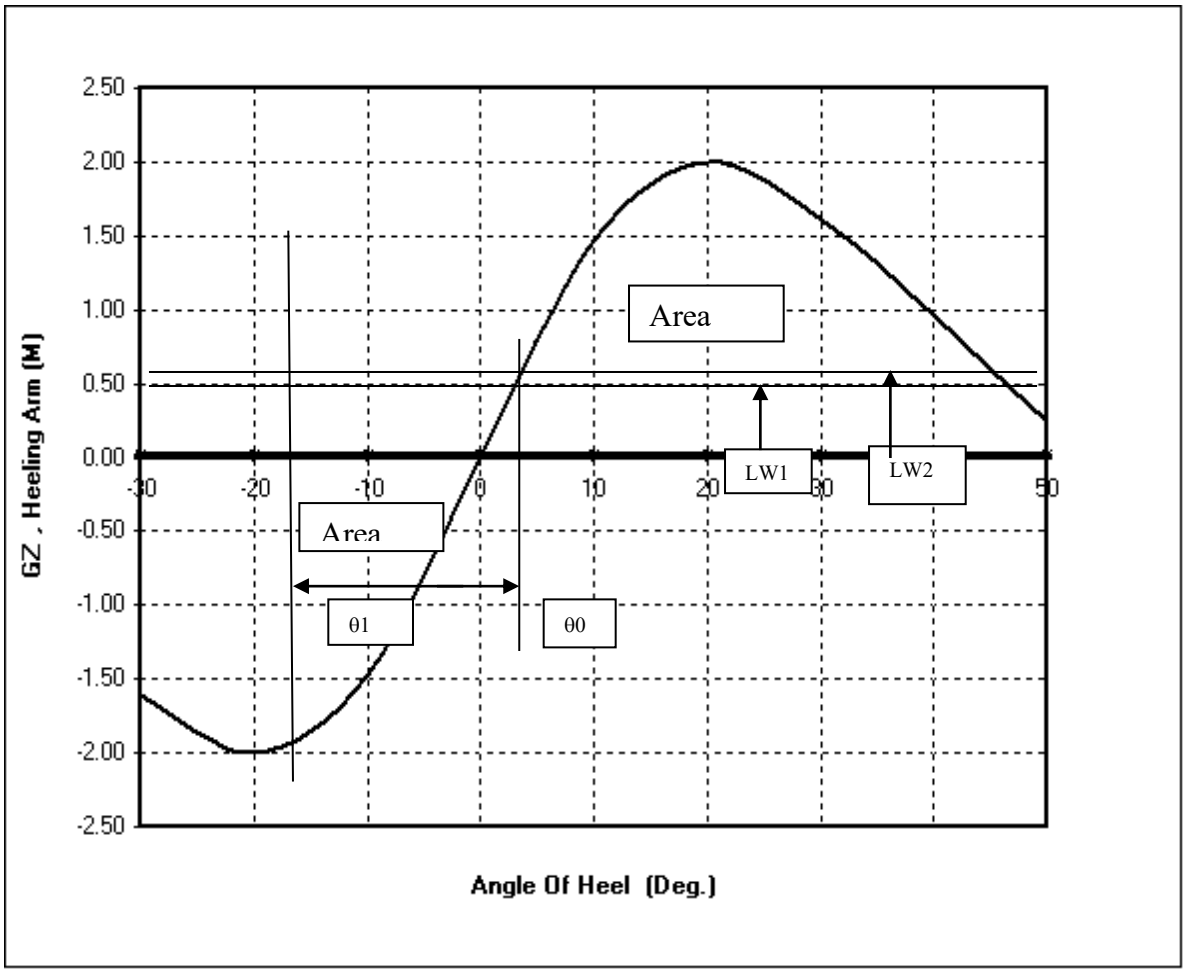
5 SEVERE WIND AND ROLLING CRITERION (WEATHER CRITERIA)

The ability of a ship to withstand the combined effect of beam wind and rolling should be demonstrated for each standard condition of loading with reference to the figure as follows:

- The Ship is subjected to a steady wind pressure acting perpendicular to the ships centerline which results in a steady wind heeling lever lw1.
- From resultant angle of equilibrium (θo) the ship is assumed to roll (θ_1) to windward owing to wave action. The angle of heel under the action of steady wind (θo) should be limited to 16 degrees or 80% of angle of deck immersion whichever is less.
- Ship is then subjected to gust wind pressure which results in heeling lever lw2.
- Under these circumstances area "b" should be greater than Area "a".

  The above criteria is represented in this booklet as follows

  (Res. Ratio = area "b"/ area "a" > 1)
- Free surface effects should be accounted for in the standard conditions of loading.



θ_2 = angle of down flooding (θ_f) or 50^0, whichever is less, where:

θ_f = angle of heel at which openings in the hull, superstructures or deckhouses which cannot be closed watertight immerse. In applying this criterion, small openings through which progressive flooding cannot take place need not be considered as open.

θ_c = angle of second intercept between wind heeling lever lw2 and GZ curves.

The wind heeling levers lw1 and lw2 referred to are constant values at all angles of inclination and are calculated as follows:

lw1 = P AZ/Disp. And

lw2 = 1.5lw1

Where:

P = 0.0514 t/m2

A = projected lateral area of the portion of the ship and deck cargo if any above the waterline (m^2)

Z = vertical distance from the center of A to the center of the underwater lateral area or approximately to a point at one half the draft (m)

Disp = Displacement (t)

The angle of roll (θ_1) should be calculated as follows:

θ_1 = 109kX1X2√(rs) (degrees)

Where:

X1 = factor as shown in table 1

X2 = factor as shown in table 2

k = factor as follows:

k = 1.0 for round-bilged ship having no bilge or bar keels;

k = 0.7 for ship having sharp bilges;

k = as shown in table 3 for a ship having bilge Keels, a bar keel or both

r = 0.73 ± 0.6 OG/d with:

OG = distance between the center of gravity and the waterline (- if it is below).

d = mean moulded draft of the ship (m)

s = factor as shown in table 4.

Table 1 - Values of coefficient X_1

B/d	**X_1**
≤ 2.4	1.00
2.5	0.98
2.6	0.96
2.7	0.95
2.8	0.93
2.9	0.91
3.0	0.90
3.1	0.88
3.2	0.86
3.4	0.82
≥ 3.5	0.80

Table 2 - Values of coefficient X2

C_B	**X_2**
≤ 0.45	0.75
0.50	0.82
0.55	0.89
0.60	0.95
0.65	0.97
≥0.70	1.00

Table 3 - Values of coefficient k

$\frac{A_K \times 100}{L \times B}$	**k**
0.0	1.00
1.0	0.98
1.5	0.95
2.0	0.88
2.5	0.79
3.0	0.74
3.5	0.72

6 FIRE FIGHTING OPERATIONS CRITERIA

Fire Fighting Operation Criterion (Class BV)

The loading conditions reported in the trim and stability booklet, with the exception of lightship, are also to be checked in order to investigate the ship's capability to support the effect of the reaction force of the water jet in the beam direction due to the monitors fitted on board. A fire-fighting ship may be considered as having sufficient stability, according to the effect of the reaction force of the water jet in the beam direction due to the monitors fitted on board, if the heeling angle of static equilibrium θ0, corresponding to the first intersection between heeling and righting arms (see Fig. 1), is less than 5°. The heeling arm may be calculated as follows:

$$b_h = \frac{\sum R_i \cdot h_i + S \cdot (T/2 - e)}{9.81\ \Delta} \cos\theta$$

Where:

bh: Heeling arm, in m, relevant to the reaction force of the water jet of the monitors fitted on board, and to the effect of transversal manoeuvring thrusters. The monitors are assumed to be oriented in beam direction parallel to the sea surface, so as to consider the most severe situation.

Ri: Reaction force, in kN, of the water jet of each monitor fitted on board (see Fig. 2).

hi: Vertical distance, in m, between the location of each monitor and half draught (see Fig. 2).

S: Thrust, in kN, relevant to manoeuvring thruster(s), if applicable (see Fig. 2).

e: Vertical distance, in m, between the manoeuvring thruster axis and keel (see Fig. 2).

Δ: Displacement, in t, relevant to the loading condition under consideration.

T: Draught, in m, corresponding to Δ (see Fig. 2).

Figure 1 : Heeling and righting arm curves

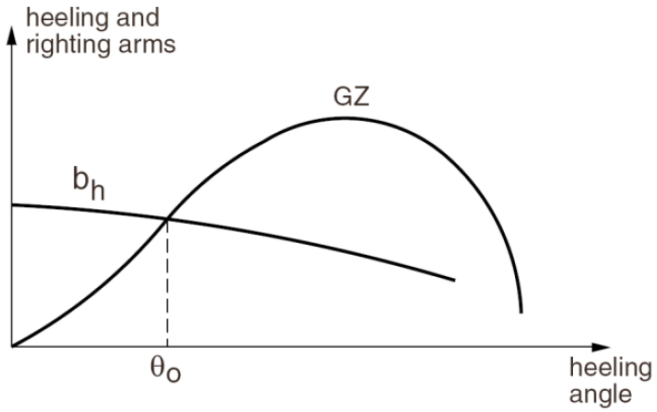
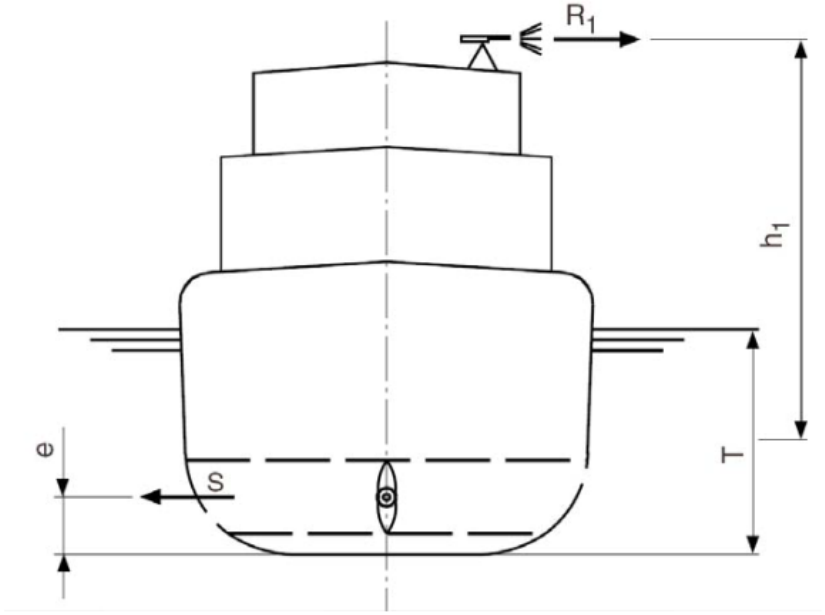


Figure 2 : Reaction force of water jet in the beam direction due to monitors



7 *CRITERIA FOR CRANE OPERATION*

RESOLUTION MSC.415(97) (adopted on 25 November 2016) AMENDMENTS TO PART B OF THE INTERNATIONAL CODE ON INTACT STABILITY, 2008 (2008 IS CODE), CHAPTER 2, SEC 2.9.3

2.9.3 Stability criteria

2.9.3.1 The stability criteria included herein, or the criteria contained in paragraphs 2.9.4, 2.9.5 or 2.9.7, as applicable shall be satisfied for all loading conditions intended for lifting with the lifting appliance and its load at the most unfavorable positions. For the purpose of this section, the lifting appliance and its load(s) and their center of gravity (COG) should be included in the displacement and center of gravity of the ship, in which case no external heeling moment/heeling lever is applied.

2.9.3.2 All loading conditions utilized during the lifting operations are to comply with the stability criteria given in sections 2.2 and 2.3 of part A. Where the ship's characteristics render compliance with section 2.2 of part A impracticable, the equivalent stability criteria given in chapter 4 of the explanatory notes to the 2008 IS Code should apply. During the lifting operation, as determined by paragraphs 2.9.1, the following stability criteria should also apply:

.1 the equilibrium heel angle, φ 1, shall not be greater than the maximum static heeling angle for which the lifting device is designed and which has been considered in the approval of the loading gear;

.2 during lifting operations in non-exposed waters, the minimum distance between the water level and the highest continuous deck enclosing the watertight hull, taking into account trim and heel at any position along the length of the ship, shall not be less than 0.50 m; and

.3 during lifting operations in exposed waters, the residual freeboard shall not be less than 1.00 m or 75% of the highest significant wave height H_S, in (m), encountered during the operation, whichever is greater.

2.9.4 Lifting operations conducted under environmental and operational limitations

2.9.4.1 For lifting conditions carried out within clearly defined limitations set forth in paragraph 2.9.4.1.1, the intact criteria set forth in paragraph 2.9.4.1.2 may be applied instead of the criteria included in paragraph 2.9.3.

a) The limits of the environmental conditions should specify at least the following:

- the maximum significant wave height
- the maximum wind speed (1 minute sustained at 10m above sea level),

The limits of the operational conditions should specify at least the following:

- the maximum duration of the lift
- limitations in ship speed
- limitations in traffic/traffic control

b) The following stability criteria apply with the lifted load is at the most unfavorable position:

- the corner of the highest continuous deck enclosing the watertight hull shall not be submerged
- $A_{RL} \geq 1,40 \times A_{HL}$

Where:

A_{RL}: The area under the net righting lever curve, corrected for crane heeling moment and for the righting moment provided by the counter ballast, if applicable, extending from the angle ϕ_1 to the angle ϕ_2, see Fig 2.9.1

A_{HL}: The area below the wind heeling lever curve due to the wind force applied to the ship at the maximum wind speed specified in a), see Fig 2.9.1

ϕ_1: Equilibrium heeling angle

ϕ_2: The lesser between the angle of downflooding (ϕ_F), the angle of vanishing stability (ϕ_R), and the second intersection of the righting lever curve with the wind heeling lever curve.

c) The area under the net righting lever curve from the equilibrium heel angle, ϕ_1, to the down flooding angle, ϕ_F, or 20°, whichever is less, shall be at least 0.03 m rad.

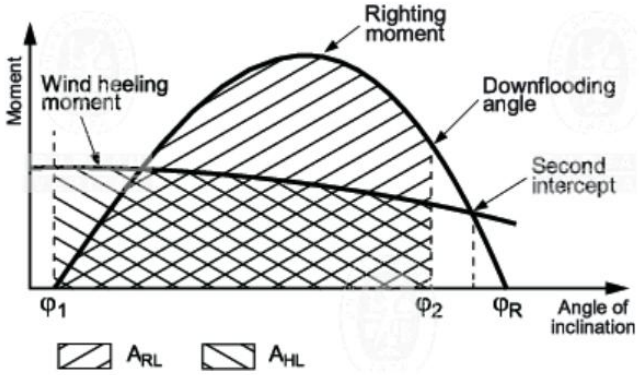


Figure 2.9.1

8 NOTES TO THE MASTER

1. Compliance with the required minimum stability criteria does not ensure immunity against capsizing, regardless of the circumstances, or absolve the master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, the weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Care should be taken to ensure that cargo, allocated to the ship is capable of being stowed, so that compliance with the stability criteria can be achieved. If necessary, the amount of cargo should be limited to the extent that ballast weight may be required.
3. In determining the sequence of tanks from which fuel oil and fresh water is to be consumed and those into which water ballast may be admitted during the voyage the master must ensure, prior to departure, that the required minimum stability criteria will be maintained throughout the voyage after making due allowance for free surface effect as may be appropriate.
4. Before a voyage commences, care should be taken to ensure that the cargo and sizeable pieces of equipment have been properly stowed or lashed, so as to minimize the possibility of both longitudinal and lateral shifting at sea, under the effect of acceleration caused by rolling and pitching.
5. Master should note that stability can be adversely affected by such influences as towing speed, beam wind on ships with large windage area, icing on top sides and deck cargo, water trapped on deck and in deck cargo, rolling characteristics and following seas.
6. Master should note that all ballast tanks used in some loading conditions shown in the stability booklet must be pressed up to avoid free surface moment which may cause dangerous situation.
7. The stability criteria contained set minimum values, but no maximum values are recommended. It is advisable to avoid excessive values of metacentric height, since these might lead to acceleration forces which could be prejudicial to the ship, its complement, its equipment and to safe carriage of the cargo.
8. The minimum bow draft of this vessel should not be less than 2.451 m to prevent waves from slamming and the minimum stern draft is not less than 2.7 m to ensure that the propeller is 100% submerged.

FSM NOTES

Max FSM considered for one transverse pair or a single centreline tank having the greatest free surface effect.

9 SAMPLE CALCULATIONS

List out the weights, LCG, VCG & FSM of each tank and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and the VCG of the vessel.

For this worked out Example, (Cond.03)

Displacement.....…… 2889.73 Tonnes
LCG .....…. 0.969 m (aft of MIDSHIP)
VCG (KG).....….. 4.901 m above Base Line

A tank, which is partially filled, adversely affects the stability of the vessel. . This is known as Free Surface Effect. So the Free surface moments of all tanks, which are not fully filled, are to be calculated and the sum of Free surface moments of all tanks put together, arrived at. In the calculations performed in this booklet, the Maximum Free surface moment is considered for all cases.

The Free surface effect is expressed as a 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

Free Surface Correction (Meters) = $\frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$

Free Surface moment for a Tank = Moment of Inertia (M^4) for all Tanks x Density of Liquid in Tank

KGo (Fluid) = KG (Solid) + Free Surface Correction.

Fixed Weight Status

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	0.40	24.100f	0.000	6.800
Total Fixed:	**1,916.45**	**0.541a**	**0.242p**	**5.803**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DWSWBFP	1.025	100.00	54.57	26.510f	0.000	3.135	39.8
NO1_DWSWB.P	1.025	50.00	19.74	20.780f	3.218p	0.786	41.3
NO1_DWSWB.S	1.025	100.00	39.49	21.049f	3.353s	1.426	0.0
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	113.5
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
Subtotals:		**53.23%**	**221.38**	**13.197f**	**3.026s**	**1.723**	**194.6**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	7.60	5.61	21.305f	3.604p	2.619	63.6
NO1_FW.S	1.000	10.00	7.38	21.331f	3.610s	2.656	63.6
NO2_FW.P	1.000	10.00	6.64	16.295f	5.444p	2.645	51.5
NO2_FW.S	1.000	10.00	6.64	16.291f	5.456s	2.645	51.5
NO3_FW.S	1.000	5.00	3.56	19.291a	6.329s	1.548	6.7
NO3_FW.P	1.000	5.00	2.70	20.394a	6.317p	1.952	6.0
Subtotals:		**8.02%**	**32.54**	**11.367f**	**0.369s**	**2.466**	**243.0**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.C	0.850	3.40	1.49	7.836f	0.132s	0.021	245.2
NO2_FO.U.P	0.850	3.40	1.50	2.381f	3.572p	0.025	198.5
NO2_FO.U.S	0.850	9.80	3.83	2.681f	3.759s	0.059	198.5
NO3_FO.P	0.850	9.80	2.20	2.682a	7.499p	2.638	2.2
NO3_FO.S	0.850	9.80	2.20	2.682a	7.501s	2.638	2.2
NO3_FO.U.P	0.850	9.80	3.14	2.693a	5.667p	0.088	33.5
NO3_FO.U.S	0.850	9.80	3.14	2.694a	5.683s	0.088	33.6
NO4_FO.U.P	0.850	9.80	3.98	8.874a	5.550p	0.084	44.1
NO4_FO.U.S	0.850	9.80	3.98	8.873a	5.566s	0.084	44.2
FODAY.P	0.850	9.80	2.20	8.082a	7.499p	2.638	2.2
FODAY.S	0.850	9.80	2.69	8.722a	7.501s	2.638	2.6
Subtotals:		**5.84%**	**30.35**	**3.791a**	**0.430s**	**0.857**	**806.8**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.997f	7.500p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.698f	7.500s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.298f	7.500p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	0.999f	7.500s	3.872	0.8

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DIRTYOIL.P	0.900	98.00	6.22	1.105a	1.499p	0.490	4.8
BILGEHOLD.S	1.000	98.00	6.91	1.105a	1.501s	0.490	5.3
SEWAGE	1.000	98.00	10.11	14.762f	0.002s	0.490	30.3
Subtotals:		**56.48%**	**72.22**	**3.785f**	**0.636p**	**2.783**	**45.2**

SCALE INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
SCLEIHBTR.P	1.100	98.00	14.47	13.401a	7.048p	3.872	3.2
SCLEIHBTR.S	1.100	98.00	14.47	13.401a	7.049s	3.872	3.2
SCALE_DISLVR.S	1.100	98.00	28.93	16.104a	7.049s	3.872	6.4
Subtotals:		**98.00%**	**57.86**	**14.752a**	**3.524s**	**3.872**	**12.8**

CORROSION INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
CORSNIHBTR.P	0.950	98.00	24.99	16.104a	7.048p	3.872	5.5
Subtotals:		**98.00%**	**24.99**	**16.104a**	**7.048p**	**3.872**	**5.5**

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
TRTDINJW.P	1.000	100.00	30.90	11.800f	2.100p	2.850	0.0
TRTDINJW.S	1.000	100.00	30.90	11.800f	2.100s	2.850	0.0
Subtotals:		**100.00%**	**61.80**	**11.800f**	**0.000**	**2.850**	**0.0**

TK ON MAIN DECK

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NITROGENTK.P	1.185	98.00	47.27	17.249a	1.050p	6.721	8.7
CHEMADCTTK.P	1.185	98.00	76.18	0.409f	2.050p	8.020	42.0
Subtotals:		**66.80%**	**123.45**	**6.352a**	**1.667p**	**7.522**	**50.7**

BRINE

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_BRINETK.C	1.550	98.00	51.55	22.652a	0.001s	4.075	66.0
NO2_BRINETK.C	1.550	98.00	49.80	20.599a	0.000	1.905	23.1
NO3_BRINETK.C	1.550	98.00	56.94	14.966a	0.000	0.653	23.9
NO4_BRINETK.P	1.550	98.00	46.94	7.598f	2.100p	2.819	12.6
NO4_BRINETK.S	1.550	98.00	46.94	7.598f	2.100s	2.819	12.6
Subtotals:		**98.00%**	**252.16**	**9.250a**	**0.000**	**2.406**	**138.1**

ACID

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_ACIDTK.C	1.160	98.00	96.53	16.266a	0.000	3.322	55.1
Subtotals:		**98.00%**	**96.53**	**16.266a**	**0.000**	**3.322**	**55.1**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**43.95%**	**973.27**	**1.812a**	**0.484s**	**3.125**	**1,552.0**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	2,866.42	0.817a	0.004s	2.045	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**2,889.73**	**0.996a**	**0.004s**	**2.036**	

All the weights onboard are listed as shown above along with their centers of gravity and Free surface moments of the tanks. The individual Vertical, Longitudinal and Transverse Moments are found out by multiplying the individual weight by corresponding VCG, LCG and TCG. The individual Vertical, Longitudinal and Transverse Moments are added up .The resultant LCG, VCG and TCG are obtained by dividing these summations by the displacement. The VCG is now to be corrected for Free Surface Effect.

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	1,393.47	1.564a	0.361s	3.201
Displacement	3,305.44	1.002a	0.012s	4.700

After working out the displacement and the centers of gravity for a particular Loading condition, the Hydrostatic particulars of the vessel at that displacement is read out from the Hydrostatics Table given. The mean draft corresponding to the displacement and the Metacentric Radius (KMt) are obtained. The Transverse metacentric height GMt of the vessel is thence calculated.
KGo is the corrected Vertical Centre of Gravity and is obtained as follows.

$$\text{Free Surface Correction (Meters)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

The trim and Metacentric Height are calculated using the Hydrostatic particulars and the values of the centers of gravity.

GMt = KMt - KGo

Trim (cm) = Displacement x (LCB - LCG) / (MCT x 100)

Floating Status

Draft FP	4.066 m	Heel	0.18s deg	GM(Solid)	4.043 m
Draft MS	4.200 m	Equil	Yes	F/S Corr.	0.370 m
Draft AP	4.335 m	Wind	Off	GM(Fluid)	3.674 m
Trim	0.27a m	Wave	No	KMT	8.743 m
LCG	1.002a m	VCG-Solid	4.700 m	TPcm	9.64
Displacement	3305.4 MT	WaterSpgr	1.025		

The Cross curves or the KN curves are used to obtain the staitical stability curves or the GZ curves of the vessel. The Intact stability of the vessel is determined based on the statical stability curves based on the requirements of the IMO regulations in this regard.
Obtain the values of KN from the Displacement, corresponding to vessel's displacement, for each angle of Heel. GZ is obtained using the relationship given in the formula below,

GZ = KN - KGo Sin θ

Unprotected Flood Points

Name	L (m)	T (m)	V (m)	Height (m)
(1) POSITION 1(S)	9.700f	5.400s	12.150	8.485
(2) POSITION 1(P)	9.700f	5.400p	12.150	8.493
(3) POSITION 2(S)	10.300f	6.300s	12.150	8.490
(4) POSITION 2(P)	10.300f	6.300p	12.150	8.500

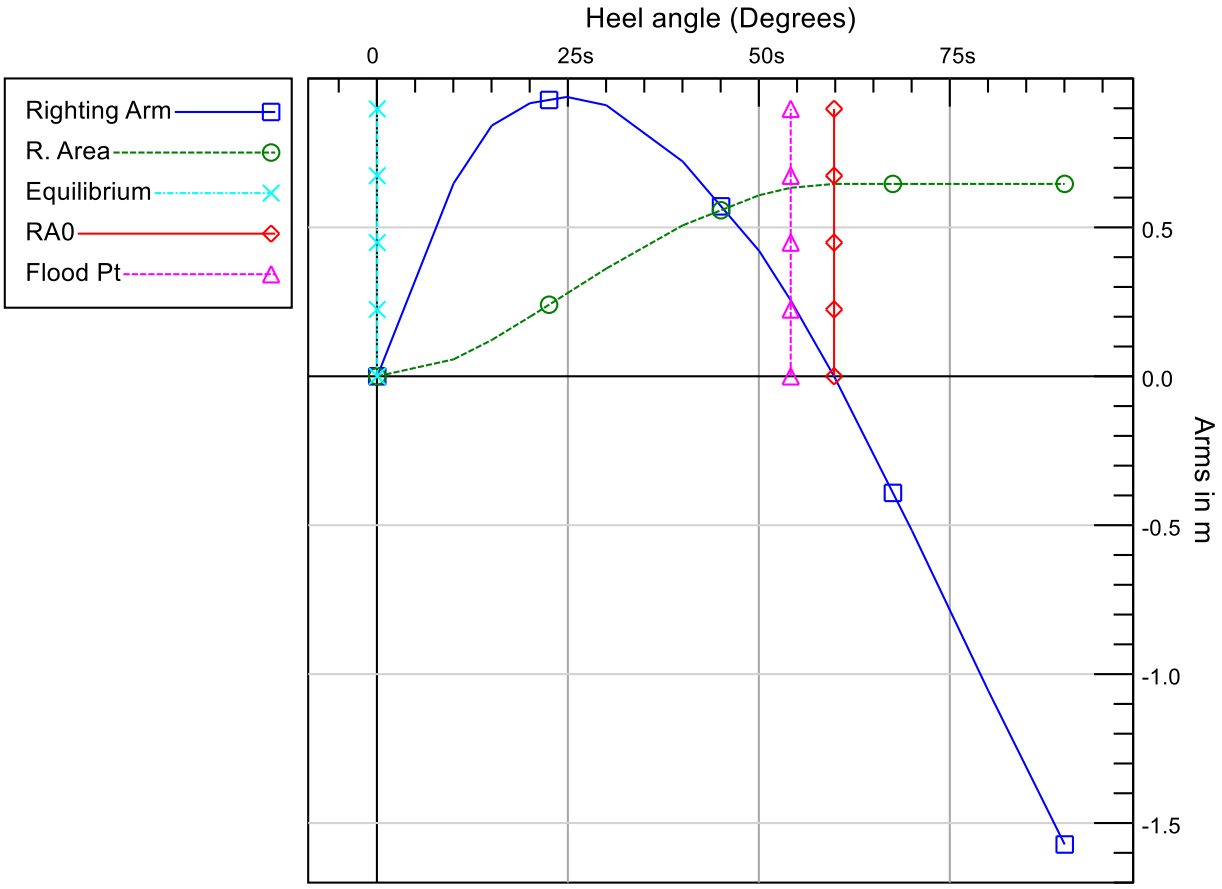
: OSV CHEMICAL CODE – CHAPTER 2 , 2.2

Free Surface Adjustment 0.537
Adjusted VCG 5.438

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Area (m-Rad)	Flood Pt Height (m)	Notes
0.04s	0.53a	3.749	0.000	0.000	8.486 (1)	Equil
10.04s	0.40a	3.672	0.647	0.056	7.266 (3)	
15.04s	0.39a	3.619	0.842	0.122	6.550 (3)	
20.04s	0.53a	3.597	0.917	0.200	5.752 (3)	
24.88s	0.79a	3.581	**0.938**	0.278	4.933 (3)	MaxRa
30.04s	1.20a	3.544	0.910	0.362	4.035 (3)	
40.04s	2.31a	3.332	0.721	0.507	2.328 (3)	
50.04s	3.70a	2.951	0.421	0.608	0.681 (3)	
54.17s	4.27a	2.766	0.255	0.633	0.000 (3)	FldPt
59.85s	5.04a	2.490	0.000	0.646	-0.932 (3)	RaZero
60.04s	5.07a	2.481	-0.010	0.646	-0.964 (3)	
70.04s	6.31a	1.946	-0.518	0.646	-2.577 (3)	
80.04s	7.39a	1.358	-1.056	0.646	-4.104 (3)	
90.04s	8.23a	0.722	-1.572	0.646	-5.491 (3)	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1552.0 MT-m was used to adjust the VCG.

: OSV CHEMICAL CODE – CHAPTER 2 , 2.2



Limit	Min/Max	Actual	Margin	Pass
(1) Area from 0.00 deg to MaxRA at 30.00	>0.0601 m-R	0.278	0.218	Yes
(2) Area from 0.00 deg to 40.00 or Flood	>0.0900 m-R	0.507	0.417	Yes
(3) Area from 30.00 deg to 40.00 or Flood	>0.0300 m-R	0.145	0.115	Yes
(4) Righting Arm at 30.00 deg	>0.200 m	0.910	0.710	Yes
(5) Absolute Angle at MaxRA	>15.00 deg	24.88	9.88	Yes
(6) GM Upright	>0.150 m	3.897	3.747	Yes

Heeling Moment Derivation

Part	LPA (m^2)	HCP (m)	Arm (m)	Pressure (MT/m^2)	Moment (m-MT)
HULL.C	218.1	2.769	4.561	0.051	51.13
DECKHOUSE	108.4	8.553	10.345	0.051	57.65
FWDMAST	13.4	16.065	17.857	0.051	12.32
ANTENNA	0.3	15.916	17.708	0.051	0.24
CRANE1	23.5	14.764	16.556	0.051	19.96
DECK	45.1	2.565	4.357	0.051	10.11
CRANE2	37.2	6.876	8.668	0.051	16.56
LIFEBOAT	10.0	6.367	8.159	0.051	4.19
CRANE3	2.9	6.727	8.519	0.051	1.26

Total wind heeling moment 173.42 to starboard

IMO SEVERE WIND & ROLLING

Free Surface Adjustment 0.484
Adjusted VCG 5.385

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
23.56p	0.71a	3.586	-0.969	0.090	-1.059	0.000	5.160 (4)	-2.455	Roll
13.56p	0.38a	3.630	-0.827	0.090	-0.917	-0.172	6.773 (4)	-0.643	
3.56p	0.52a	3.741	-0.254	0.090	-0.344	-0.289	8.089 (4)	0.752	
1.01s	0.53a	3.749	0.060	0.090	-0.030	-0.304	8.384 (3)	1.119	SSWEq
1.44s	0.53a	3.748	0.090	0.090	0.000	-0.304	8.335 (3)	1.057	Equil
6.44s	0.49a	3.719	0.425	0.090	0.335	-0.289	7.734 (3)	0.350	
9.07s	0.43a	3.686	0.591	0.090	0.501	-0.270	7.395 (3)	0.000	Deck Imm.
16.00s	0.41a	3.613	0.871	0.090	0.781	-0.191	6.402 (3)	-1.046	
16.44s	0.41a	3.611	0.880	0.090	0.789	-0.185	6.333 (3)	-1.122	
25.30s	0.82a	3.579	0.955	0.090	**0.865**	-0.053	4.860 (3)	-2.804	MaxRa
26.44s	0.90a	3.573	0.954	0.090	0.864	-0.036	4.661 (3)	-3.039	
36.44s	1.87a	3.431	0.829	0.090	0.739	0.107	2.933 (3)	-5.133	
46.44s	3.19a	3.100	0.581	0.090	0.491	0.216	1.273 (3)	-7.206	
50.00s	3.69a	2.952	0.459	0.090	0.369	0.243	0.688 (3)	-7.912	
54.17s	4.26a	2.765	0.295	0.090	0.205	0.264	0.000 (3)	-8.706	FldPt
56.44s	4.58a	2.658	0.198	0.090	0.108	0.270	-0.375 (3)	-9.124	
58.84s	4.90a	2.541	0.090	0.090	0.000	0.272	-0.768 (3)	-9.548	RaZero
66.44s	5.88a	2.143	-0.281	0.090	-0.371	0.272	-2.003 (3)	-10.792	

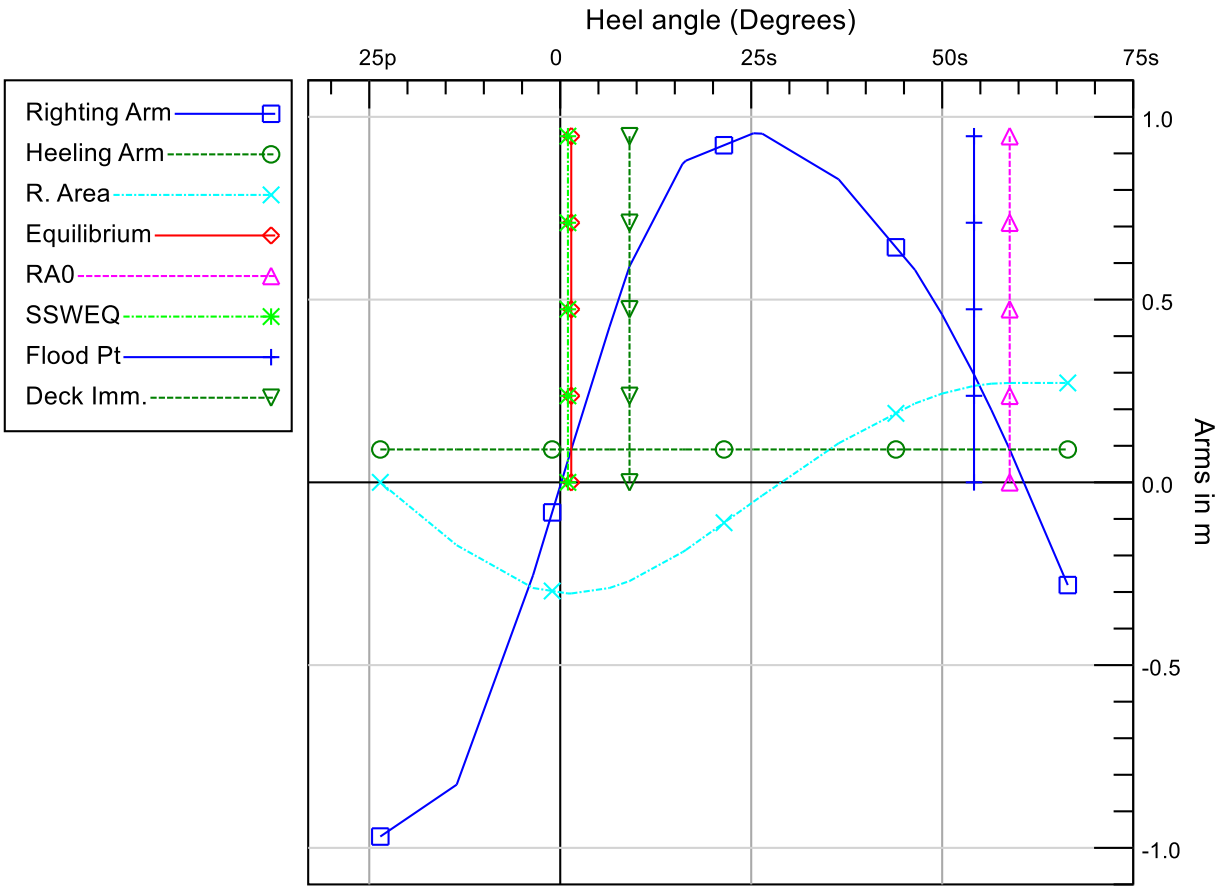
Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.

A Free Surface Moment of 1398.7 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
wind heeling arms derived from this moment (in m-MT):
Stbd heeling moment = 260.13

Roll angle is 24.47
Equilibrium for load condition without gust is 1.01s

IMO SEVERE WIND & ROLLING



Limit	Min/Max	Actual	Margin	Pass
(1) Res. Ratio from Roll to Abs 50.00 deg or Flood	>1.000	1.799	0.799	Yes
(2) Angle from SSWEQ to 80% of Dk. Imm. or Abs 16.00 deg	>0.00 deg	6.25	6.25	Yes

Heeling Moment Report

FIFI HEELING MOMENT CALCULATION			
Reaction Force of the Fire Monitor	=	1.530	MT
Number of Fire Monitors	=	2	
Draft of the loading condition	=	3.75	m
D/2	=	1.875	m
Height of firefighting monitor above baseline	=	19.600	m
Moment due to Fire Fighting operation	=	54.239	MT-m
Reaction Force of the Bow Thruster	=	18	MT
D/2	=	1.875	m
Height of Bow Thruster above baseline	=	1.200	m
Moment due to Bow Thruster operation	=	12.15	MT-m
Total Heeling Moment	=	66.389	Tm

Heeling moment: 66.39 MT-m to starboard

FIRE FIGHTING CRITERIA

Free Surface Adjustment 0.525
Adjusted VCG 5.426

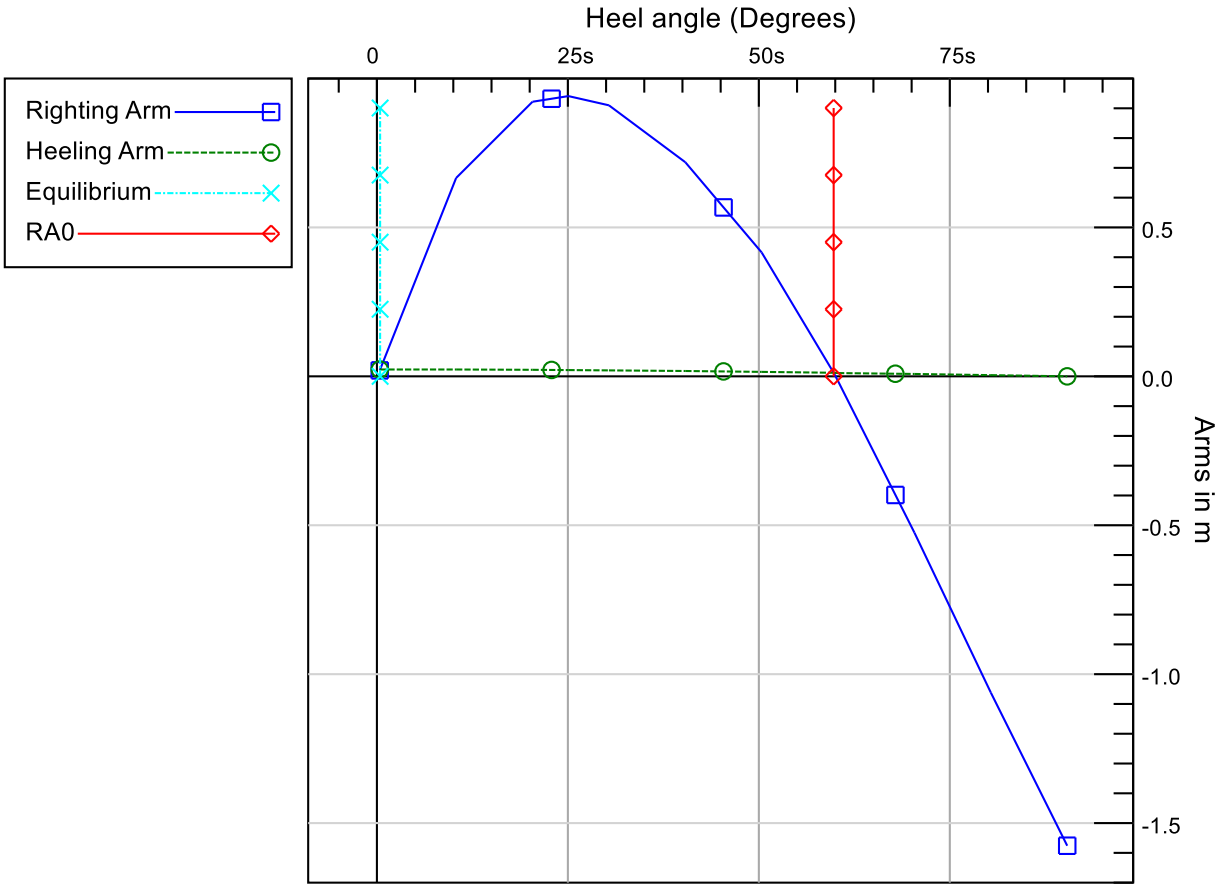
Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Notes
0.37s	0.53a	3.749	0.020	0.023	-0.003	
0.42s	0.53a	3.749	0.023	0.023	0.000	Equil
10.37s	0.40a	3.667	0.666	0.023	0.643	
20.37s	0.54a	3.596	0.922	0.022	0.900	
25.07s	0.80a	3.580	0.941	0.021	**0.920**	MaxRa
30.37s	1.23a	3.540	0.910	0.020	0.891	
40.37s	2.36a	3.322	0.719	0.018	0.701	
50.37s	3.74a	2.937	0.416	0.015	0.401	
59.79s	5.03a	2.493	0.012	0.012	0.000	RaZero
60.37s	5.11a	2.464	-0.016	0.011	-0.027	
70.37s	6.35a	1.928	-0.525	0.008	-0.532	
80.37s	7.42a	1.338	-1.062	0.004	-1.066	
90.37s	8.25a	0.700	-1.576	0.000	-1.576	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1517.3 MT-m was used to adjust the VCG.

Note:

Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Stbd heeling mmt = 66.39Cos(heel) + 0.00Sin(heel) <= 90 deg
Stbd heeling mmt = 0.00 > 90 deg

FIRE FIGHTING CRITERIA



Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	0.42	4.58	Yes

10 SYMBOLS AND SIGN CONVENTIONS

AP	at	Frame 0
FP	at	57.400 Fwd of AP
Midship	at	100mm AFT of FR.48

Note: The Baseline of the vessel is on the upper side of the keel.

List of Symbols

Symbol	Unit	Description
LOA	m	Length Overall
LBP	m	Length Between Perpendiculars
B(Mld)	m	Moulded Breadth
D(Mld)	m	Moulded Depth
TPCm	T/cm	Tones per 1 cm immersion
MCTCm	T-m/cm	Moment to change Trim by 1 cm
Draft (AP)	m	Draft at A.P (moulded)
Draft (FP)	m	Draft at F.P (moulded)
Draft (Mean)	m	[Draft (AP) + Draft (FP)] x 0.5
VCG (KG)	m	Vertical Center of Gravity above Base Line Before correcting for free surface
KGo	m	Vertical Center of Gravity above Base Line after correcting for free surface
KB	m	Vertical Center of Buoyancy from Base Line
KMt	m	Transverse Metacenter above Base Line
GoMt	m	Transverse Metacenter above KGo
GZ	m	Righting Arm
FSM	T-m	Free Surface Moment
Sp.gr.		Specific Gravity

Sign Convention

LCB, LCG, LCF	Reference is MIDSHIP (100mm AFT of FR.48) Fwd of MIDSHIP (number)f, Aft of MIDSHIP (number)a
TRIM BY AFT	(number) a
TRIM BY FWD	(number) f
HEEL	“S” Starboard side / “P” Port side

11 METRIC CONVERSION TABLE

MULTIPLY BY	TO CONVERT FROM	TO OBTAIN	
0.03937	MILLIMETERS	INCHES	25.400
0.39370	CENTIMETERS	INCHES	2.5400
3.28080	METERS	FEET	0.3048
2.20460	KILOGRAMS	POUNDS	0.4540
0.00098	KILOGRAMS	TONS (2240 LB)	1016.10
0.98420	METRIC TONNES (TONNES OF 1000 KG)	TONS (2240 LB)	1.0160
2.49980	METRIC TONNES PER CM OF IMMERSION	TONS PER INCH OF IMMERSION	0.4000
8.20140	MOMENT TO CHANGE TRIM BY 1 CM (TONNE METER UNITS)	MOMENT TO CHANGE TRIM BY 1 INCH (TON FOOT UNITS)	0.122
187.976	METER RADIANS	FEET DEGREES	0.0053
	TO OBTAIN	TO CONVERT FROM	MULTIPLY

RELATION BETWEEN WEIGHT AND VOLUME

1000 cubic millimeters	1 Cubic Centimeter
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

12 LOADING CONDITIONS

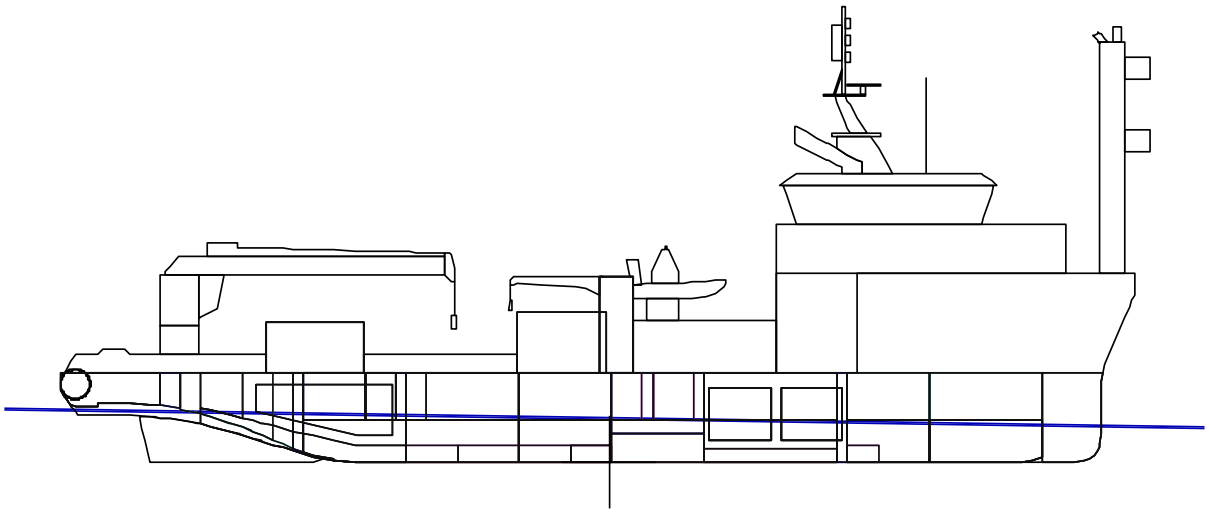
12.1 CONDITION 1: LIGHTSHIP (NOT A SAILING CONDITION)

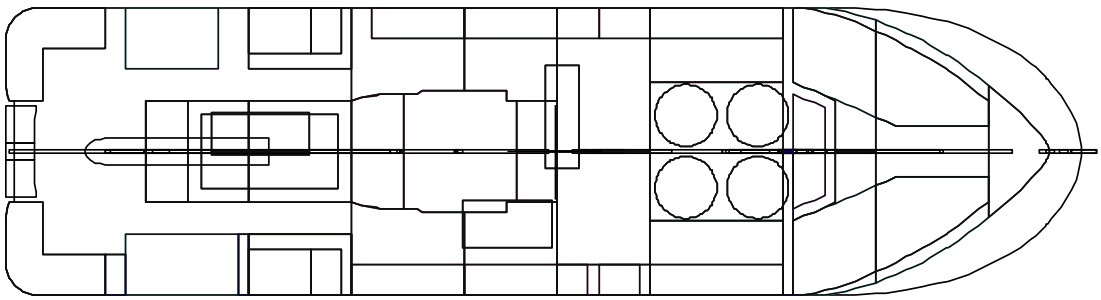
Floating Status

Draft FP	2.220 m	Heel	2.75p deg	GM(Solid)	5.031 m
Draft MS	2.663 m	Equil	Yes	F/S Corr.	0.000 m
Draft AP	3.106 m	Wind	Off	GM(Fluid)	5.031 m
Trim	0.88a m	Wave	No	KMT	10.816 m
LCG	0.592a m	VCG-Solid	5.792 m	TPcm	8.64
Displacement	1912.0 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Displacement	1,911.97	0.592a	0.243p	5.792





Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
Total Weight:	**1,911.97**	**0.592a**	**0.243p**	**5.792**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	1,888.67	0.384a	0.458p	1.438	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**1,911.97**	**0.660a**	**0.453p**	**1.433**	

Unprotected Flood Points

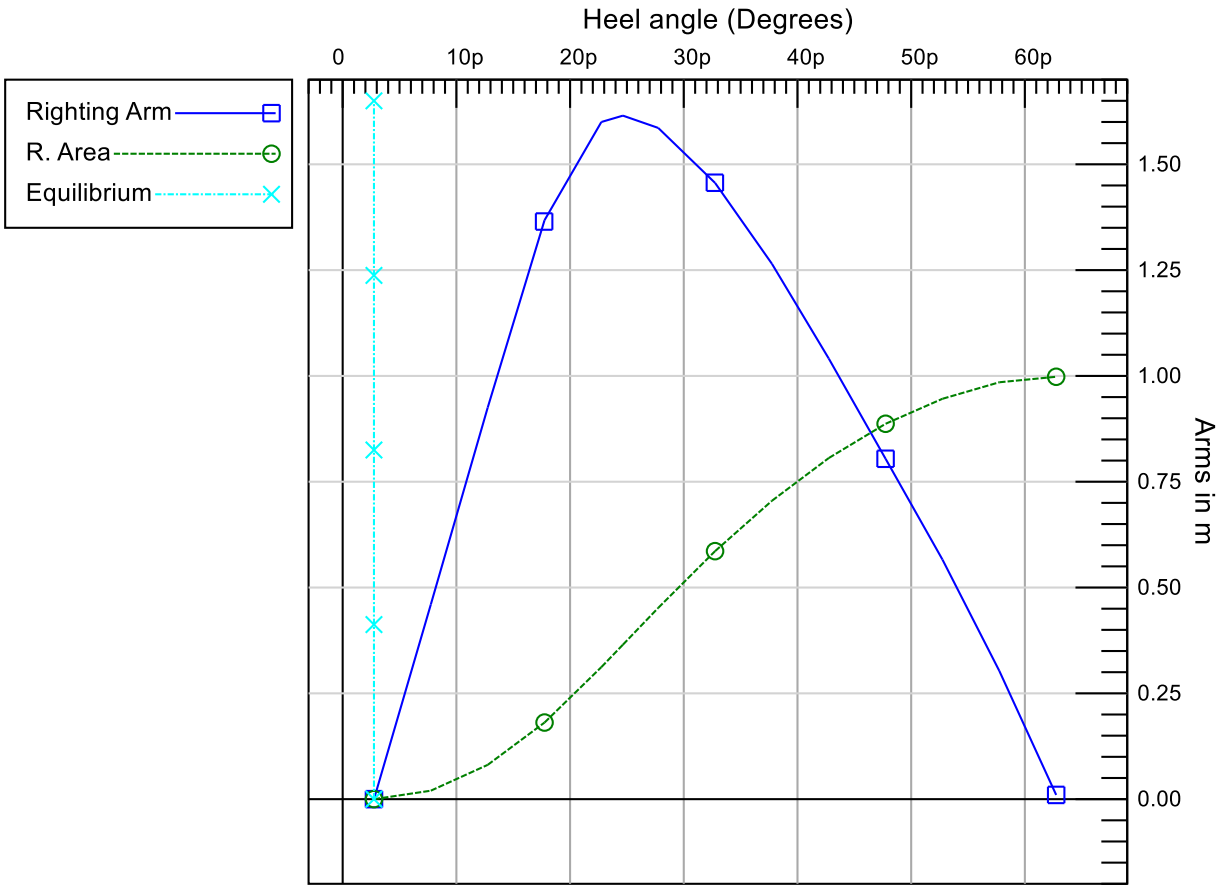
Name	L (m)	T (m)	V (m)	Height (m)
(1) POSITION 1(S)	9.700f	5.400s	12.150	9.884
(2) POSITION 1(P)	9.700f	5.400p	12.150	9.365
(3) POSITION 2(S)	10.300f	6.300s	12.150	9.936
(4) POSITION 2(P)	10.300f	6.300p	12.150	9.331

: OSV CHEMICAL CODE – CHAPTER 2 , 2.2

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Area (m-Rad)	Flood Pt Height (m)	Notes
2.75p	0.88a	2.660	0.000	0.000	9.331 (4)	Equil
7.75p	0.72a	2.613	0.460	0.020	8.705 (4)	
12.75p	0.50a	2.532	0.925	0.080	8.018 (4)	
17.75p	0.26a	2.411	1.368	0.181	7.285 (4)	
22.75p	0.06a	2.244	1.600	0.312	6.535 (4)	
24.65p	0.01a	2.169	**1.615**	0.365	6.249 (4)	MaxRa
27.75p	0.03f	2.035	1.586	0.452	5.777 (4)	
32.75p	0.03f	1.791	1.456	0.586	5.014 (4)	
37.75p	0.06a	1.514	1.265	0.705	4.246 (4)	
42.75p	0.24a	1.206	1.041	0.806	3.482 (4)	
47.75p	0.52a	0.869	0.804	0.887	2.730 (4)	
52.75p	0.91a	0.502	0.566	0.946	2.000 (4)	
57.75p	1.38a	0.122	0.303	0.985	1.280 (4)	
62.75p	1.89a	-0.264	0.010	0.998	0.564 (4)	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.

: OSV CHEMICAL CODE – CHAPTER 2 , 2.2



Limit	Min/Max	Actual	Margin	Pass
(1) Area from 0.00 deg to MaxRA at 24.65	>0.0604 m-R	0.365	0.305	Yes
(2) Area from 0.00 deg to 40.00 or Flood	>0.0900 m-R	0.806	0.716	Yes
(3) Area from 30.00 deg to 40.00 or Flood	>0.0300 m-R	0.220	0.190	Yes
(4) Righting Arm at 30.00 deg	>0.200 m	1.456	1.256	Yes
(5) Absolute Angle at MaxRA	>15.00 deg	24.65	9.65	Yes
(6) GM Upright	>0.150 m	5.004	4.854	Yes

Heeling Moment Derivation

Part	LPA (m^2)	HCP (m)	Arm (m)	Pressure (MT/m^2)	Moment (m-MT)
HULL.C	284.5	3.096	4.396	0.051	64.29
SKEG.C	0.0	0.015	1.315	0.051	0.00
DECKHOUSE	108.4	9.654	10.954	0.051	61.04
FWDMAST	13.4	17.166	18.466	0.051	12.74
ANTENNA	0.3	17.017	18.317	0.051	0.25
CRANE1	23.5	15.865	17.165	0.051	20.70
DECK	45.1	3.666	4.966	0.051	11.52
CRANE2	37.2	7.977	9.277	0.051	17.72
LIFEBOAT	10.0	7.468	8.768	0.051	4.50
CRANE3	2.9	7.828	9.128	0.051	1.35

Total wind heeling moment 194.12 to port

IMO SEVERE WIND & ROLLING

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
26.90s	0.03f	2.074	-2.031	0.152	-2.183	0.000	5.906 (3)	-1.153	Roll
21.90s	0.09a	2.275	-2.031	0.152	-2.184	-0.191	6.663 (3)	-0.536	
16.90s	0.30a	2.435	-1.760	0.152	-1.912	-0.371	7.412 (3)	0.037	
11.90s	0.54a	2.548	-1.319	0.152	-1.471	-0.520	8.138 (3)	0.622	
6.90s	0.76a	2.624	-0.861	0.152	-1.013	-0.629	8.816 (3)	1.229	
1.90s	0.89a	2.663	-0.408	0.152	-0.561	-0.697	9.430 (3)	1.878	
3.10p	0.88a	2.658	0.030	0.152	-0.122	-0.727	9.290 (4)	1.715	
3.89p	0.86a	2.653	0.101	0.152	-0.051	-0.728	9.195 (4)	1.608	SSWEq
4.44p	0.84a	2.649	0.152	0.152	0.000	-0.728	9.127 (4)	1.537	Equil
8.10p	0.71a	2.609	0.492	0.152	0.340	-0.718	8.659 (4)	1.082	
13.10p	0.49a	2.525	0.957	0.152	0.805	-0.668	7.969 (4)	0.479	
16.00p	0.35a	2.459	1.220	0.152	1.067	-0.620	7.546 (4)	0.140	
17.23p	0.28a	2.426	1.325	0.152	1.172	-0.596	7.364 (4)	0.000	Deck Imm.
18.10p	0.24a	2.401	1.395	0.152	1.243	-0.578	7.234 (4)	-0.097	
23.10p	0.05a	2.231	1.605	0.152	1.453	-0.459	6.483 (4)	-0.677	
24.67p	0.01a	2.169	1.615	0.152	**1.462**	-0.419	6.246 (4)	-0.865	MaxRa
28.10p	0.04f	2.020	1.580	0.152	1.428	-0.332	5.725 (4)	-1.306	
33.10p	0.03f	1.773	1.444	0.152	1.292	-0.212	4.961 (4)	-1.924	
38.10p	0.07a	1.494	1.250	0.152	1.098	-0.108	4.193 (4)	-2.543	
43.10p	0.26a	1.184	1.025	0.152	0.873	-0.022	3.430 (4)	-3.189	
48.10p	0.55a	0.844	0.788	0.152	0.635	0.044	2.679 (4)	-3.842	
50.00p	0.68a	0.707	0.697	0.152	0.545	0.064	2.399 (4)	-4.093	

IMO SEVERE WIND & ROLLING

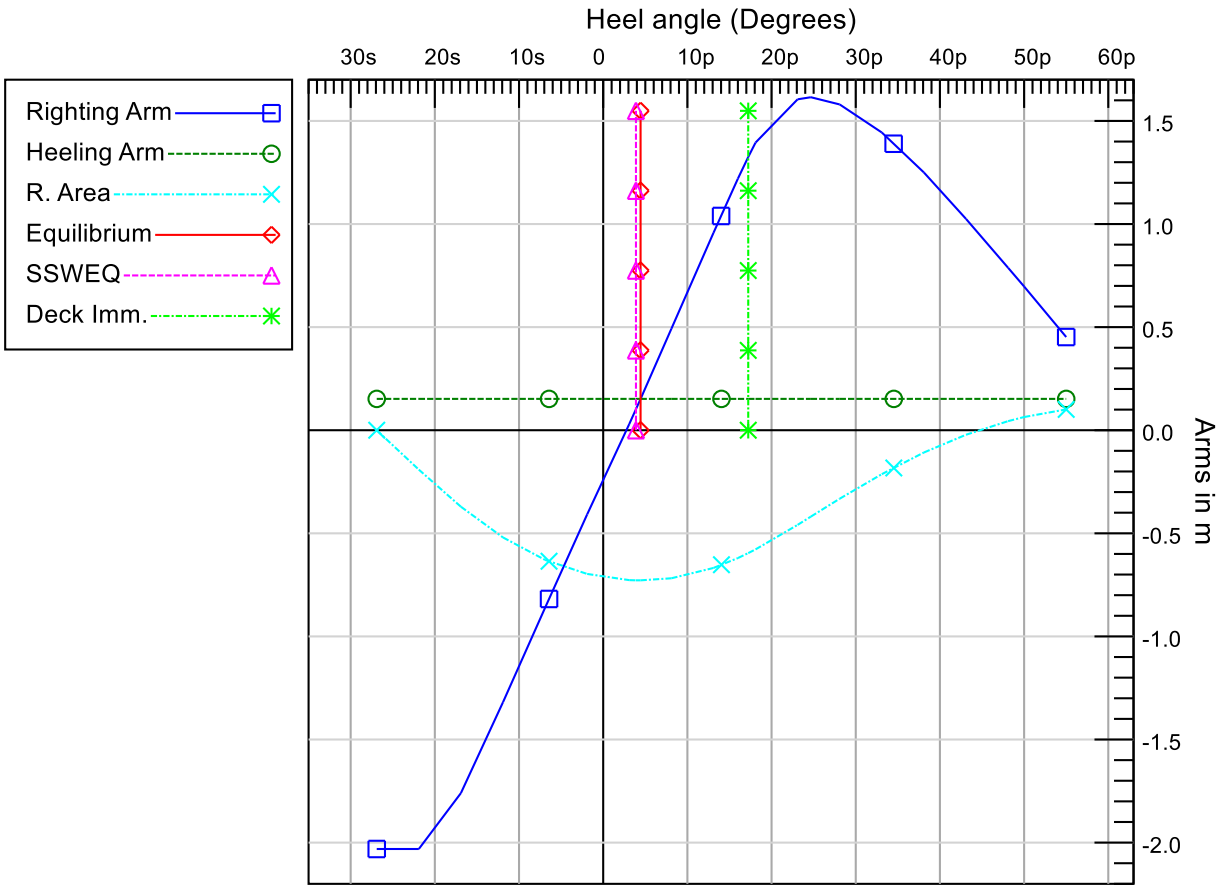
Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
55.00p	1.12a	0.332	0.452	0.152	0.300	0.101	1.676 (4)	-4.757	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.

Note:
Residual Righting Arms shown above are in excess of the
wind heeling arms derived from this moment (in m-MT):
Port heeling moment = 291.17

Roll angle is 30.80
Equilibrium for load condition without gust is 3.89p

IMO SEVERE WIND & ROLLING



Limit	Min/Max	Actual	Margin	Pass
(1) Res. Ratio from Roll to Abs 50.00 deg or Flood	>1.000	1.088	0.088	Yes
(2) Angle from SSWEQ to 80% of Dk. Imm. or Abs 16.00 deg	>0.00 deg	9.89	9.89	Yes

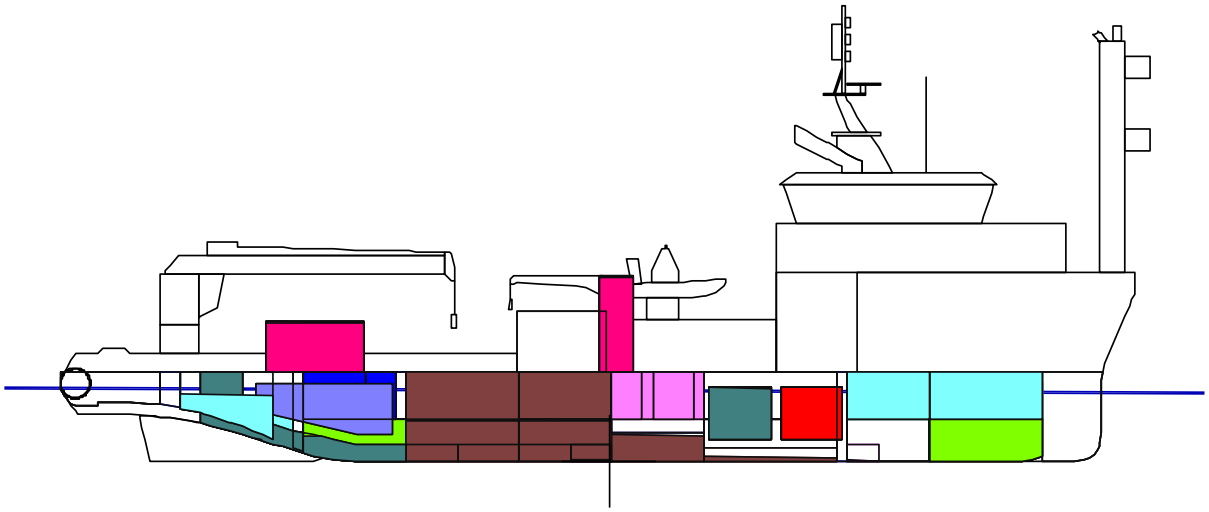
12.2 CONDITION NO.2: FULLY LOADED DEPARTURE

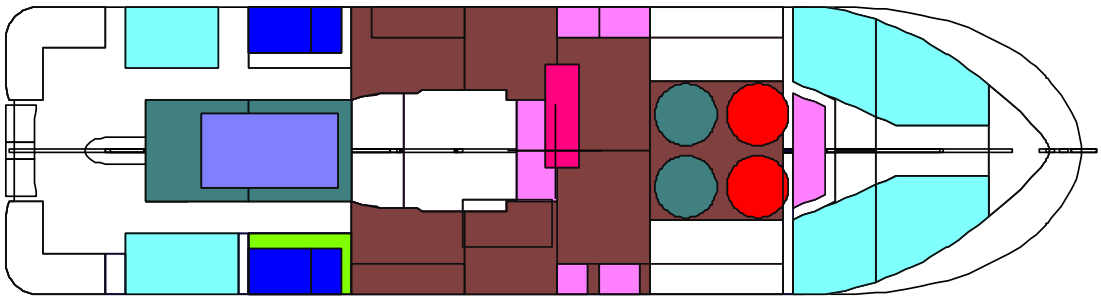
Floating Status

Draft FP	4.061 m	Heel	0.16s deg	GM(Solid)	4.045 m
Draft MS	4.200 m	Equil	Yes	F/S Corr.	0.413 m
Draft AP	4.339 m	Wind	Off	GM(Fluid)	3.632 m
Trim	0.28a m	Wave	No	KMT	8.743 m
LCG	1.013a m	VCG-Solid	4.698 m	TPcm	9.64
Displacement	3305.7 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	1,393.70	1.591a	0.358s	3.197
Displacement	3,305.67	1.013a	0.010s	4.698





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	88.68	21.33	FRESH WATER	325.38	80.21
FUEL OIL	303.47	58.38	MISC	51.30	40.12
SCALE INHIBITOR	57.86	98.00	CORROSION INHIBITOR	24.99	98.00
TREATED INJECTION WATER	61.80	100.00	TK ON MAIN DECK	123.45	66.80
BRINE	252.16	98.00	ACID	96.53	98.00

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	4.00	24.100f	0.000	6.800
Total Fixed:	**1,920.05**	**0.495a**	**0.242p**	**5.805**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_DWSWB.S	1.025	100.00	39.49	21.049f	3.353s	1.426	0.0
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
Subtotals:		**21.33%**	**88.68**	**1.184f**	**5.120s**	**1.385**	**194.6**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	76.00	56.11	21.476f	3.707p	3.617	63.6
NO1_FW.S	1.000	100.00	73.83	21.497f	3.747s	3.946	63.6
NO2_FW.P	1.000	100.00	66.44	16.374f	5.508p	3.914	0.0
NO2_FW.S	1.000	100.00	66.44	16.374f	5.508s	3.914	0.0
NO3_FW.S	1.000	50.00	35.59	20.969a	6.514s	2.843	25.0
NO3_FW.P	1.000	50.00	26.97	21.828a	6.506p	3.064	20.5
Subtotals:		**80.21%**	**325.38**	**11.165f**	**0.384s**	**3.683**	**172.6**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.C	0.850	34.00	14.87	9.315f	0.052s	0.145	274.4
NO2_FO.U.P	0.850	34.00	14.99	2.856f	3.822p	0.223	198.5
NO2_FO.U.S	0.850	98.00	38.33	2.940f	4.277s	0.579	198.5
NO3_FO.P	0.850	98.00	21.99	2.604a	7.500p	3.872	2.2
NO3_FO.S	0.850	98.00	21.99	2.604a	7.500s	3.872	2.2
NO3_FO.U.P	0.850	98.00	31.43	2.571a	6.465p	0.947	2.2
NO3_FO.U.S	0.850	98.00	31.43	2.571a	6.465s	0.947	2.2
NO4_FO.U.P	0.850	98.00	39.79	8.640a	6.400p	0.926	2.7
NO4_FO.U.S	0.850	98.00	39.79	8.640a	6.401s	0.926	2.7
FODAY.P	0.850	98.00	21.99	8.004a	7.500p	3.872	2.2
FODAY.S	0.850	98.00	26.88	8.606a	7.500s	3.872	2.6
Subtotals:		**58.38%**	**303.47**	**3.549a**	**0.475s**	**1.715**	**690.2**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.999f	7.500p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.699f	7.500s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.299f	7.500p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	1.000f	7.500s	3.872	0.8
DIRTYOIL.P	0.900	9.80	0.62	1.124a	1.478p	0.049	4.8
BILGEHOLD.S	1.000	9.80	0.69	1.124a	1.521s	0.049	5.3
SEWAGE	1.000	9.80	1.01	14.751f	0.084s	0.049	30.3
Subtotals:		**40.12%**	**51.30**	**2.966f**	**0.912p**	**3.699**	**45.2**

SCALE INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
SCLEIHBTR.P	1.100	98.00	14.47	13.400a	7.048p	3.872	3.2
SCLEIHBTR.S	1.100	98.00	14.47	13.400a	7.049s	3.872	3.2
SCALE_DISLVR.S	1.100	98.00	28.93	16.102a	7.049s	3.872	6.4
Subtotals:		**98.00%**	**57.86**	**14.751a**	**3.525s**	**3.872**	**12.8**

CORROSION INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
CORSNIHBTR.P	0.950	98.00	24.99	16.102a	7.048p	3.872	5.5
Subtotals:		**98.00%**	**24.99**	**16.102a**	**7.048p**	**3.872**	**5.5**

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
TRTDINJW.P	1.000	100.00	30.90	11.800f	2.100p	2.850	0.0
TRTDINJW.S	1.000	100.00	30.90	11.800f	2.100s	2.850	0.0
Subtotals:		**100.00%**	**61.80**	**11.800f**	**0.000**	**2.850**	**0.0**

TK ON MAIN DECK

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NITROGENTK.P	1.185	98.00	47.27	17.245a	1.049p	6.721	8.7
CHEMADCTTK.P	1.185	98.00	76.18	0.410f	2.048p	8.020	42.0
Subtotals:		**66.80%**	**123.45**	**6.350a**	**1.666p**	**7.522**	**50.7**

BRINE

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_BRINETK.C	1.550	98.00	51.55	22.651a	0.004s	4.075	66.0
NO2_BRINETK.C	1.550	98.00	49.80	20.599a	0.001s	1.905	23.2
NO3_BRINETK.C	1.550	98.00	56.94	14.966a	0.001s	0.653	24.0
NO4_BRINETK.P	1.550	98.00	46.94	7.599f	2.099p	2.819	12.6
NO4_BRINETK.S	1.550	98.00	46.94	7.599f	2.101s	2.819	12.6
Subtotals:		**98.00%**	**252.16**	**9.249a**	**0.002s**	**2.406**	**138.3**

ACID

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_ACIDTK.C	1.160	98.00	96.53	16.257a	0.002s	3.322	55.6
Subtotals:		**98.00%**	**96.53**	**16.257a**	**0.002s**	**3.322**	**55.6**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**62.57%**	**1,385.62**	**1.731a**	**0.360s**	**3.164**	**1,365.7**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	3,282.37	0.869a	0.017s	2.290	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**3,305.67**	**1.025a**	**0.017s**	**2.281**	

Unprotected Flood Points

Name	L (m)	T (m)	V (m)	Height (m)
(1) POSITION 1(S)	9.700f	5.400s	12.150	7.982
(2) POSITION 1(P)	9.700f	5.400p	12.150	8.012
(3) POSITION 2(S)	10.300f	6.300s	12.150	7.982
(4) POSITION 2(P)	10.300f	6.300p	12.150	8.017

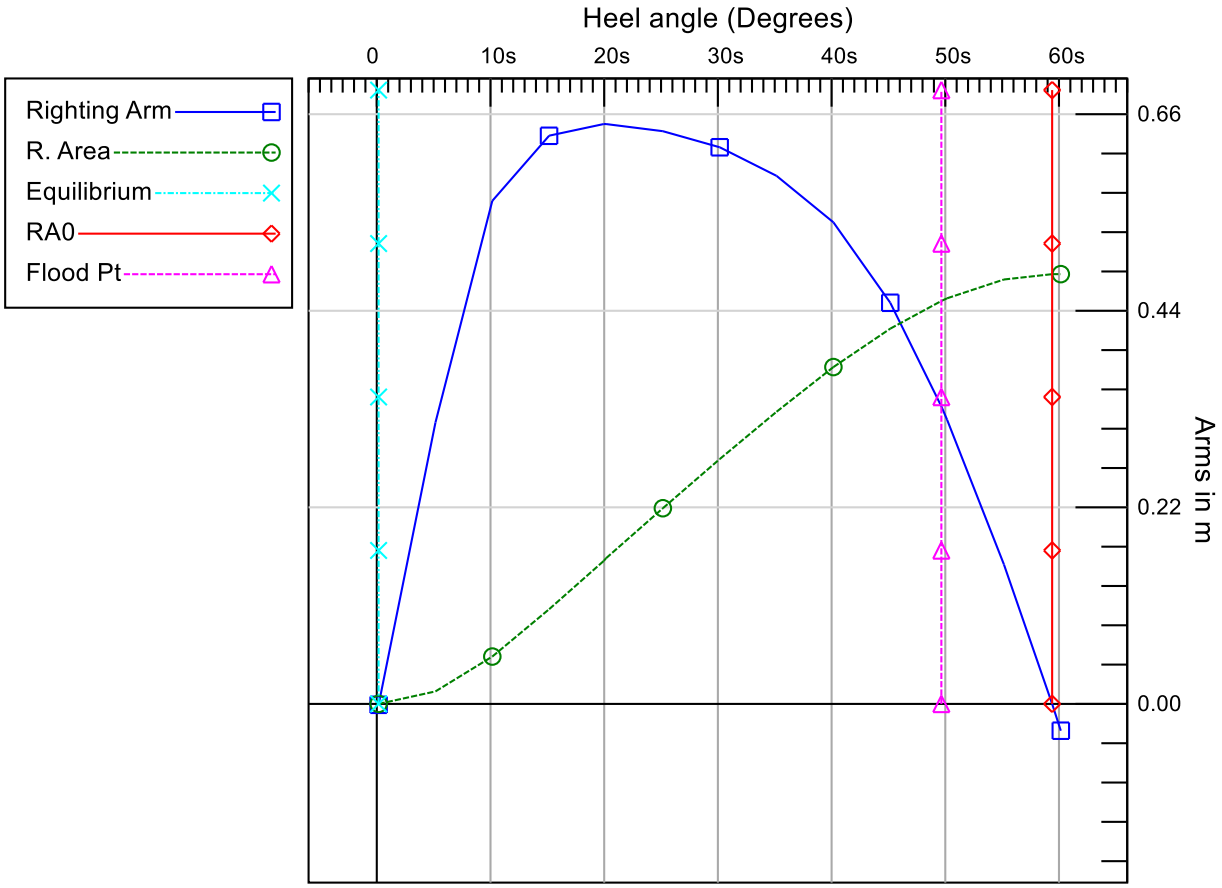
: OSV CHEMICAL CODE – CHAPTER 2 , 2.2

Free Surface Adjustment 0.413
Adjusted VCG 5.111

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Area (m-Rad)	Flood Pt Height (m)	Notes
0.16s	0.28a	4.200	-0.001	0.000	7.982 (1)	Equil
5.16s	0.27a	4.181	0.315	0.014	7.402 (3)	
10.16s	0.30a	4.147	0.563	0.053	6.754 (3)	
15.16s	0.49a	4.161	0.636	0.106	6.007 (3)	
19.96s	0.81a	4.189	**0.649**	0.160	5.226 (3)	MaxRa
20.16s	0.83a	4.190	0.649	0.163	5.193 (3)	
25.16s	1.30a	4.222	0.641	0.219	4.327 (3)	
30.16s	1.88a	4.240	0.623	0.274	3.433 (3)	
35.16s	2.58a	4.226	0.591	0.327	2.536 (3)	
40.16s	3.39a	4.164	0.539	0.377	1.658 (3)	
45.16s	4.24a	4.064	0.449	0.420	0.786 (3)	
49.65s	4.99a	3.948	0.333	0.451	0.000 (3)	FldPt
50.16s	5.07a	3.934	0.318	0.454	-0.089 (3)	
55.16s	5.87a	3.777	0.156	0.475	-0.964 (3)	
59.40s	6.50a	3.626	0.000	0.481	-1.702 (3)	RaZero
60.16s	6.62a	3.597	-0.030	0.481	-1.834 (3)	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1365.7 MT-m was used to adjust the VCG.

: OSV CHEMICAL CODE – CHAPTER 2 , 2.2



Limit	Min/Max	Actual	Margin	Pass
(1) Area from 0.00 deg to MaxRA at 19.96	>0.0650 m-R	0.160	0.095	Yes
(2) Area from 0.00 deg to 40.00 or Flood	>0.0900 m-R	0.377	0.287	Yes
(3) Area from 30.00 deg to 40.00 or Flood	>0.0300 m-R	0.103	0.073	Yes
(4) Righting Arm at 30.00 deg	>0.200 m	0.623	0.423	Yes
(5) Absolute Angle at MaxRA	>15.00 deg	19.96	4.96	Yes
(6) GM Upright	>0.150 m	3.632	3.482	Yes

Heeling Moment Derivation

Part	LPA (m^2)	HCP (m)	Arm (m)	Pressure (MT/m^2)	Moment (m-MT)
HULL.C	189.9	2.712	4.693	0.051	45.82
DECKHOUSE	108.4	8.118	10.099	0.051	56.28
FWDMAST	13.4	15.630	17.611	0.051	12.15
ANTENNA	0.3	15.481	17.462	0.051	0.24
CRANE1	23.5	14.329	16.310	0.051	19.66
DECK	45.1	2.131	4.111	0.051	9.54
CRANE2	37.2	6.441	8.422	0.051	16.09
LIFEBOAT	10.0	5.932	7.913	0.051	4.06
CRANE3	2.9	6.293	8.273	0.051	1.23

Total wind heeling moment 165.07 to starboard

IMO SEVERE WIND & ROLLING

Free Surface Adjustment 0.413

Adjusted VCG 5.111

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
21.93p	0.98a	4.202	-0.672	0.075	-0.747	0.000	4.891 (4)	-2.938	Roll
16.93p	0.60a	4.170	-0.669	0.075	-0.744	-0.065	5.725 (4)	-1.859	
11.93p	0.35a	4.148	-0.626	0.075	-0.701	-0.128	6.499 (4)	-0.880	
6.93p	0.26a	4.165	-0.452	0.075	-0.527	-0.183	7.183 (4)	-0.056	
1.93p	0.28a	4.197	-0.138	0.075	-0.212	-0.216	7.783 (4)	0.670	
1.02s	0.28a	4.199	0.050	0.075	-0.025	-0.222	7.886 (3)	0.803	SSWEq
1.41s	0.28a	4.198	0.075	0.075	0.000	-0.223	7.842 (3)	0.746	Equil
3.07s	0.28a	4.193	0.180	0.075	0.105	-0.221	7.652 (3)	0.503	
6.54s	0.27a	4.169	0.399	0.075	0.324	-0.208	7.232 (3)	0.000	Deck Imm.
8.07s	0.27a	4.155	0.485	0.075	0.410	-0.198	7.039 (3)	-0.225	
13.07s	0.39a	4.151	0.613	0.075	0.539	-0.157	6.329 (3)	-1.091	
16.00s	0.54a	4.165	0.637	0.075	0.562	-0.129	5.874 (3)	-1.667	
18.07s	0.67a	4.177	0.644	0.075	0.569	-0.108	5.541 (3)	-2.095	
20.04s	0.82a	4.190	0.646	0.075	**0.571**	-0.088	5.212 (3)	-2.518	MaxRa
23.07s	1.09a	4.209	0.643	0.075	0.568	-0.058	4.694 (3)	-3.192	
28.07s	1.62a	4.235	0.629	0.075	0.554	-0.009	3.809 (3)	-4.358	
33.07s	2.27a	4.237	0.604	0.075	0.529	0.038	2.910 (3)	-5.570	
38.07s	3.04a	4.196	0.561	0.075	0.486	0.082	2.024 (3)	-6.799	
43.07s	3.88a	4.109	0.490	0.075	0.415	0.122	1.152 (3)	-8.013	
48.07s	4.73a	3.992	0.375	0.075	0.300	0.154	0.278 (3)	-9.179	

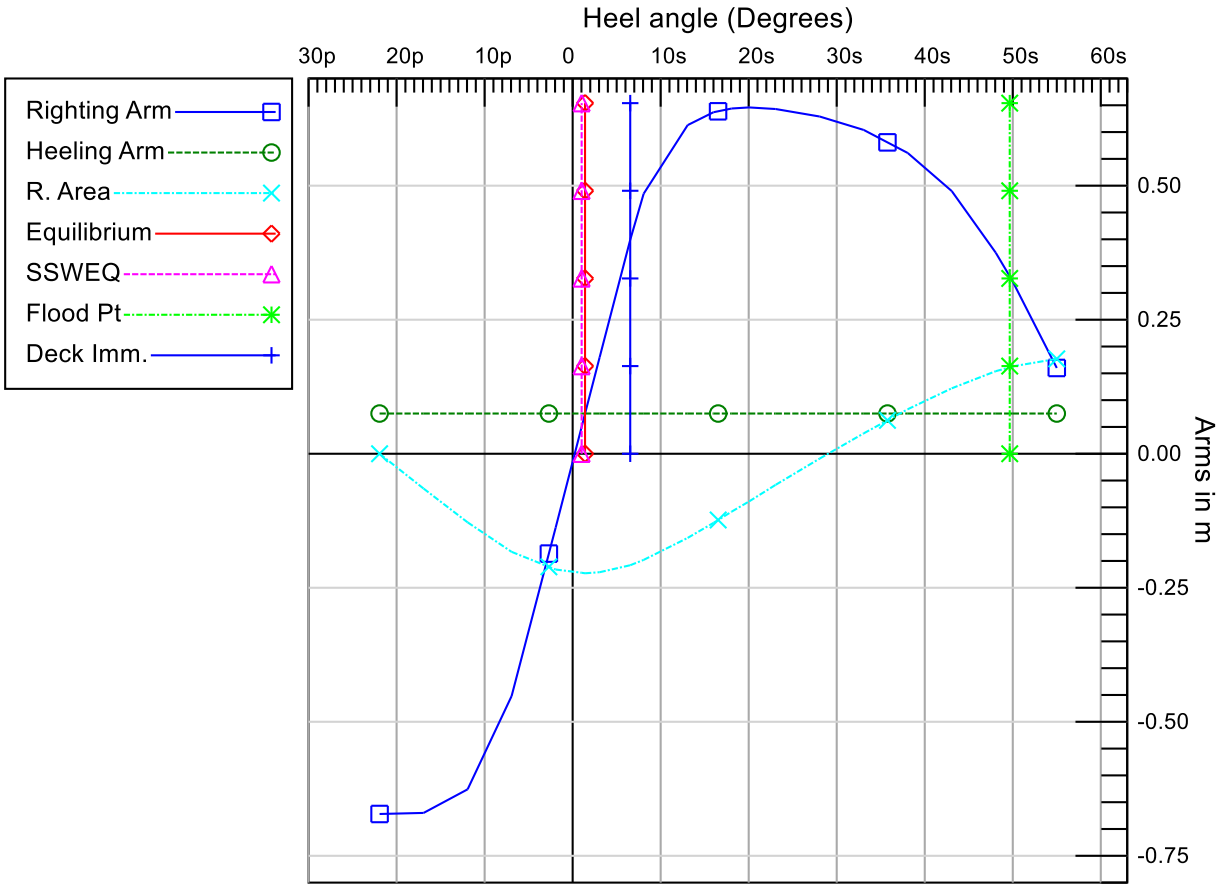
Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
49.65s	4.98a	3.948	0.331	0.075	0.256	0.161	0.000 (3)	-9.534	FldPt
50.00s	5.04a	3.938	0.321	0.075	0.246	0.163	-0.061 (3)	-9.613	
55.00s	5.84a	3.783	0.160	0.075	0.085	0.177	-0.936 (3)	-10.683	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1363.9 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
wind heeling arms derived from this moment (in m-MT):
Stbd heeling moment = 247.60

Roll angle is 22.90
Equilibrium for load condition without gust is 1.02s

IMO SEVERE WIND & ROLLING



Limit	Min/Max	Actual	Margin	Pass
(1) Res. Ratio from Roll to Abs 50.00 deg or Flood	>1.000	1.725	0.725	Yes
(2) Angle from SSWEQ to 80% of Dk. Imm. or Abs 16.00 deg	>0.00 deg	4.21	4.21	Yes

Heeling Moment Report

FIFI HEELING MOMENT CALCULATION			
Reaction Force of the Fire Monitor	=	1.530	MT
Number of Fire Monitors	=	2	
Draft of the loading condition	=	4.2	m
D/2	=	2.1	m
Height of firefighting monitor above baseline	=	19.600	m
Moment due to Fire Fighting operation	=	53.550	MT-m
Reaction Force of the Bow Thruster	=	18	MT
D/2	=	2.1	m
Height of Bow Thruster above baseline	=	1.200	m
Moment due to Bow Thruster operation	=	16.2	MT-m
Total Heeling Moment	=	69.750	Tm

Heeling moment: 69.75 MT-m to starboard

FIRE FIGHTING CRITERIA

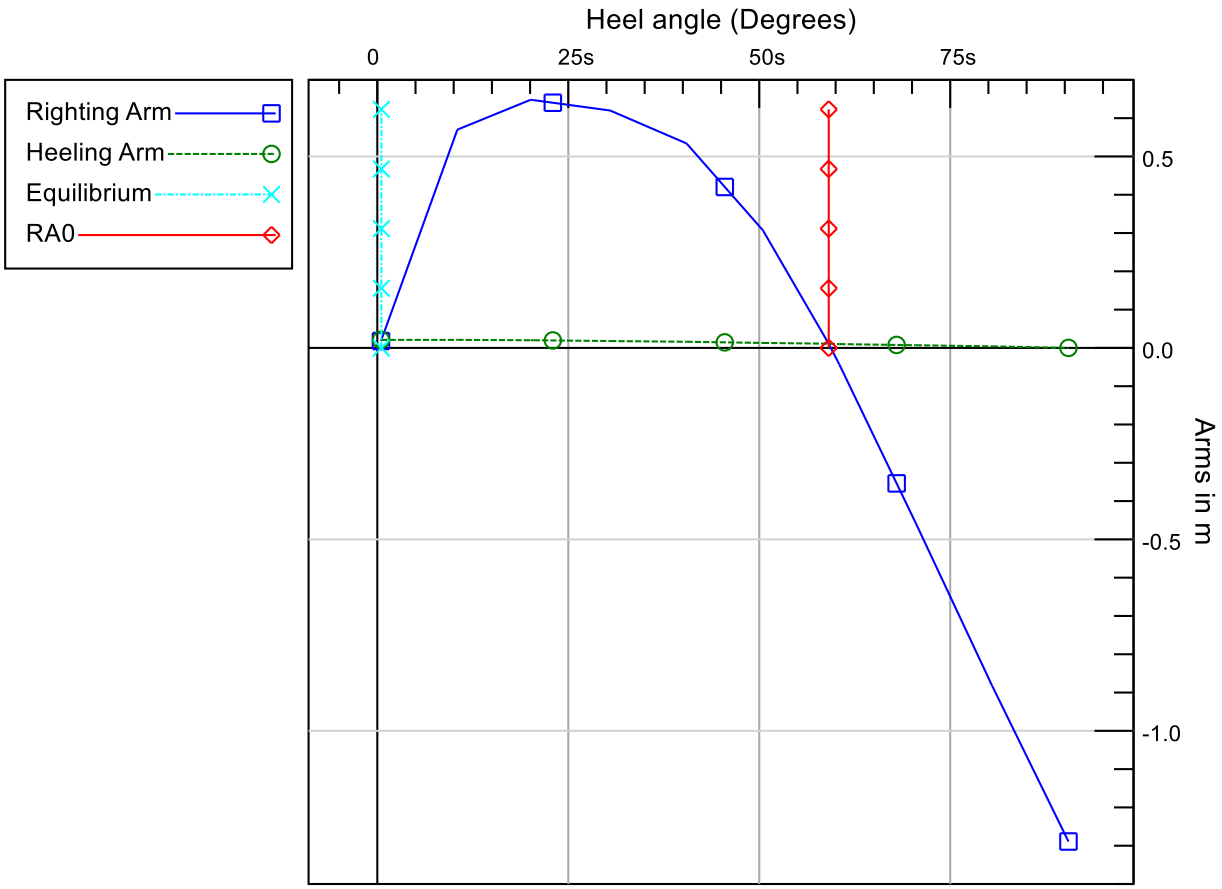
Free Surface Adjustment 0.413
Adjusted VCG 5.111

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Notes
0.48s	0.28a	4.200	0.018	0.021	-0.003	
0.53s	0.28a	4.200	0.021	0.021	0.000	Equil
10.48s	0.30a	4.147	0.570	0.021	0.549	
20.07s	0.82a	4.190	0.648	0.020	**0.628**	MaxRa
20.48s	0.86a	4.193	0.647	0.020	0.628	
30.48s	1.92a	4.240	0.620	0.018	0.602	
40.48s	3.44a	4.158	0.534	0.016	0.518	
50.48s	5.12a	3.925	0.307	0.013	0.294	
59.10s	6.46a	3.637	0.011	0.011	0.000	RaZero
60.48s	6.66a	3.585	-0.043	0.010	-0.053	
70.48s	8.00a	3.151	-0.457	0.007	-0.464	
80.48s	9.11a	2.616	-0.882	0.003	-0.886	
90.48s	9.95a	1.997	-1.289	0.000	-1.289	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1366.5 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Stbd heeling mmt = 69.75Cos(heel) + 0.00Sin(heel) <= 90 deg
Stbd heeling mmt = 0.00 > 90 deg

FIRE FIGHTING CRITERIA



Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	0.53	4.47	Yes

12.2.1 MAX TRANSVERSE CRANE OPERATION

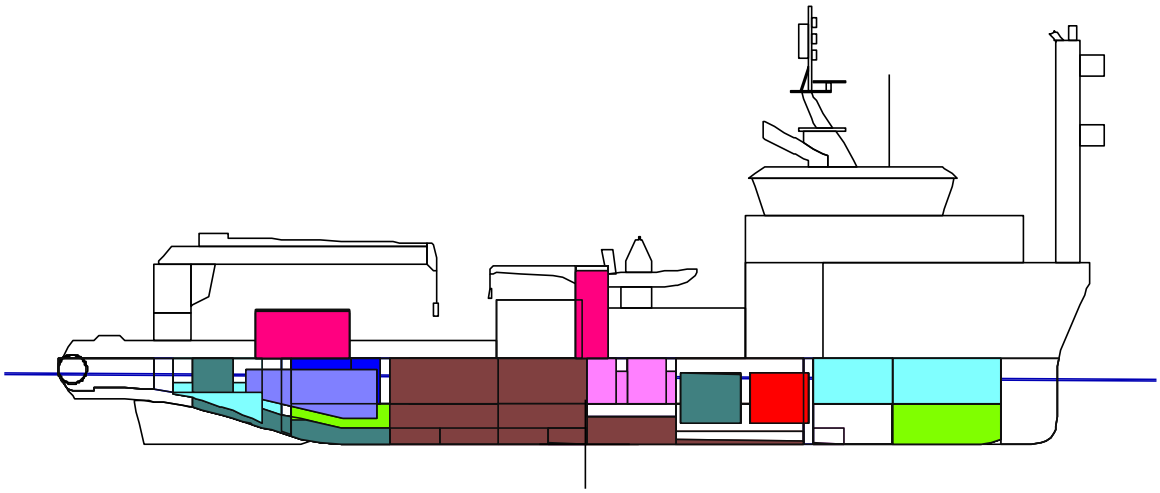
Max. Transverse moment – Lifting 3.500t @ 35.00 m radius (35.530 m boom length –Side Lifting to port side)

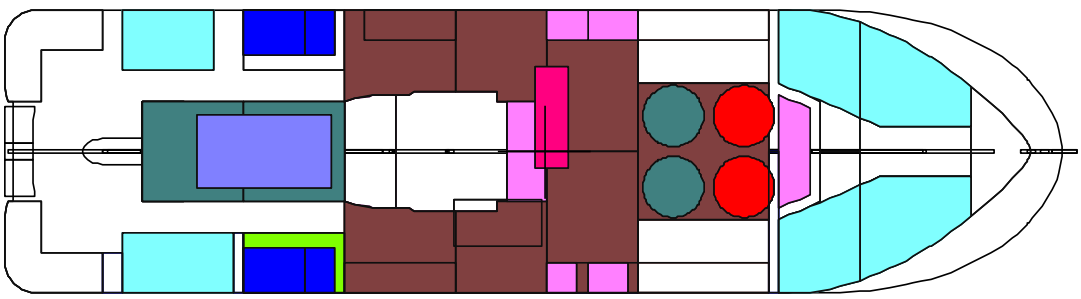
Floating Status

Draft FP	4.021 m	Heel	3.06p deg	GM(Solid)	4.040 m
Draft MS	4.198 m	Equil	Yes	F/S Corr.	0.370 m
Draft AP	4.375 m	Wind	Off	GM(Fluid)	3.669 m
Trim	0.35a m	Wave	No	KMT	8.736 m
LCG	1.115a m	VCG-Solid	4.702 m	TPcm	9.64
Displacement	3309.2 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	1,397.20	1.830a	0.181p	3.210
Displacement	3,309.17	1.115a	0.217p	4.702





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	88.68	21.33	FRESH WATER	325.38	80.21
FUEL OIL	303.47	58.38	MISC	51.30	40.12
SCALE INHIBITOR	57.86	98.00	CORROSION INHIBITOR	24.99	98.00
TREATED INJECTION WATER	61.80	100.00	TK ON MAIN DECK	123.45	66.80
BRINE	252.16	98.00	ACID	96.53	98.00

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	4.00	24.100f	0.000	6.800
CRANE BOOM FINAL	32.55	25.100a	23.235p	11.190
CRANE BOOM INITIAL	-32.55	17.438a	6.000p	11.504
LIFTED WEIGHT FINAL	3.50	25.100a	41.000p	11.190
Total Fixed:	**1,923.55**	**0.669a**	**0.608p**	**5.809**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_DWSWB.S	1.025	100.00	39.49	21.049f	3.353s	1.426	0.0
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
Subtotals:		**21.33%**	**88.68**	**1.184f**	**5.120s**	**1.385**	**194.6**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	76.00	56.11	21.445f	3.768p	3.618	63.6
NO1_FW.S	1.000	100.00	73.83	21.497f	3.747s	3.946	63.6
NO2_FW.P	1.000	100.00	66.44	16.374f	5.508p	3.914	0.0
NO2_FW.S	1.000	100.00	66.44	16.374f	5.508s	3.914	0.0
NO3_FW.S	1.000	50.00	35.59	20.972a	6.475s	2.844	24.9
NO3_FW.P	1.000	50.00	26.97	21.830a	6.549p	3.066	20.6
Subtotals:		**80.21%**	**325.38**	**11.160f**	**0.366s**	**3.683**	**172.7**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.C	0.850	34.00	14.87	9.292f	0.987p	0.171	275.6

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_FO.U.P	0.850	34.00	14.99	2.894f	4.538p	0.241	198.5
NO2_FO.U.S	0.850	98.00	38.33	2.940f	4.275s	0.579	198.5
NO3_FO.P	0.850	98.00	21.99	2.605a	7.505p	3.872	2.1
NO3_FO.S	0.850	98.00	21.99	2.605a	7.495s	3.872	2.1
NO3_FO.U.P	0.850	98.00	31.43	2.572a	6.469p	0.947	2.2
NO3_FO.U.S	0.850	98.00	31.43	2.572a	6.462s	0.947	2.2
NO4_FO.U.P	0.850	98.00	39.79	8.641a	6.404p	0.926	2.7
NO4_FO.U.S	0.850	98.00	39.79	8.641a	6.397s	0.926	2.7
FODAY.P	0.850	98.00	21.99	8.005a	7.505p	3.872	2.1
FODAY.S	0.850	98.00	26.88	8.608a	7.495s	3.872	2.5
Subtotals:		**58.38%**	**303.47**	**3.549a**	**0.385s**	**1.717**	**691.1**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.998f	7.505p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.699f	7.495s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.299f	7.505p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	0.999f	7.495s	3.872	0.8
DIRTYOIL.P	0.900	9.80	0.62	1.130a	1.912p	0.060	4.8
BILGEHOLD.S	1.000	9.80	0.69	1.130a	1.088s	0.060	5.3
SEWAGE	1.000	9.80	1.01	14.742f	1.414p	0.083	30.3
Subtotals:		**40.12%**	**51.30**	**2.965f**	**0.958p**	**3.700**	**45.2**

SCALE INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
SCLEIHBTR.P	1.100	98.00	14.47	13.401a	7.060p	3.872	2.2
SCLEIHBTR.S	1.100	98.00	14.47	13.401a	7.037s	3.872	2.2
SCALE_DISLVR.S	1.100	98.00	28.93	16.102a	7.037s	3.872	4.4
Subtotals:		**98.00%**	**57.86**	**14.751a**	**3.513s**	**3.872**	**8.8**

CORROSION INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
CORSNIHBTR.P	0.950	98.00	24.99	16.102a	7.060p	3.872	3.8
Subtotals:		**98.00%**	**24.99**	**16.102a**	**7.060p**	**3.872**	**3.8**

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
TRTDINJW.P	1.000	100.00	30.90	11.800f	2.100p	2.850	0.0
TRTDINJW.S	1.000	100.00	30.90	11.800f	2.100s	2.850	0.0

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Subtotals:		**100.00%**	**61.80**	**11.800f**	**0.000**	**2.850**	**0.0**

TK ON MAIN DECK

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NITROGENTK.P	1.185	98.00	47.27	17.245a	1.060p	6.721	7.0
CHEMADCTTK.P	1.185	98.00	76.18	0.410f	2.077p	8.020	24.3
Subtotals:		**66.80%**	**123.45**	**6.351a**	**1.688p**	**7.523**	**31.3**

BRINE

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_BRINETK.C	1.550	98.00	51.55	22.651a	0.039p	4.075	10.8
NO2_BRINETK.C	1.550	98.00	49.80	20.601a	0.026p	1.906	15.3
NO3_BRINETK.C	1.550	98.00	56.94	14.968a	0.024p	0.654	17.3
NO4_BRINETK.P	1.550	98.00	46.94	7.598f	2.113p	2.820	7.9
NO4_BRINETK.S	1.550	98.00	46.94	7.598f	2.087s	2.820	7.9
Subtotals:		**98.00%**	**252.16**	**9.250a**	**0.024p**	**2.407**	**59.3**

ACID

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_ACIDTK.C	1.160	98.00	96.53	16.255a	0.024p	3.322	19.1
Subtotals:		**98.00%**	**96.53**	**16.255a**	**0.024p**	**3.322**	**19.1**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**62.57%**	**1,385.62**	**1.733a**	**0.325s**	**3.165**	**1,225.9**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	3,285.87	0.975a	0.348p	2.302	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**3,309.17**	**1.130a**	**0.346p**	**2.293**	

Heeling Moment Report

Draft	4.200	m					Resultant	1.0000
WIND	20	Knots					(Sin 90Deg)	
Part	**Wind area**	**Effective area**	**Centre of Area from sea-level**	**Ch**	**Lever**	**Cs**	**Wind Force**	**Moment**
	(m2)	**(m2)**	**m**		**(m)**		**(N)**	**(m-MT)**
HULL.C	189.900	189.900	2.593	1.000	4.693	1.000	12261.749	5.866
DECKHOUSE	108.400	108.400	7.999	1.000	10.099	1.000	6999.334	7.206
FWDMAST	13.400	13.400	15.511	1.100	17.611	1.000	951.754	1.709
ANTENNA	0.300	0.300	15.362	1.100	17.462	1.000	21.308	0.038
CRANE1	23.500	23.500	14.210	1.000	16.310	1.500	2276.075	3.784
DECK	45.100	45.100	2.011	1.000	4.111	1.000	2912.085	1.220
CRANE2	37.200	37.200	6.322	1.000	8.422	1.500	3602.978	3.093
LIFEBOAT	10.000	10.000	5.813	1.000	7.913	1.000	645.695	0.521
CRANE3	2.900	2.900	6.173	1.000	8.273	1.500	280.877	0.237
TOTAL WIND HMMT								**23.673**

Heeling moment: 23.64 MT-m to port

IS CODE CRANE OPERATION CRITERIA

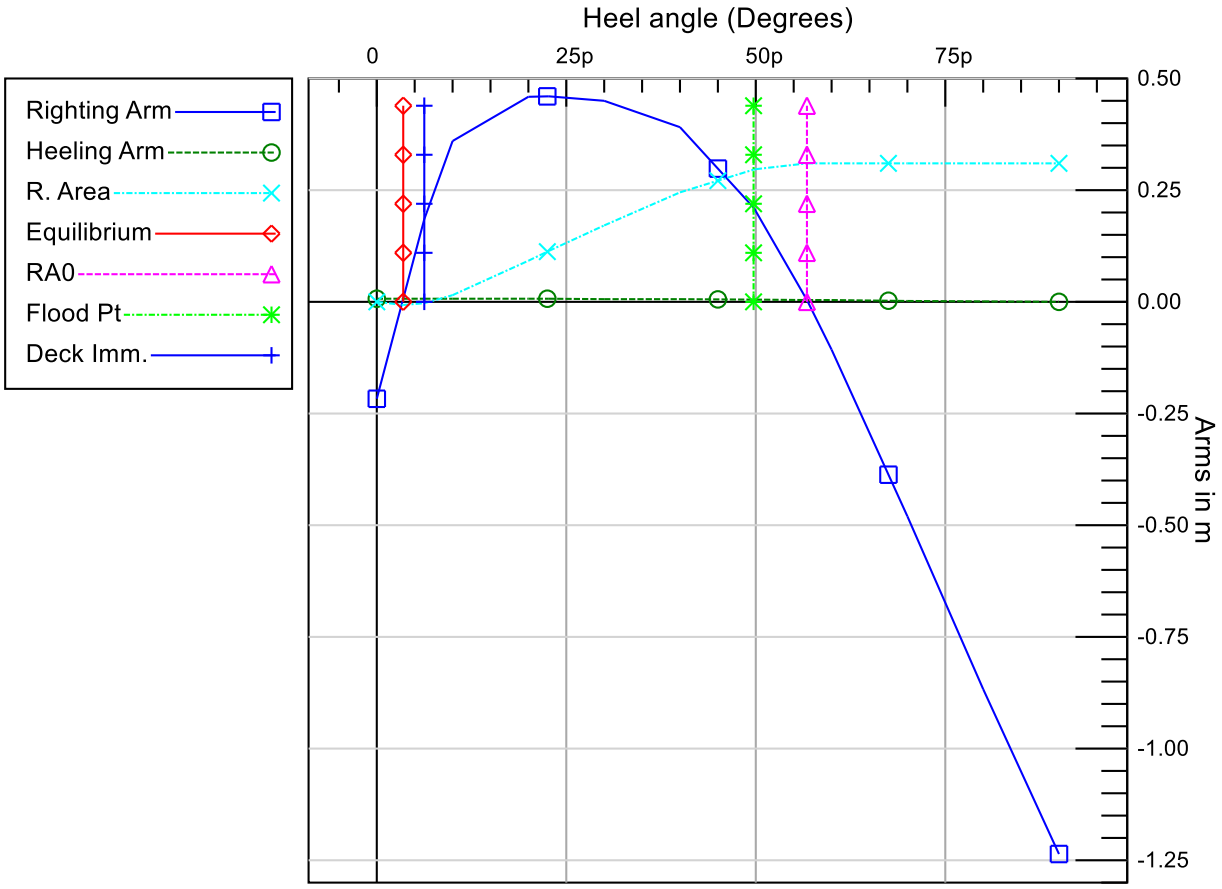
Free Surface Adjustment 0.369
Adjusted VCG 5.071

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
0.00	0.35a	4.198	-0.217	0.007	-0.224	0.000	8.012 (1,2)	0.903	
3.49p	0.35a	4.190	0.007	0.007	0.000	-0.007	7.618 (4)	0.403	Equil
6.27p	0.35a	4.170	0.185	0.007	0.178	-0.002	7.281 (4)	0.000	Deck Imm.
10.00p	0.38a	4.148	0.360	0.007	0.352	0.015	6.792 (4)	-0.588	
20.00p	0.94a	4.197	0.459	0.007	0.452	0.092	5.234 (4)	-2.583	
22.70p	1.18a	4.216	0.460	0.007	**0.454**	0.114	4.772 (4)	-3.186	MaxRa
30.00p	2.00a	4.252	0.450	0.006	0.444	0.171	3.475 (4)	-4.916	
40.00p	3.51a	4.183	0.391	0.005	0.386	0.244	1.696 (4)	-7.390	
49.71p	5.16a	3.966	0.211	0.005	0.206	0.296	0.000 (4)	-9.705	FldPt
50.00p	5.21a	3.957	0.203	0.005	0.199	0.297	-0.051 (4)	-9.771	
56.75p	6.28a	3.743	0.004	0.004	0.000	0.310	-1.231 (4)	-11.229	RaZero
60.00p	6.76a	3.624	-0.106	0.004	-0.110	0.310	-1.797 (4)	-11.881	
70.00p	8.11a	3.196	-0.480	0.002	-0.483	0.310	-3.490 (4)	-13.645	
80.00p	9.23a	2.665	-0.867	0.001	-0.868	0.310	-5.054 (4)	-14.999	
90.00p	10.08a	2.049	-1.236	0.000	-1.236	0.310	-6.449 (4)	-15.887	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1221.0 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling moment = 23.67Cos(heel) + 0.00Sin(heel)

IS CODE CRANE OPERATION CRITERIA



Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	3.49	1.51	Yes
(2) Abs. Ratio from 0.00 deg to Flood or RAzero	>1.400	55.304	53.904	Yes
(3) Area from Equilibrium to 20.00 deg or Flood	>0.0300 m-R	0.099	0.069	Yes
(4) Absolute Angle at Deck Immersion	>0.00 deg	6.27	6.27	Yes

12.2.2 MAX VERTICAL CRANE OPERATION

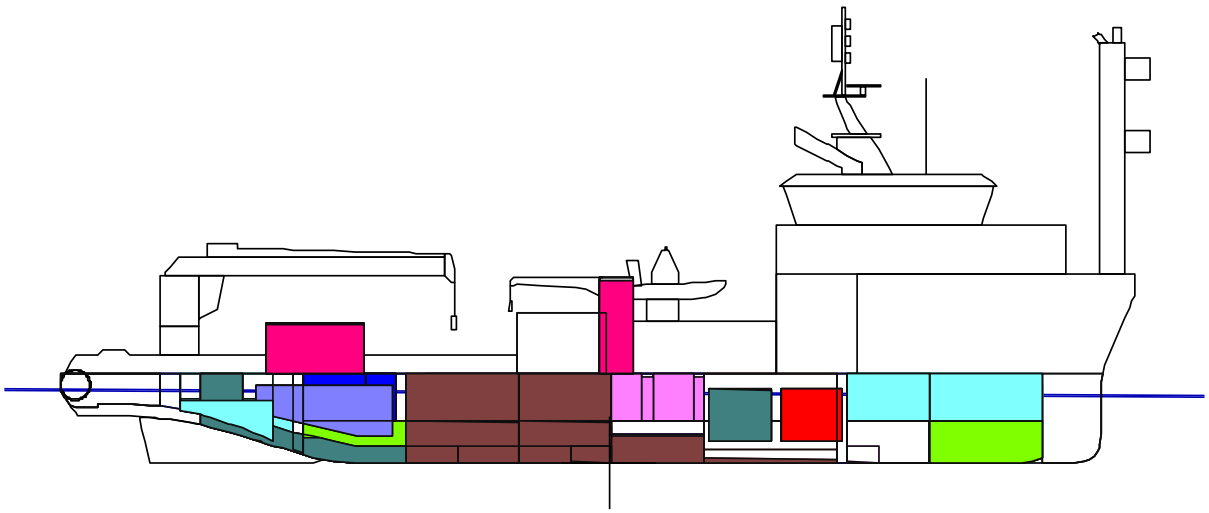
Max. Vertical moment – Lifting 3.500t @ 9.600 m radius (35.530 m boom length - Side Lifting to port side)

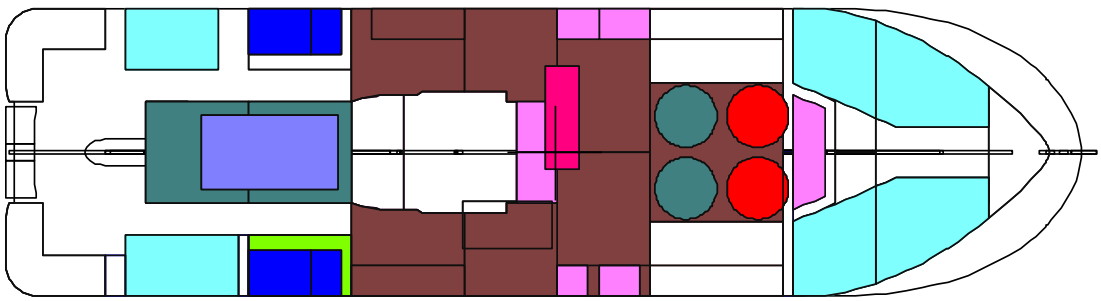
Floating Status

Draft FP	4.022 m	Heel	0.81p deg	GM(Solid)	3.834 m
Draft MS	4.198 m	Equil	Yes	F/S Corr.	0.413 m
Draft AP	4.375 m	Wind	Off	GM(Fluid)	3.421 m
Trim	0.35a m	Wave	No	KMT	8.738 m
LCG	1.114a m	VCG-Solid	4.905 m	TPcm	9.64
Displacement	3309.2 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	1,397.20	1.829a	0.202s	3.692
Displacement	3,309.17	1.114a	0.055p	4.905





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	88.68	21.33	FRESH WATER	325.38	80.21
FUEL OIL	303.47	58.38	MISC	51.30	40.12
SCALE INHIBITOR	57.86	98.00	CORROSION INHIBITOR	24.99	98.00
TREATED INJECTION WATER	61.80	100.00	TK ON MAIN DECK	123.45	66.80
BRINE	252.16	98.00	ACID	96.53	98.00

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	4.00	24.100f	0.000	6.800
CRANE BOOM FINAL	32.55	25.100a	10.535p	28.218
CRANE BOOM INITIAL	-32.55	17.438a	6.000p	11.504
LIFTED WEIGHT FINAL	3.50	25.100a	15.600p	45.245
Total Fixed:	**1,923.55**	**0.669a**	**0.347p**	**6.159**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_DWSWB.S	1.025	100.00	39.49	21.049f	3.353s	1.426	0.0
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
Subtotals:		**21.33%**	**88.68**	**1.184f**	**5.120s**	**1.385**	**194.6**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	76.00	56.11	21.465f	3.726p	3.617	63.6
NO1_FW.S	1.000	100.00	73.83	21.497f	3.747s	3.946	63.6
NO2_FW.P	1.000	100.00	66.44	16.374f	5.508p	3.914	0.0
NO2_FW.S	1.000	100.00	66.44	16.374f	5.508s	3.914	0.0
NO3_FW.S	1.000	50.00	35.59	20.972a	6.502s	2.843	24.9
NO3_FW.P	1.000	50.00	26.97	21.830a	6.519p	3.065	20.5
Subtotals:		**80.21%**	**325.38**	**11.163f**	**0.379s**	**3.683**	**172.6**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.C	0.850	34.00	14.87	9.292f	0.262p	0.147	274.5

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_FO.U.P	0.850	34.00	14.99	2.862f	4.034p	0.224	198.5
NO2_FO.U.S	0.850	98.00	38.33	2.940f	4.276s	0.579	198.5
NO3_FO.P	0.850	98.00	21.99	2.605a	7.501p	3.872	2.2
NO3_FO.S	0.850	98.00	21.99	2.605a	7.499s	3.872	2.2
NO3_FO.U.P	0.850	98.00	31.43	2.572a	6.466p	0.947	2.2
NO3_FO.U.S	0.850	98.00	31.43	2.572a	6.464s	0.947	2.2
NO4_FO.U.P	0.850	98.00	39.79	8.641a	6.402p	0.926	2.7
NO4_FO.U.S	0.850	98.00	39.79	8.641a	6.400s	0.926	2.7
FODAY.P	0.850	98.00	21.99	8.005a	7.501p	3.872	2.2
FODAY.S	0.850	98.00	26.88	8.608a	7.498s	3.872	2.6
Subtotals:		**58.38%**	**303.47**	**3.551a**	**0.448s**	**1.715**	**690.3**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.998f	7.501p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.699f	7.498s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.299f	7.501p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	0.999f	7.498s	3.872	0.8
DIRTYOIL.P	0.900	9.80	0.62	1.130a	1.609p	0.050	4.8
BILGEHOLD.S	1.000	9.80	0.69	1.130a	1.391s	0.050	5.3
SEWAGE	1.000	9.80	1.01	14.748f	0.425p	0.052	30.3
Subtotals:		**40.12%**	**51.30**	**2.966f**	**0.927p**	**3.699**	**45.2**

SCALE INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
SCLEIHBTR.P	1.100	98.00	14.47	13.401a	7.052p	3.872	3.2
SCLEIHBTR.S	1.100	98.00	14.47	13.401a	7.045s	3.872	3.2
SCALE_DISLVR.S	1.100	98.00	28.93	16.102a	7.045s	3.872	6.4
Subtotals:		**98.00%**	**57.86**	**14.751a**	**3.521s**	**3.872**	**12.8**

CORROSION INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
CORSNIHBTR.P	0.950	98.00	24.99	16.102a	7.052p	3.872	5.5
Subtotals:		**98.00%**	**24.99**	**16.102a**	**7.052p**	**3.872**	**5.5**

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
TRTDINJW.P	1.000	100.00	30.90	11.800f	2.100p	2.850	0.0
TRTDINJW.S	1.000	100.00	30.90	11.800f	2.100s	2.850	0.0

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Subtotals:		**100.00%**	**61.80**	**11.800f**	**0.000**	**2.850**	**0.0**

TK ON MAIN DECK

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NITROGENTK.P	1.185	98.00	47.27	17.246a	1.053p	6.721	8.7
CHEMADCTTK.P	1.185	98.00	76.18	0.410f	2.058p	8.020	42.0
Subtotals:		**66.80%**	**123.45**	**6.351a**	**1.673p**	**7.522**	**50.7**

BRINE

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_BRINETK.C	1.550	98.00	51.55	22.651a	0.018p	4.075	65.4
NO2_BRINETK.C	1.550	98.00	49.80	20.599a	0.007p	1.905	24.4
NO3_BRINETK.C	1.550	98.00	56.94	14.966a	0.006p	0.653	25.1
NO4_BRINETK.P	1.550	98.00	46.94	7.598f	2.104p	2.819	12.6
NO4_BRINETK.S	1.550	98.00	46.94	7.598f	2.096s	2.819	12.6
Subtotals:		**98.00%**	**252.16**	**9.249a**	**0.008p**	**2.406**	**140.1**

ACID

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_ACIDTK.C	1.160	98.00	96.53	16.259a	0.009p	3.322	54.7
Subtotals:		**98.00%**	**96.53**	**16.259a**	**0.009p**	**3.322**	**54.7**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**62.57%**	**1,385.62**	**1.733a**	**0.349s**	**3.164**	**1,366.6**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	3,285.87	0.974a	0.093p	2.294	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**3,309.17**	**1.129a**	**0.092p**	**2.285**	

Heeling Moment Report

Draft	4.200	m					Resultant	1.0000
WIND	20	Knots					(Sin 90Deg)	
Part	**Wind area**	**Effective area**	**Centre of Area from sea-level**	**Ch**	**Lever**	**Cs**	**Wind Force**	**Moment**
	(m2)	**(m2)**	**m**		**(m)**		**(N)**	**(m-MT)**
HULL.C	189.900	189.900	2.593	1.000	4.693	1.000	12261.749	5.866
DECKHOUSE	108.400	108.400	7.999	1.000	10.099	1.000	6999.334	7.206
FWDMAST	13.400	13.400	15.511	1.100	17.611	1.000	951.754	1.709
ANTENNA	0.300	0.300	15.362	1.100	17.462	1.000	21.308	0.038
CRANE1	23.500	23.500	14.210	1.000	16.310	1.500	2276.075	3.784
DECK	45.100	45.100	2.011	1.000	4.111	1.000	2912.085	1.220
CRANE2	37.200	37.200	6.322	1.000	8.422	1.500	3602.978	3.093
LIFEBOAT	10.000	10.000	5.813	1.000	7.913	1.000	645.695	0.521
CRANE3	2.900	2.900	6.173	1.000	8.273	1.500	280.877	0.237
TOTAL WIND HMMT								**23.673**

Heeling moment: 23.67 MT-m to port

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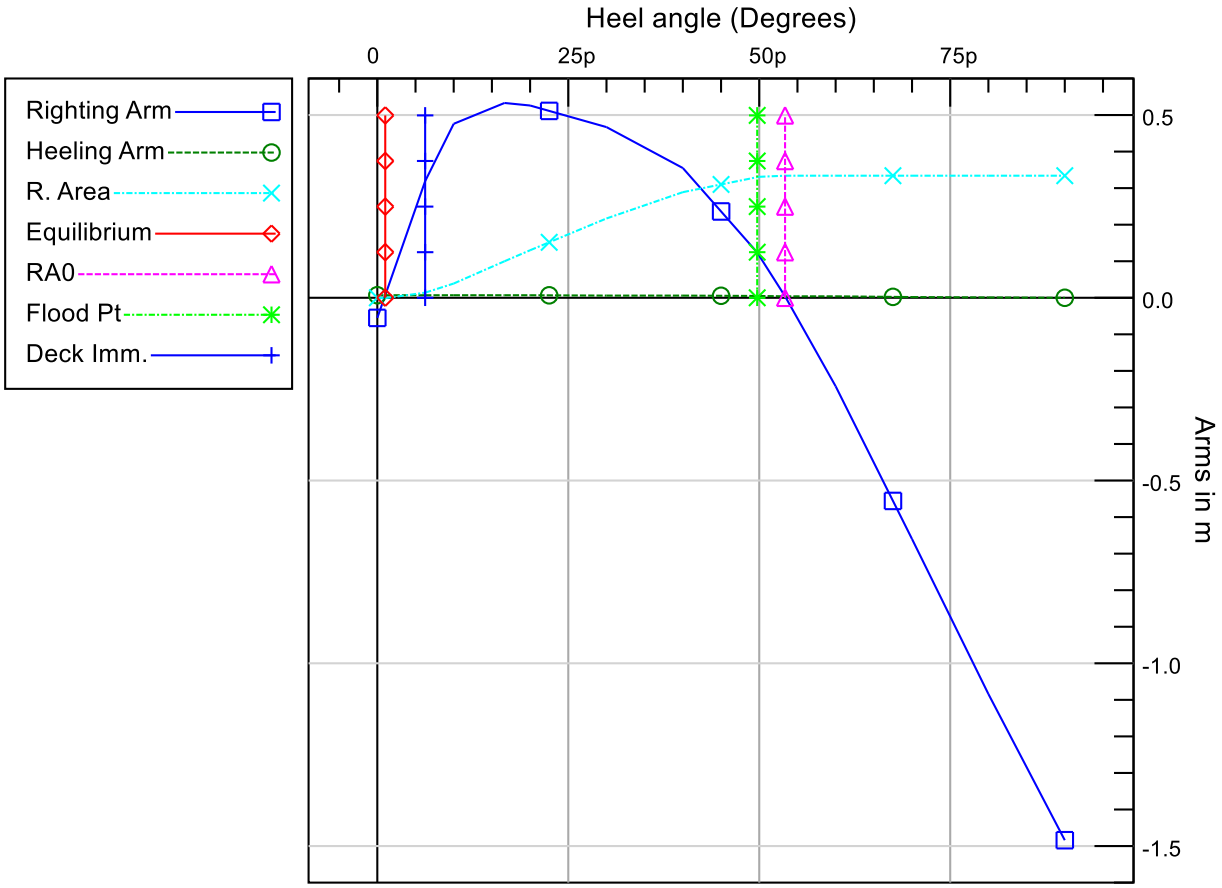
Free Surface Adjustment 0.411
Adjusted VCG 5.316

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
0.00	0.36a	4.198	-0.055	0.007	-0.062	0.000	8.012 (1,2)	0.903	
1.04p	0.36a	4.197	0.007	0.007	0.000	-0.001	7.900 (4)	0.761	Equil
6.26p	0.35a	4.170	0.319	0.007	0.312	0.014	7.282 (4)	0.000	Deck Imm.
10.00p	0.39a	4.148	0.476	0.007	0.469	0.039	6.793 (4)	-0.588	
16.66p	0.70a	4.174	0.533	0.007	**0.527**	0.100	5.785 (4)	-1.868	MaxRa
20.00p	0.95a	4.197	0.526	0.007	0.520	0.130	5.235 (4)	-2.586	
30.00p	2.01a	4.253	0.467	0.006	0.460	0.217	3.476 (4)	-4.925	
40.00p	3.54a	4.185	0.355	0.005	0.349	0.289	1.699 (4)	-7.408	
49.73p	5.21a	3.968	0.124	0.005	0.119	0.330	0.000 (4)	-9.738	FldPt
50.00p	5.25a	3.961	0.115	0.005	0.111	0.331	-0.047 (4)	-9.799	
53.38p	5.81a	3.860	0.004	0.004	0.000	0.334	-0.638 (4)	-10.551	RaZero
60.00p	6.82a	3.629	-0.242	0.004	-0.246	0.334	-1.791 (4)	-11.919	
70.00p	8.18a	3.201	-0.660	0.002	-0.662	0.334	-3.483 (4)	-13.690	
80.00p	9.30a	2.671	-1.084	0.001	-1.085	0.334	-5.046 (4)	-15.044	
90.00p	10.14a	2.055	-1.484	0.000	-1.484	0.334	-6.442 (4)	-15.927	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1359.4 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling moment = 23.67Cos(heel) + 0.00Sin(heel)

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Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	1.04	3.96	Yes
(2) Abs. Ratio from 0.00 deg to Flood or RAzero	>1.400	61.503	60.103	Yes
(3) Area from Equilibrium to 20.00 deg or Flood	>0.0300 m-R	0.131	0.101	Yes
(4) Absolute Angle at Deck Immersion	>0.00 deg	6.26	6.26	Yes

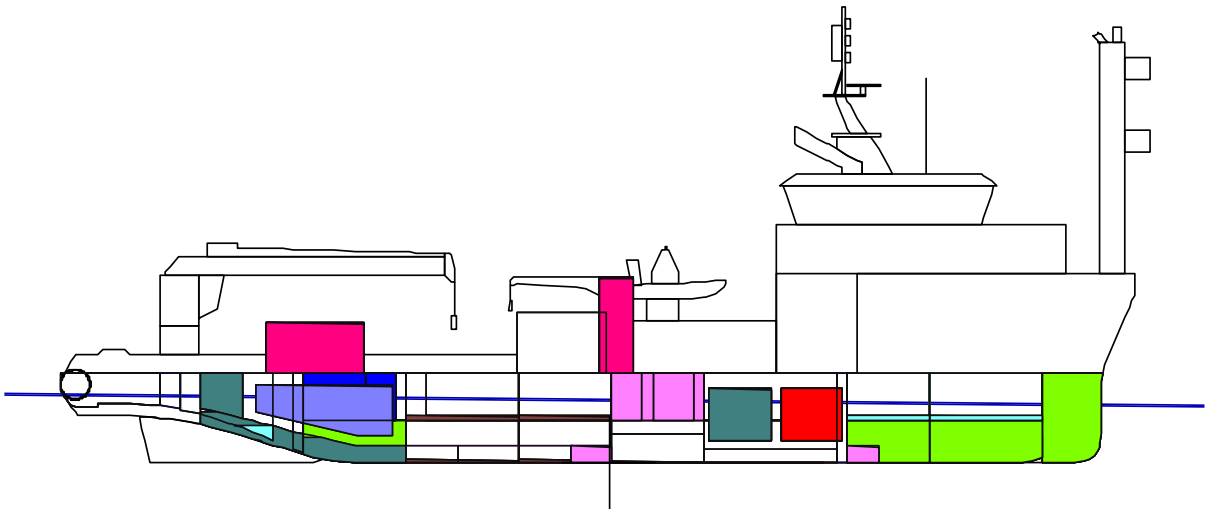
12.3 CONDITION NO.3: FULLY LOADED ARRIVAL

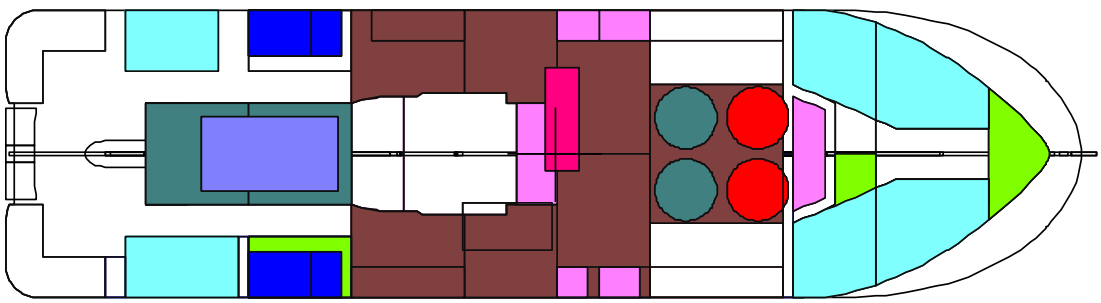
Floating Status

Draft FP	3.485 m	Heel	0.04s deg	GM(Solid)	4.434 m
Draft MS	3.750 m	Equil	Yes	F/S Corr.	0.537 m
Draft AP	4.015 m	Wind	Off	GM(Fluid)	3.897 m
Trim	0.53a m	Wave	No	KMT	9.335 m
LCG	0.969a m	VCG-Solid	4.901 m	TPcm	9.56
Displacement	2889.7 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	977.75	1.707a	0.482s	3.159
Displacement	2,889.73	0.969a	0.002s	4.901





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	221.38	53.23	FRESH WATER	32.54	8.02
FUEL OIL	30.35	5.84	MISC	72.22	56.48
SCALE INHIBITOR	57.86	98.00	CORROSION INHIBITOR	24.99	98.00
TREATED INJECTION WATER	61.80	100.00	TK ON MAIN DECK	123.45	66.80
BRINE	252.16	98.00	ACID	96.53	98.00

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	0.40	24.100f	0.000	6.800
Total Fixed:	**1,916.45**	**0.541a**	**0.242p**	**5.803**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DWSWBFP	1.025	100.00	54.57	26.510f	0.000	3.135	39.8
NO1_DWSWB.P	1.025	50.00	19.74	20.780f	3.218p	0.786	41.3
NO1_DWSWB.S	1.025	100.00	39.49	21.049f	3.353s	1.426	0.0
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	113.5
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
Subtotals:		**53.23%**	**221.38**	**13.197f**	**3.026s**	**1.723**	**194.6**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	7.60	5.61	21.305f	3.604p	2.619	63.6
NO1_FW.S	1.000	10.00	7.38	21.331f	3.610s	2.656	63.6
NO2_FW.P	1.000	10.00	6.64	16.295f	5.444p	2.645	51.5
NO2_FW.S	1.000	10.00	6.64	16.291f	5.456s	2.645	51.5
NO3_FW.S	1.000	5.00	3.56	19.291a	6.329s	1.548	6.7
NO3_FW.P	1.000	5.00	2.70	20.394a	6.317p	1.952	6.0
Subtotals:		**8.02%**	**32.54**	**11.367f**	**0.369s**	**2.466**	**243.0**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.C	0.850	3.40	1.49	7.836f	0.132s	0.021	245.2

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_FO.U.P	0.850	3.40	1.50	2.381f	3.572p	0.025	198.5
NO2_FO.U.S	0.850	9.80	3.83	2.681f	3.759s	0.059	198.5
NO3_FO.P	0.850	9.80	2.20	2.682a	7.499p	2.638	2.2
NO3_FO.S	0.850	9.80	2.20	2.682a	7.501s	2.638	2.2
NO3_FO.U.P	0.850	9.80	3.14	2.693a	5.667p	0.088	33.5
NO3_FO.U.S	0.850	9.80	3.14	2.694a	5.683s	0.088	33.6
NO4_FO.U.P	0.850	9.80	3.98	8.874a	5.550p	0.084	44.1
NO4_FO.U.S	0.850	9.80	3.98	8.873a	5.566s	0.084	44.2
FODAY.P	0.850	9.80	2.20	8.082a	7.499p	2.638	2.2
FODAY.S	0.850	9.80	2.69	8.722a	7.501s	2.638	2.6
Subtotals:		**5.84%**	**30.35**	**3.791a**	**0.430s**	**0.857**	**806.8**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.997f	7.500p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.698f	7.500s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.298f	7.500p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	0.999f	7.500s	3.872	0.8
DIRTYOIL.P	0.900	98.00	6.22	1.105a	1.499p	0.490	4.8
BILGEHOLD.S	1.000	98.00	6.91	1.105a	1.501s	0.490	5.3
SEWAGE	1.000	98.00	10.11	14.762f	0.002s	0.490	30.3
Subtotals:		**56.48%**	**72.22**	**3.785f**	**0.636p**	**2.783**	**45.2**

SCALE INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
SCLEIHBTR.P	1.100	98.00	14.47	13.401a	7.048p	3.872	3.2
SCLEIHBTR.S	1.100	98.00	14.47	13.401a	7.049s	3.872	3.2
SCALE_DISLVR.S	1.100	98.00	28.93	16.104a	7.049s	3.872	6.4
Subtotals:		**98.00%**	**57.86**	**14.752a**	**3.524s**	**3.872**	**12.8**

CORROSION INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
CORSNIHBTR.P	0.950	98.00	24.99	16.104a	7.048p	3.872	5.5
Subtotals:		**98.00%**	**24.99**	**16.104a**	**7.048p**	**3.872**	**5.5**

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
TRTDINJW.P	1.000	100.00	30.90	11.800f	2.100p	2.850	0.0
TRTDINJW.S	1.000	100.00	30.90	11.800f	2.100s	2.850	0.0

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Subtotals:		**100.00%**	**61.80**	**11.800f**	**0.000**	**2.850**	**0.0**

TK ON MAIN DECK

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NITROGENTK.P	1.185	98.00	47.27	17.249a	1.050p	6.721	8.7
CHEMADCTTK.P	1.185	98.00	76.18	0.409f	2.050p	8.020	42.0
Subtotals:		**66.80%**	**123.45**	**6.352a**	**1.667p**	**7.522**	**50.7**

BRINE

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_BRINETK.C	1.550	98.00	51.55	22.652a	0.001s	4.075	66.0
NO2_BRINETK.C	1.550	98.00	49.80	20.599a	0.000	1.905	23.1
NO3_BRINETK.C	1.550	98.00	56.94	14.966a	0.000	0.653	23.9
NO4_BRINETK.P	1.550	98.00	46.94	7.598f	2.100p	2.819	12.6
NO4_BRINETK.S	1.550	98.00	46.94	7.598f	2.100s	2.819	12.6
Subtotals:		**98.00%**	**252.16**	**9.250a**	**0.000**	**2.406**	**138.1**

ACID

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_ACIDTK.C	1.160	98.00	96.53	16.266a	0.000	3.322	55.1
Subtotals:		**98.00%**	**96.53**	**16.266a**	**0.000**	**3.322**	**55.1**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**43.95%**	**973.27**	**1.812a**	**0.484s**	**3.125**	**1,552.0**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	2,866.42	0.817a	0.004s	2.045	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**2,889.73**	**0.996a**	**0.004s**	**2.036**	

Unprotected Flood Points

Name	L (m)	T (m)	V (m)	Height (m)
(1) POSITION 1(S)	9.700f	5.400s	12.150	8.485
(2) POSITION 1(P)	9.700f	5.400p	12.150	8.493
(3) POSITION 2(S)	10.300f	6.300s	12.150	8.490
(4) POSITION 2(P)	10.300f	6.300p	12.150	8.500

: OSV CHEMICAL CODE – CHAPTER 2 , 2.2

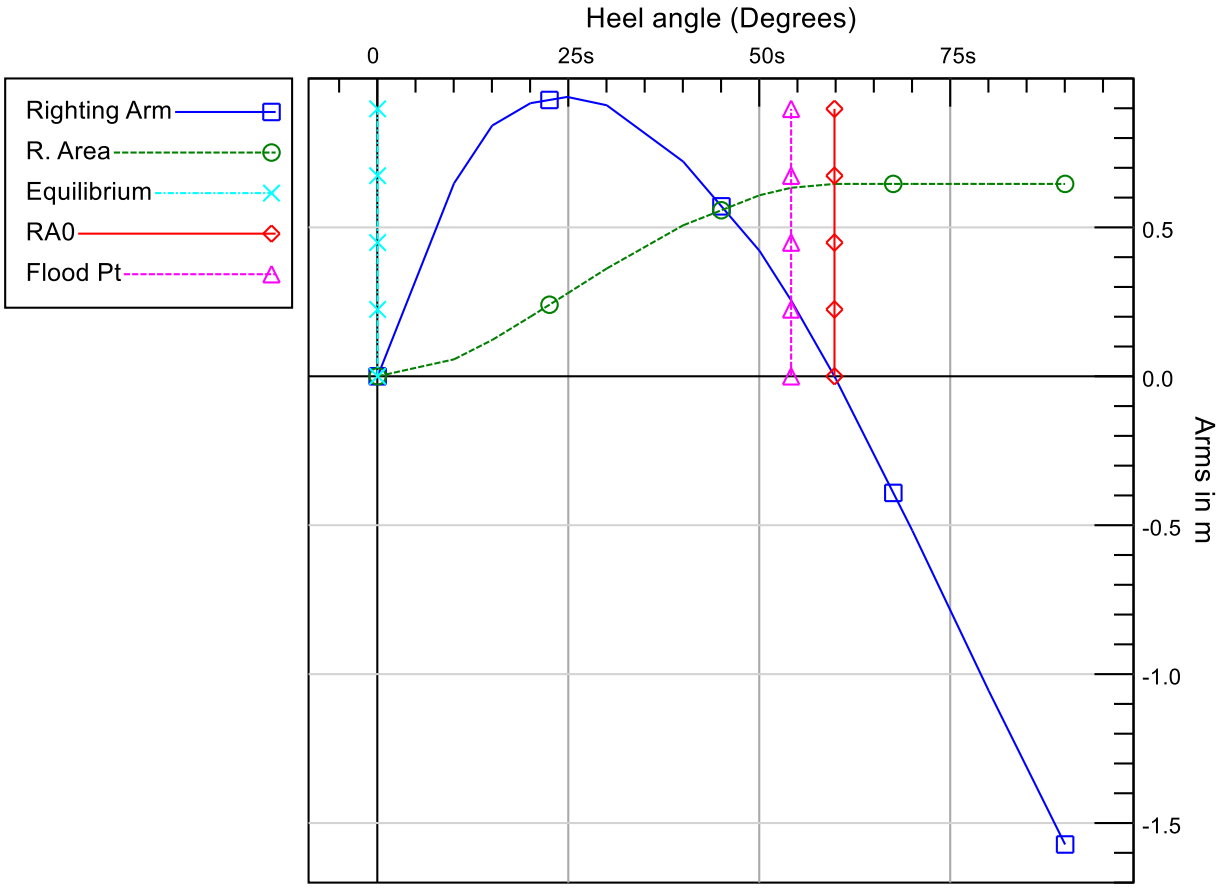
Free Surface Adjustment 0.537

Adjusted VCG 5.438

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Area (m-Rad)	Flood Pt Height (m)	Notes
0.04s	0.53a	3.749	0.000	0.000	8.486 (1)	Equil
10.04s	0.40a	3.672	0.647	0.056	7.266 (3)	
15.04s	0.39a	3.619	0.842	0.122	6.550 (3)	
20.04s	0.53a	3.597	0.917	0.200	5.752 (3)	
24.88s	0.79a	3.581	**0.938**	0.278	4.933 (3)	MaxRa
30.04s	1.20a	3.544	0.910	0.362	4.035 (3)	
40.04s	2.31a	3.332	0.721	0.507	2.328 (3)	
50.04s	3.70a	2.951	0.421	0.608	0.681 (3)	
54.17s	4.27a	2.766	0.255	0.633	0.000 (3)	FldPt
59.85s	5.04a	2.490	0.000	0.646	-0.932 (3)	RaZero
60.04s	5.07a	2.481	-0.010	0.646	-0.964 (3)	
70.04s	6.31a	1.946	-0.518	0.646	-2.577 (3)	
80.04s	7.39a	1.358	-1.056	0.646	-4.104 (3)	
90.04s	8.23a	0.722	-1.572	0.646	-5.491 (3)	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1552.0 MT-m was used to adjust the VCG.

: OSV CHEMICAL CODE – CHAPTER 2 , 2.2



Limit	Min/Max	Actual	Margin	Pass
(1) Area from 0.00 deg to MaxRA at 30.00	>0.0601 m-R	0.278	0.218	Yes
(2) Area from 0.00 deg to 40.00 or Flood	>0.0900 m-R	0.507	0.417	Yes
(3) Area from 30.00 deg to 40.00 or Flood	>0.0300 m-R	0.145	0.115	Yes
(4) Righting Arm at 30.00 deg	>0.200 m	0.910	0.710	Yes
(5) Absolute Angle at MaxRA	>15.00 deg	24.88	9.88	Yes
(6) GM Upright	>0.150 m	3.897	3.747	Yes

Heeling Moment Derivation

Part	LPA (m^2)	HCP (m)	Arm (m)	Pressure (MT/m^2)	Moment (m-MT)
HULL.C	218.1	2.769	4.561	0.051	51.13
DECKHOUSE	108.4	8.553	10.345	0.051	57.65
FWDMAST	13.4	16.065	17.857	0.051	12.32
ANTENNA	0.3	15.916	17.708	0.051	0.24
CRANE1	23.5	14.764	16.556	0.051	19.96
DECK	45.1	2.565	4.357	0.051	10.11
CRANE2	37.2	6.876	8.668	0.051	16.56
LIFEBOAT	10.0	6.367	8.159	0.051	4.19
CRANE3	2.9	6.727	8.519	0.051	1.26

Total wind heeling moment 173.42 to starboard

IMO SEVERE WIND & ROLLING

Free Surface Adjustment 0.484

Adjusted VCG 5.385

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
23.56p	0.71a	3.586	-0.969	0.090	-1.059	0.000	5.160 (4)	-2.455	Roll
13.56p	0.38a	3.630	-0.827	0.090	-0.917	-0.172	6.773 (4)	-0.643	
3.56p	0.52a	3.741	-0.254	0.090	-0.344	-0.289	8.089 (4)	0.752	
1.01s	0.53a	3.749	0.060	0.090	-0.030	-0.304	8.384 (3)	1.119	SSWEq
1.44s	0.53a	3.748	0.090	0.090	0.000	-0.304	8.335 (3)	1.057	Equil
6.44s	0.49a	3.719	0.425	0.090	0.335	-0.289	7.734 (3)	0.350	
9.07s	0.43a	3.686	0.591	0.090	0.501	-0.270	7.395 (3)	0.000	Deck Imm.
16.00s	0.41a	3.613	0.871	0.090	0.781	-0.191	6.402 (3)	-1.046	
16.44s	0.41a	3.611	0.880	0.090	0.789	-0.185	6.333 (3)	-1.122	
25.30s	0.82a	3.579	0.955	0.090	**0.865**	-0.053	4.860 (3)	-2.804	MaxRa
26.44s	0.90a	3.573	0.954	0.090	0.864	-0.036	4.661 (3)	-3.039	
36.44s	1.87a	3.431	0.829	0.090	0.739	0.107	2.933 (3)	-5.133	
46.44s	3.19a	3.100	0.581	0.090	0.491	0.216	1.273 (3)	-7.206	
50.00s	3.69a	2.952	0.459	0.090	0.369	0.243	0.688 (3)	-7.912	
54.17s	4.26a	2.765	0.295	0.090	0.205	0.264	0.000 (3)	-8.706	FldPt
56.44s	4.58a	2.658	0.198	0.090	0.108	0.270	-0.375 (3)	-9.124	
58.84s	4.90a	2.541	0.090	0.090	0.000	0.272	-0.768 (3)	-9.548	RaZero
66.44s	5.88a	2.143	-0.281	0.090	-0.371	0.272	-2.003 (3)	-10.792	

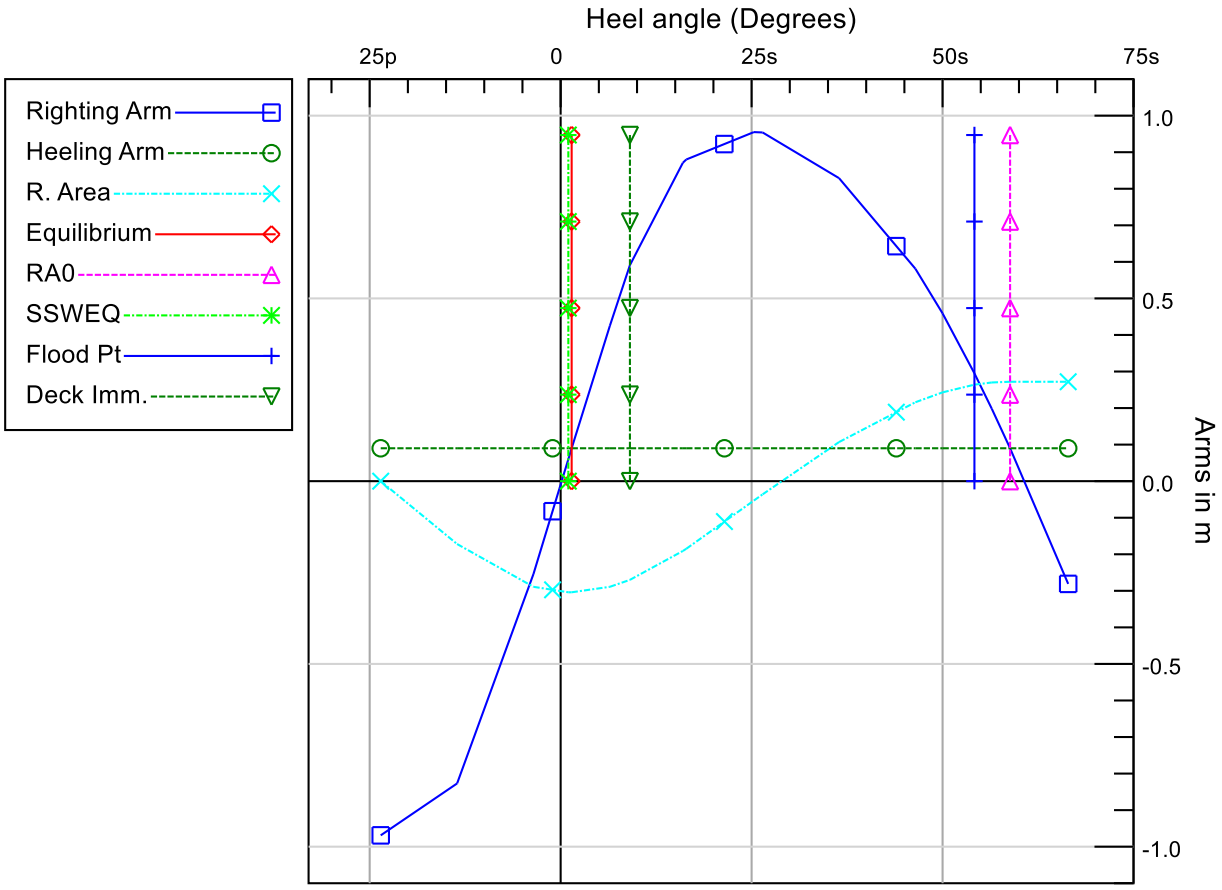
Weight and C.G. used above include tank loads.

The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1398.7 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
wind heeling arms derived from this moment (in m-MT):
Stbd heeling moment = 260.13

Roll angle is 24.47
Equilibrium for load condition without gust is 1.01s

IMO SEVERE WIND & ROLLING



Limit	Min/Max	Actual	Margin	Pass
(1) Res. Ratio from Roll to Abs 50.00 deg or Flood	>1.000	1.799	0.799	Yes
(2) Angle from SSWEQ to 80% of Dk. Imm. or Abs 16.00 deg	>0.00 deg	6.25	6.25	Yes

Heeling Moment Report

FIFI HEELING MOMENT CALCULATION			
Reaction Force of the Fire Monitor	=	1.530	MT
Number of Fire Monitors	=	2	
Draft of the loading condition	=	3.75	m
D/2	=	1.875	m
Height of firefighting monitor above baseline	=	19.600	m
Moment due to Fire Fighting operation	=	54.239	MT-m
Reaction Force of the Bow Thruster	=	18	MT
D/2	=	1.875	m
Height of Bow Thruster above baseline	=	1.200	m
Moment due to Bow Thruster operation	=	12.15	MT-m
Total Heeling Moment	=	66.389	Tm

Heeling moment: 66.39 MT-m to starboard

FIRE FIGHTING CRITERIA

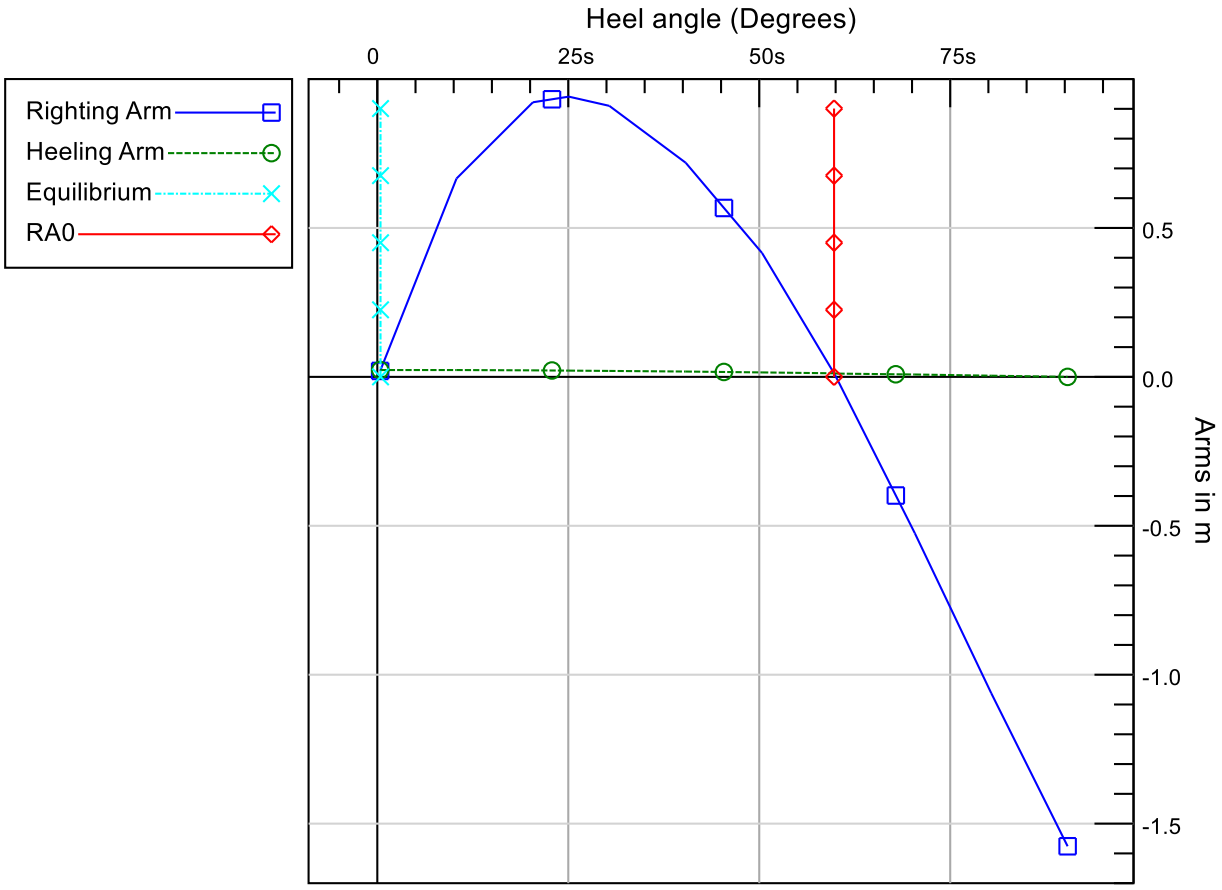
Free Surface Adjustment 0.525
Adjusted VCG 5.426

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Notes
0.37s	0.53a	3.749	0.020	0.023	-0.003	
0.42s	0.53a	3.749	0.023	0.023	0.000	Equil
10.37s	0.40a	3.667	0.666	0.023	0.643	
20.37s	0.54a	3.596	0.922	0.022	0.900	
25.07s	0.80a	3.580	0.941	0.021	**0.920**	MaxRa
30.37s	1.23a	3.540	0.910	0.020	0.891	
40.37s	2.36a	3.322	0.719	0.018	0.701	
50.37s	3.74a	2.937	0.416	0.015	0.401	
59.79s	5.03a	2.493	0.012	0.012	0.000	RaZero
60.37s	5.11a	2.464	-0.016	0.011	-0.027	
70.37s	6.35a	1.928	-0.525	0.008	-0.532	
80.37s	7.42a	1.338	-1.062	0.004	-1.066	
90.37s	8.25a	0.700	-1.576	0.000	-1.576	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1517.3 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Stbd heeling mmt = 66.39Cos(heel) + 0.00Sin(heel) <= 90 deg
Stbd heeling mmt = 0.00 > 90 deg

FIRE FIGHTING CRITERIA



Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	0.42	4.58	Yes

12.3.1 MAX TRANSVERSE CRANE OPERATION

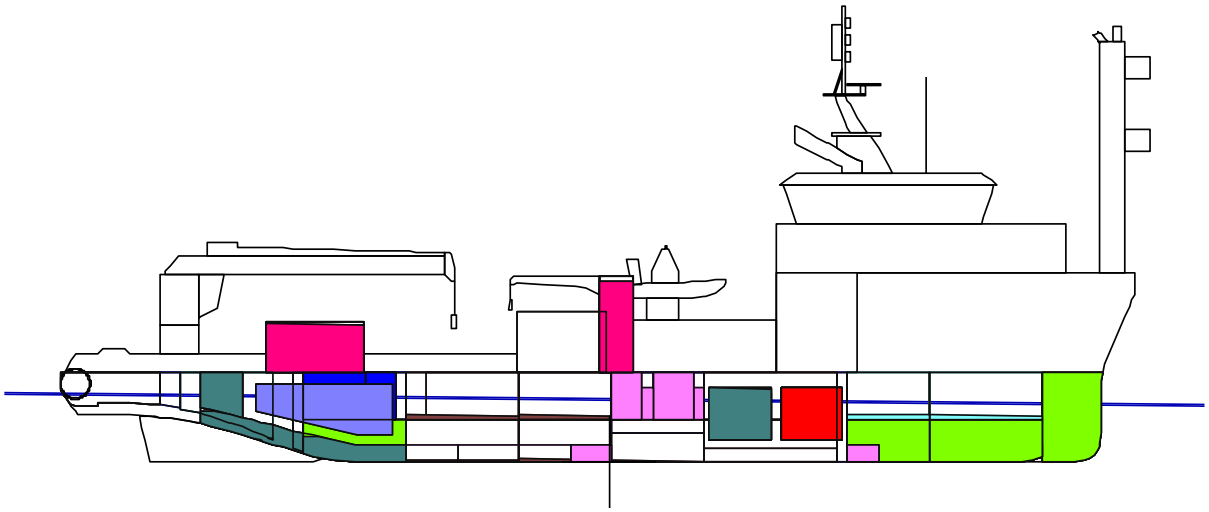
Max. Transverse moment – Lifting 3.500t @ 35.00 m radius (35.530 m boom length –Side Lifting to port side)

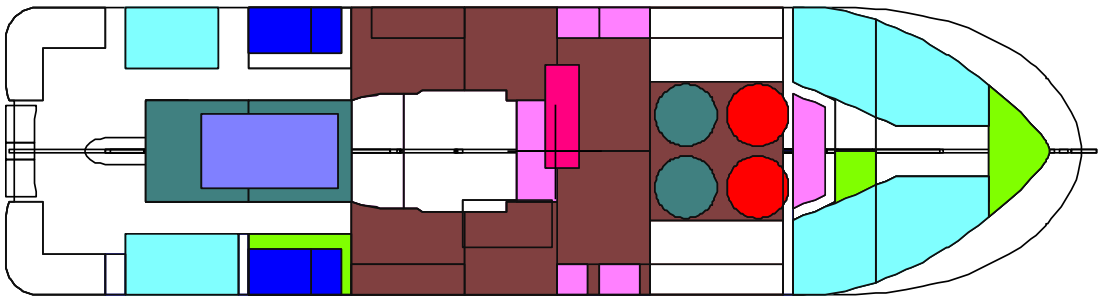
Floating Status

Draft FP	3.446 m	Heel	3.34p deg	GM(Solid)	4.407 m
Draft MS	3.747 m	Equil	Yes	F/S Corr.	0.402 m
Draft AP	4.047 m	Wind	Off	GM(Fluid)	4.005 m
Trim	0.60a m	Wave	No	KMT	9.305 m
LCG	1.084a m	VCG-Solid	4.906 m	TPcm	9.54
Displacement	2893.2 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	981.25	2.044a	0.292p	3.178
Displacement	2,893.23	1.084a	0.260p	4.906





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	221.38	53.23	FRESH WATER	32.54	8.02
FUEL OIL	30.35	5.84	MISC	72.22	56.48
SCALE INHIBITOR	57.86	98.00	CORROSION INHIBITOR	24.99	98.00
TREATED INJECTION WATER	61.80	100.00	TK ON MAIN DECK	123.45	66.80
BRINE	252.16	98.00	ACID	96.53	98.00

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	0.40	24.100f	0.000	6.800
CRANE BOOM FINAL	32.55	25.100a	23.235p	11.190
CRANE BOOM INITIAL	-32.55	17.438a	6.000p	11.504
LIFTED WEIGHT FINAL	3.50	25.100a	41.000p	11.190
Total Fixed:	**1,919.95**	**0.716a**	**0.609p**	**5.807**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DWSWBFP	1.025	100.00	54.57	26.510f	0.000	3.135	39.8
NO1_DWSWB.P	1.025	50.00	19.74	20.734f	3.291p	0.788	41.3
NO1_DWSWB.S	1.025	100.00	39.49	21.049f	3.353s	1.426	0.0
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	113.5
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
Subtotals:		**53.23%**	**221.38**	**13.193f**	**3.019s**	**1.723**	**194.6**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	7.60	5.61	21.025f	4.070p	2.635	63.6
NO1_FW.S	1.000	10.00	7.38	21.515f	3.270s	2.664	63.6
NO2_FW.P	1.000	10.00	6.64	16.154f	5.910p	2.660	52.5
NO2_FW.S	1.000	10.00	6.64	16.417f	5.002s	2.657	51.1
NO3_FW.S	1.000	5.00	3.56	19.308a	6.221s	1.552	6.6
NO3_FW.P	1.000	5.00	2.70	20.387a	6.484p	1.957	6.2
Subtotals:		**8.02%**	**32.54**	**11.356f**	**0.002p**	**2.477**	**243.5**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.C	0.850	3.40	1.49	8.760f	3.035p	0.056	13.0
NO2_FO.U.P	0.850	3.40	1.50	2.857f	6.492p	0.071	198.5
NO2_FO.U.S	0.850	9.80	3.83	2.641f	1.817s	0.106	198.5
NO3_FO.P	0.850	9.80	2.20	2.693a	7.557p	2.639	2.2
NO3_FO.S	0.850	9.80	2.20	2.693a	7.442s	2.639	2.2
NO3_FO.U.P	0.850	9.80	3.14	2.772a	6.339p	0.109	34.2
NO3_FO.U.S	0.850	9.80	3.14	2.651a	5.090s	0.104	28.8
NO4_FO.U.P	0.850	9.80	3.98	8.855a	6.248p	0.105	49.5
NO4_FO.U.S	0.850	9.80	3.98	8.942a	4.956s	0.102	37.1
FODAY.P	0.850	9.80	2.20	8.093a	7.557p	2.639	2.2
FODAY.S	0.850	9.80	2.69	8.738a	7.442s	2.640	2.7
Subtotals:		**5.84%**	**30.35**	**3.742a**	**0.435p**	**0.876**	**568.9**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.997f	7.506p	3.872	1.5
FOAM.S	1.145	98.00	13.30	3.698f	7.494s	3.872	1.2
HYDOIL.P	0.924	98.00	10.73	1.298f	7.506p	3.872	0.9
LUBOIL.S	0.924	98.00	8.05	0.999f	7.494s	3.872	0.7
DIRTYOIL.P	0.900	98.00	6.22	1.102a	1.521p	0.490	4.8
BILGEHOLD.S	1.000	98.00	6.91	1.102a	1.479s	0.490	5.3
SEWAGE	1.000	98.00	10.11	14.765f	0.047p	0.491	30.3
Subtotals:		**56.48%**	**72.22**	**3.786f**	**0.651p**	**2.784**	**44.7**

SCALE INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
SCLEIHBTR.P	1.100	98.00	14.47	13.401a	7.060p	3.872	1.9
SCLEIHBTR.S	1.100	98.00	14.47	13.401a	7.036s	3.872	1.9
SCALE_DISLVR.S	1.100	98.00	28.93	16.103a	7.037s	3.872	3.8
Subtotals:		**98.00%**	**57.86**	**14.752a**	**3.512s**	**3.872**	**7.7**

CORROSION INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
CORSNIHBTR.P	0.950	98.00	24.99	16.103a	7.060p	3.872	3.3
Subtotals:		**98.00%**	**24.99**	**16.103a**	**7.060p**	**3.872**	**3.3**

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
TRTDINJW.P	1.000	100.00	30.90	11.800f	2.100p	2.850	0.0

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
TRTDINJW.S	1.000	100.00	30.90	11.800f	2.100s	2.850	0.0
Subtotals:		**100.00%**	**61.80**	**11.800f**	**0.000**	**2.850**	**0.0**

TK ON MAIN DECK

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NITROGENTK.P	1.185	98.00	47.27	17.249a	1.060p	6.721	6.2
CHEMADCTTK.P	1.185	98.00	76.18	0.409f	2.079p	8.020	21.3
Subtotals:		**66.80%**	**123.45**	**6.352a**	**1.689p**	**7.523**	**27.5**

BRINE

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_BRINETK.C	1.550	98.00	51.55	22.651a	0.040p	4.075	9.5
NO2_BRINETK.C	1.550	98.00	49.80	20.601a	0.028p	1.906	14.4
NO3_BRINETK.C	1.550	98.00	56.94	14.968a	0.026p	0.654	16.7
NO4_BRINETK.P	1.550	98.00	46.94	7.597f	2.114p	2.820	7.1
NO4_BRINETK.S	1.550	98.00	46.94	7.597f	2.086s	2.820	7.1
Subtotals:		**98.00%**	**252.16**	**9.251a**	**0.025p**	**2.407**	**54.7**

ACID

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_ACIDTK.C	1.160	98.00	96.53	16.261a	0.023p	3.322	17.3
Subtotals:		**98.00%**	**96.53**	**16.261a**	**0.023p**	**3.322**	**17.3**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**43.95%**	**973.27**	**1.811a**	**0.429s**	**3.127**	**1,162.3**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	2,869.92	0.936a	0.430p	2.060	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**2,893.23**	**1.115a**	**0.427p**	**2.052**	

Heeling Moment Report

Draft	3.750	m					Resultant	1.0000
WIND	20	Knots					(Sin 90Deg)	
Part	**Wind area**	**Effective area**	**Centre of Area from sea-level**	**Ch**	**Lever**	**Cs**	**Wind Force**	**Moment**
	(m2)	**(m2)**	**m**		**(m)**		**(N)**	**(m-MT)**
HULL.C	218.100	218.100	2.461	1.000	4.561	1.000	14082.608	6.547
DECKHOUSE	108.400	108.400	8.245	1.000	10.345	1.000	6999.334	7.381
FWDMAST	13.400	13.400	15.757	1.100	17.857	1.000	951.754	1.732
ANTENNA	0.300	0.300	15.608	1.100	17.708	1.000	21.308	0.038
CRANE1	23.500	23.500	14.456	1.000	16.556	1.500	2276.075	3.841
DECK	45.100	45.100	2.257	1.000	4.357	1.000	2912.085	1.293
CRANE2	37.200	37.200	6.568	1.000	8.668	1.500	3602.978	3.184
LIFEBOAT	10.000	10.000	6.059	1.000	8.159	1.000	645.695	0.537
CRANE3	2.900	2.900	6.419	1.000	8.519	1.500	280.877	0.244
TOTAL WIND HMMT								**24.799**

Heeling moment: 24.799 MT-m to port

IS CODE CRANE OPERATION CRITERIA

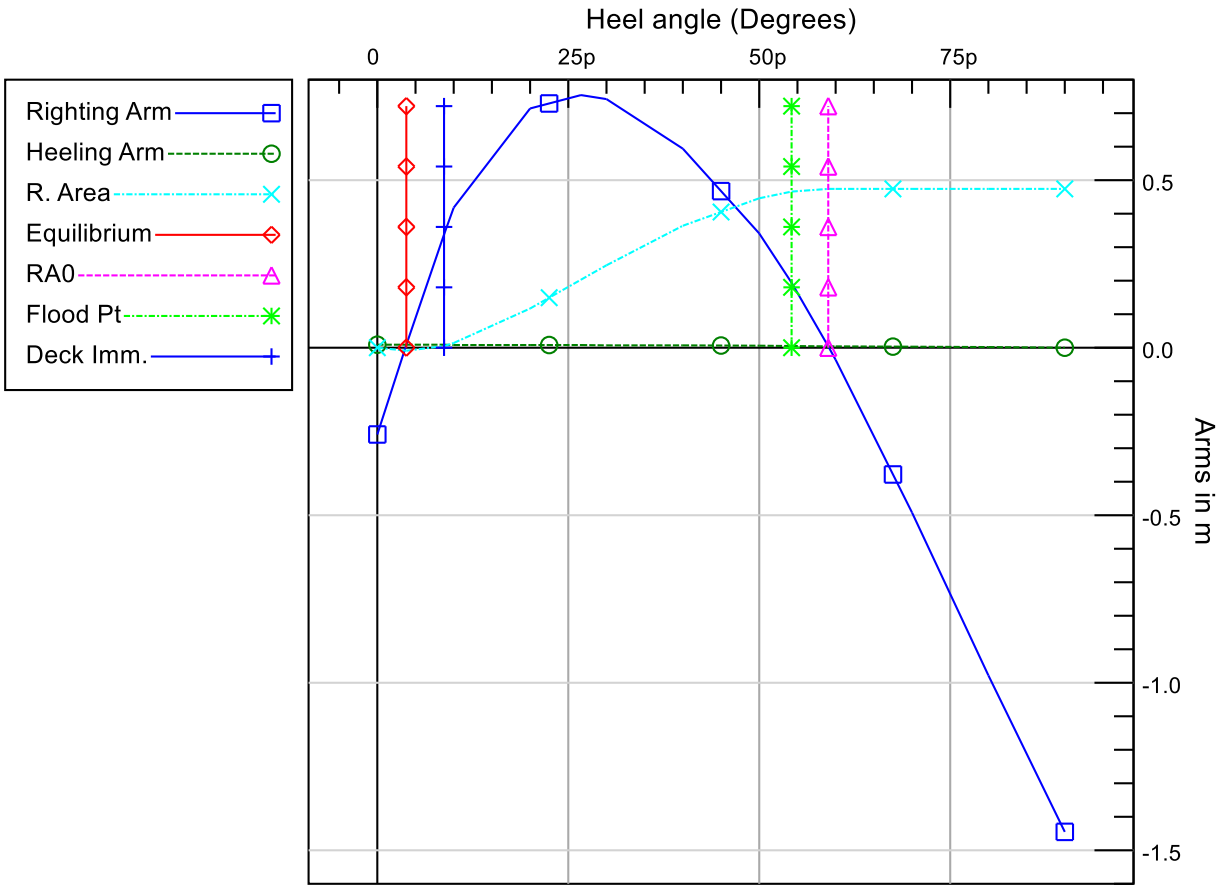
Free Surface Adjustment 0.400
Adjusted VCG 5.305

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
0.00	0.61a	3.748	-0.259	0.009	-0.268	0.000	8.505 (1,2)	1.211	
3.80p	0.60a	3.738	0.009	0.009	0.000	-0.009	8.075 (4)	0.676	Equil
8.74p	0.52a	3.690	0.338	0.008	0.329	0.005	7.455 (4)	0.000	Deck Imm.
10.00p	0.49a	3.672	0.417	0.008	0.409	0.014	7.287 (4)	-0.168	
20.00p	0.64a	3.603	0.714	0.008	0.706	0.117	5.775 (4)	-1.832	
26.66p	1.05a	3.581	0.754	0.008	**0.746**	0.203	4.637 (4)	-3.162	MaxRa
30.00p	1.34a	3.555	0.742	0.007	0.734	0.246	4.056 (4)	-3.864	
40.00p	2.46a	3.349	0.594	0.007	0.588	0.364	2.346 (4)	-5.989	
50.00p	3.86a	2.972	0.341	0.006	0.336	0.446	0.699 (4)	-8.051	
54.23p	4.45a	2.782	0.194	0.005	0.189	0.466	0.000 (4)	-8.871	FldPt
59.03p	5.10a	2.551	0.004	0.004	0.000	0.474	-0.789 (4)	-9.753	RaZero
60.00p	5.23a	2.503	-0.036	0.004	-0.041	0.474	-0.947 (4)	-9.924	
70.00p	6.49a	1.969	-0.491	0.003	-0.494	0.474	-2.558 (4)	-11.528	
80.00p	7.56a	1.382	-0.978	0.001	-0.980	0.474	-4.086 (4)	-12.794	
90.00p	8.40a	0.746	-1.445	0.000	-1.445	0.474	-5.473 (4)	-13.660	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1156.6 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling moment = 24.80Cos(heel) + 0.00Sin(heel)

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Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	3.80	1.20	Yes
(2) Abs. Ratio from 0.00 deg to Flood or RAzero	>1.400	67.967	66.567	Yes
(3) Area from Equilibrium to 20.00 deg or Flood	>0.0300 m-R	0.126	0.096	Yes
(4) Absolute Angle at Deck Immersion	>0.00 deg	8.74	8.74	Yes

12.3.2 MAX VERTICAL CRANE OPERATION

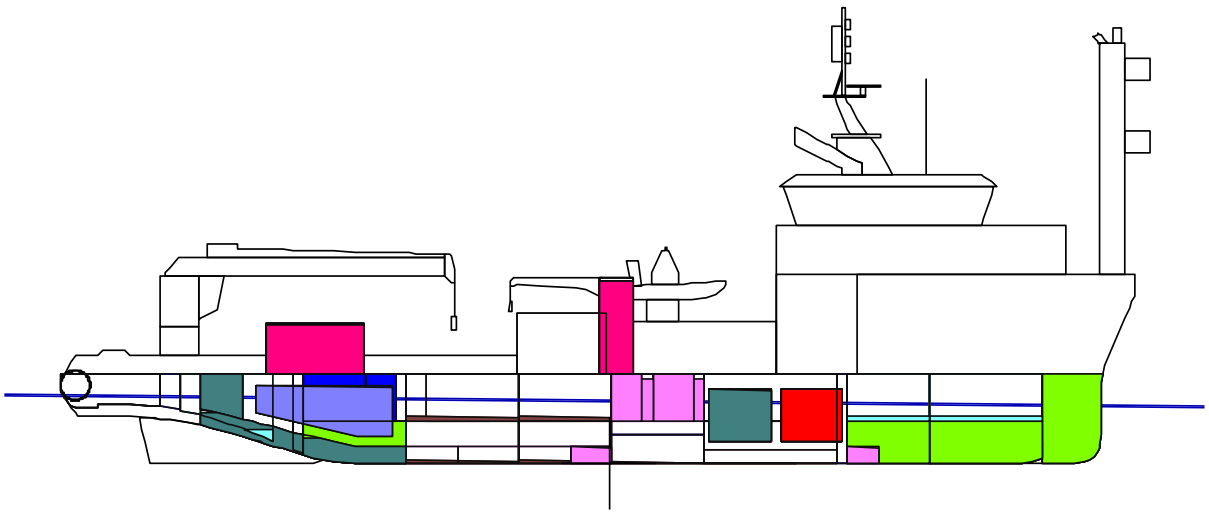
Max. Vertical moment – Lifting 3.500t @ 9.600 m radius (35.530 m boom length - Side Lifting to port side)

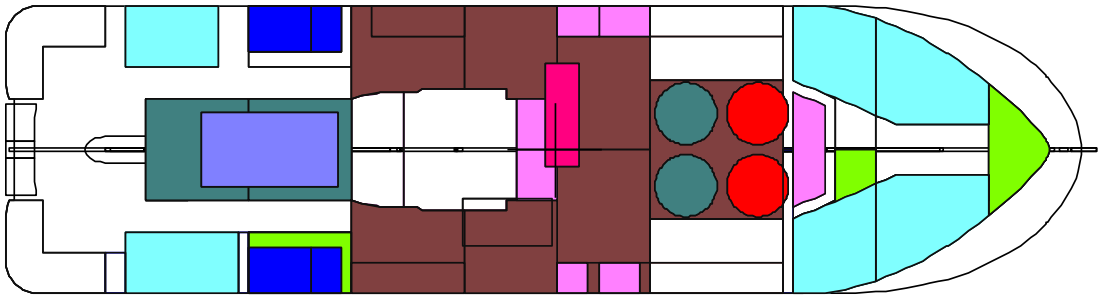
Floating Status

Draft FP	3.444 m	Heel	1.01p deg	GM(Solid)	4.198 m
Draft MS	3.748 m	Equil	Yes	F/S Corr.	0.475 m
Draft AP	4.052 m	Wind	Off	GM(Fluid)	3.723 m
Trim	0.61a m	Wave	No	KMT	9.335 m
LCG	1.085a m	VCG-Solid	5.138 m	TPcm	9.56
Displacement	2893.2 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	981.25	2.045a	0.252s	3.864
Displacement	2,893.23	1.085a	0.075p	5.138





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	221.38	53.23	FRESH WATER	32.54	8.02
FUEL OIL	30.35	5.84	MISC	72.22	56.48
SCALE INHIBITOR	57.86	98.00	CORROSION INHIBITOR	24.99	98.00
TREATED INJECTION WATER	61.80	100.00	TK ON MAIN DECK	123.45	66.80
BRINE	252.16	98.00	ACID	96.53	98.00

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	0.40	24.100f	0.000	6.800
CRANE BOOM FINAL	32.55	25.100a	10.535p	28.218
CRANE BOOM INITIAL	-32.55	17.438a	6.000p	11.504
LIFTED WEIGHT FINAL	3.50	25.100a	15.600p	45.245
Total Fixed:	**1,919.95**	**0.716a**	**0.347p**	**6.158**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DWSWBFP	1.025	100.00	54.57	26.510f	0.000	3.135	39.8
NO1_DWSWB.P	1.025	50.00	19.74	20.764f	3.241p	0.786	41.3
NO1_DWSWB.S	1.025	100.00	39.49	21.049f	3.353s	1.426	0.0
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	113.5
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
Subtotals:		**53.23%**	**221.38**	**13.196f**	**3.024s**	**1.723**	**194.6**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	7.60	5.61	21.205f	3.751p	2.621	63.6
NO1_FW.S	1.000	10.00	7.38	21.378f	3.507s	2.656	63.6
NO2_FW.P	1.000	10.00	6.64	16.245f	5.590p	2.647	51.8
NO2_FW.S	1.000	10.00	6.64	16.324f	5.317s	2.646	51.4
NO3_FW.S	1.000	5.00	3.56	19.297a	6.297s	1.548	6.7
NO3_FW.P	1.000	5.00	2.70	20.388a	6.365p	1.952	6.0
Subtotals:		**8.02%**	**32.54**	**11.357f**	**0.254s**	**2.467**	**243.1**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.C	0.850	3.40	1.49	8.250f	2.130p	0.033	91.9
NO2_FO.U.P	0.850	3.40	1.50	2.505f	5.352p	0.037	198.5
NO2_FO.U.S	0.850	9.80	3.83	2.616f	2.979s	0.066	198.5
NO3_FO.P	0.850	9.80	2.20	2.694a	7.517p	2.638	2.2
NO3_FO.S	0.850	9.80	2.20	2.694a	7.483s	2.638	2.2
NO3_FO.U.P	0.850	9.80	3.14	2.734a	5.869p	0.090	34.8
NO3_FO.U.S	0.850	9.80	3.14	2.697a	5.490s	0.090	32.3
NO4_FO.U.P	0.850	9.80	3.98	8.892a	5.757p	0.086	45.9
NO4_FO.U.S	0.850	9.80	3.98	8.918a	5.365s	0.086	42.3
FODAY.P	0.850	9.80	2.20	8.094a	7.517p	2.638	2.2
FODAY.S	0.850	9.80	2.69	8.740a	7.483s	2.638	2.6
Subtotals:		**5.84%**	**30.35**	**3.790a**	**0.032s**	**0.860**	**653.3**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.997f	7.502p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.698f	7.498s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.298f	7.502p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	0.999f	7.498s	3.872	0.8
DIRTYOIL.P	0.900	98.00	6.22	1.104a	1.511p	0.490	4.8
BILGEHOLD.S	1.000	98.00	6.91	1.104a	1.489s	0.490	5.3
SEWAGE	1.000	98.00	10.11	14.763f	0.035p	0.490	30.3
Subtotals:		**56.48%**	**72.22**	**3.785f**	**0.645p**	**2.784**	**45.2**

SCALE INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
SCLEIHBTR.P	1.100	98.00	14.47	13.401a	7.052p	3.872	3.2
SCLEIHBTR.S	1.100	98.00	14.47	13.401a	7.045s	3.872	3.2
SCALE_DISLVR.S	1.100	98.00	28.93	16.104a	7.045s	3.872	6.4
Subtotals:		**98.00%**	**57.86**	**14.753a**	**3.520s**	**3.872**	**12.8**

CORROSION INHIBITOR

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
CORSNIHBTR.P	0.950	98.00	24.99	16.104a	7.052p	3.872	5.5
Subtotals:		**98.00%**	**24.99**	**16.104a**	**7.052p**	**3.872**	**5.5**

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
TRTDINJW.P	1.000	100.00	30.90	11.800f	2.100p	2.850	0.0
TRTDINJW.S	1.000	100.00	30.90	11.800f	2.100s	2.850	0.0
Subtotals:		**100.00%**	**61.80**	**11.800f**	**0.000**	**2.850**	**0.0**

TK ON MAIN DECK

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NITROGENTK.P	1.185	98.00	47.27	17.250a	1.053p	6.721	8.7
CHEMADCTTK.P	1.185	98.00	76.18	0.409f	2.060p	8.020	42.0
Subtotals:		**66.80%**	**123.45**	**6.353a**	**1.674p**	**7.522**	**50.7**

BRINE

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_BRINETK.C	1.550	98.00	51.55	22.652a	0.022p	4.075	55.6
NO2_BRINETK.C	1.550	98.00	49.80	20.599a	0.009p	1.905	24.8
NO3_BRINETK.C	1.550	98.00	56.94	14.966a	0.008p	0.653	25.5
NO4_BRINETK.P	1.550	98.00	46.94	7.597f	2.105p	2.819	12.6
NO4_BRINETK.S	1.550	98.00	46.94	7.597f	2.095s	2.819	12.6
Subtotals:		**98.00%**	**252.16**	**9.250a**	**0.010p**	**2.406**	**131.1**

ACID

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_ACIDTK.C	1.160	98.00	96.53	16.267a	0.009p	3.322	39.3
Subtotals:		**98.00%**	**96.53**	**16.267a**	**0.009p**	**3.322**	**39.3**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**43.95%**	**973.27**	**1.813a**	**0.462s**	**3.125**	**1,375.7**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	2,869.92	0.940a	0.131p	2.049	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**2,893.23**	**1.118a**	**0.130p**	**2.041**	

Heeling Moment Report

Draft	3.750	m					Resultant	1.0000
WIND	20	Knots					(Sin 90Deg)	
Part	**Wind area**	**Effective area**	**Centre of Area from sea-level**	**Ch**	**Lever**	**Cs**	**Wind Force**	**Moment**
	(m2)	**(m2)**	**m**		**(m)**		**(N)**	**(m-MT)**
HULL.C	218.100	218.100	2.461	1.000	4.561	1.000	14082.608	6.547
DECKHOUSE	108.400	108.400	8.245	1.000	10.345	1.000	6999.334	7.381
FWDMAST	13.400	13.400	15.757	1.100	17.857	1.000	951.754	1.732
ANTENNA	0.300	0.300	15.608	1.100	17.708	1.000	21.308	0.038
CRANE1	23.500	23.500	14.456	1.000	16.556	1.500	2276.075	3.841
DECK	45.100	45.100	2.257	1.000	4.357	1.000	2912.085	1.293
CRANE2	37.200	37.200	6.568	1.000	8.668	1.500	3602.978	3.184
LIFEBOAT	10.000	10.000	6.059	1.000	8.159	1.000	645.695	0.537
CRANE3	2.900	2.900	6.419	1.000	8.519	1.500	280.877	0.244
TOTAL WIND HMMT								**24.799**

Heeling moment: 24.79 MT-m to port

IS CODE CRANE OPERATION CRITERIA

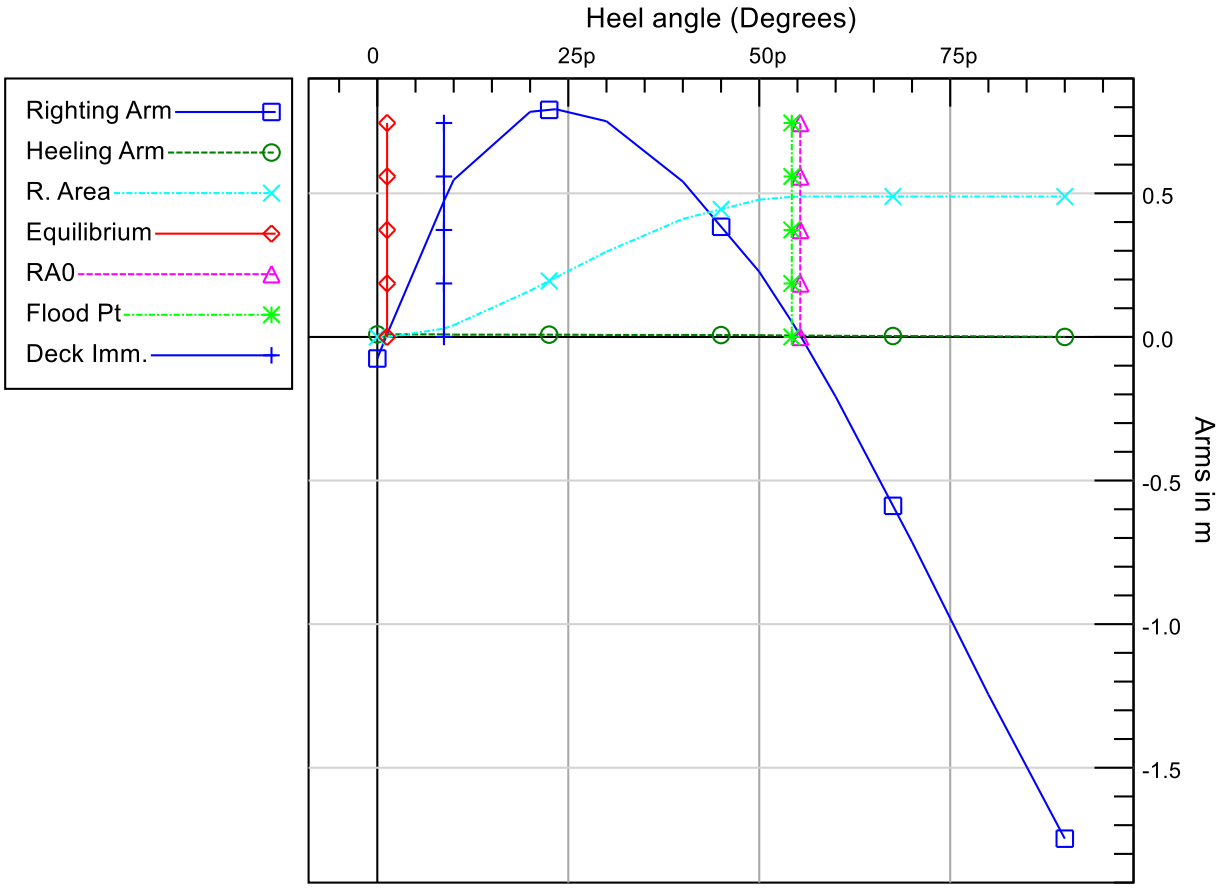
Free Surface Adjustment 0.467
Adjusted VCG 5.605

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
0.00	0.61a	3.748	-0.075	0.009	-0.083	0.000	8.505 (1,2)	1.209	
1.28p	0.61a	3.746	0.009	0.009	0.000	-0.001	8.369 (4)	1.038	Equil
8.72p	0.52a	3.690	0.473	0.008	0.465	0.029	7.458 (4)	0.000	Deck Imm.
10.00p	0.49a	3.672	0.547	0.008	0.538	0.040	7.288 (4)	-0.170	
20.00p	0.65a	3.603	0.784	0.008	0.776	0.161	5.776 (4)	-1.835	
23.35p	0.83a	3.594	0.793	0.008	**0.786**	0.207	5.210 (4)	-2.490	MaxRa
30.00p	1.35a	3.555	0.750	0.007	0.743	0.297	4.057 (4)	-3.870	
40.00p	2.49a	3.351	0.541	0.007	0.534	0.411	2.349 (4)	-6.002	
50.00p	3.90a	2.974	0.227	0.006	0.221	0.478	0.702 (4)	-8.074	
54.26p	4.50a	2.785	0.053	0.005	0.048	0.488	0.000 (4)	-8.904	FldPt
55.37p	4.65a	2.733	0.005	0.005	0.000	0.489	-0.183 (4)	-9.114	RaZero
60.00p	5.28a	2.507	-0.208	0.004	-0.212	0.489	-0.942 (4)	-9.955	
70.00p	6.55a	1.973	-0.714	0.003	-0.717	0.489	-2.552 (4)	-11.566	
80.00p	7.62a	1.387	-1.244	0.001	-1.246	0.489	-4.079 (4)	-12.833	
90.00p	8.46a	0.750	-1.747	0.000	-1.747	0.489	-5.467 (4)	-13.695	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 1351.7 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling moment = 24.80Cos(heel) + 0.00Sin(heel)

IS CODE CRANE OPERATION CRITERIA



Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	1.28	3.72	Yes
(2) Abs. Ratio from 0.00 deg to Flood or RAzero	>1.400	71.187	69.787	Yes
(3) Area from Equilibrium to 20.00 deg or Flood	>0.0300 m-R	0.162	0.132	Yes
(4) Absolute Angle at Deck Immersion	>0.00 deg	8.72	8.72	Yes

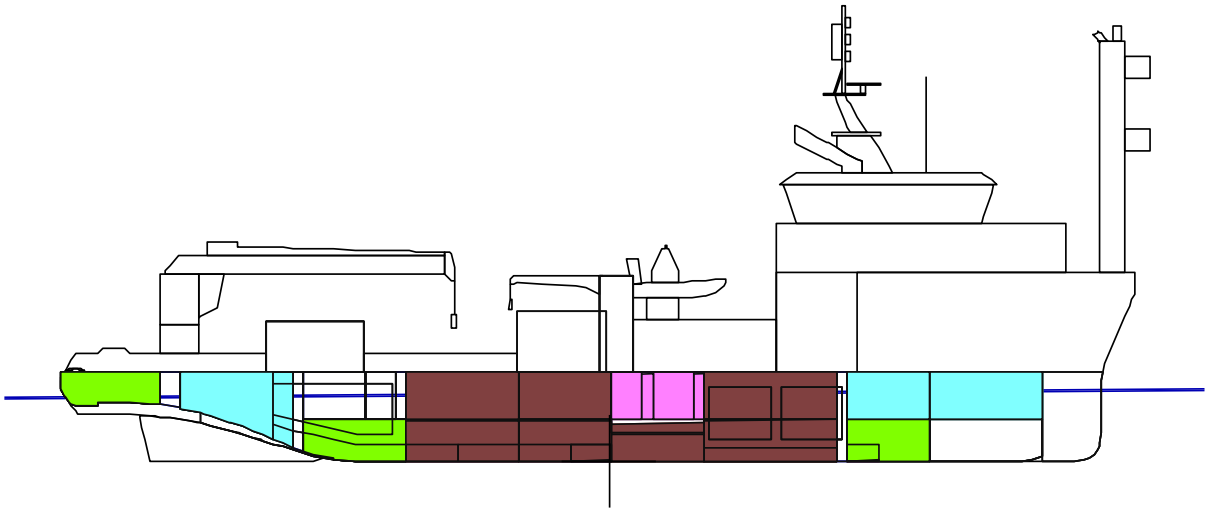
12.4 CONDITION NO.4: BALLAST DEPARTURE (WITHOUT CARGO)

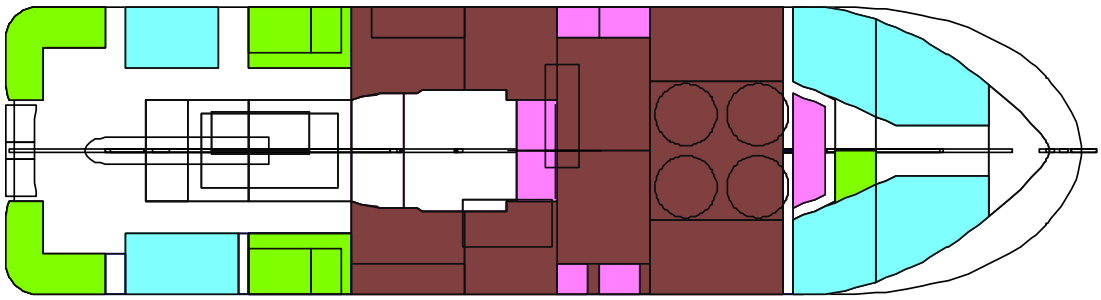
Floating Status

Draft FP	4.211 m	Heel	0.50p deg	GM(Solid)	4.335 m
Draft MS	4.038 m	Equil	Yes	F/S Corr.	0.232 m
Draft AP	3.864 m	Wind	Off	GM(Fluid)	4.103 m
Trim	0.35f m	Wave	No	KMT	8.934 m
LCG	0.013f m	VCG-Solid	4.599 m	TPcm	9.54
Displacement	3110.4 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	1,198.40	0.978f	0.286s	2.696
Displacement	3,110.38	0.013f	0.039p	4.599





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	223.93	53.85	FRESH WATER	405.66	100.00
FUEL OIL	509.44	98.00	MISC	51.30	40.12

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	4.00	24.100f	0.000	6.800
Total Fixed:	**1,920.05**	**0.495a**	**0.242p**	**5.805**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	0.0
NO3_DWSWB.P	1.025	100.00	49.20	14.761a	6.538p	1.351	0.0
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
DWSWBAP.P	1.025	100.00	33.58	29.636a	6.202p	4.439	0.0
DWSWBAP.S	1.025	100.00	33.58	29.636a	6.202s	4.439	0.0
Subtotals:		**53.85%**	**223.93**	**11.087a**	**1.247s**	**2.247**	**0.0**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	100.00	73.83	21.497f	3.747p	3.946	63.6
NO1_FW.S	1.000	100.00	73.83	21.497f	3.747s	3.946	63.6
NO2_FW.P	1.000	100.00	66.44	16.374f	5.508p	3.914	0.0
NO2_FW.S	1.000	100.00	66.44	16.374f	5.508s	3.914	0.0
NO3_FW.S	1.000	100.00	71.18	21.378a	6.556s	3.689	0.0
NO3_FW.P	1.000	100.00	53.94	22.110a	6.554p	3.828	0.0
Subtotals:		**100.00%**	**405.66**	**6.497f**	**0.279s**	**3.875**	**127.2**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.P	0.850	98.00	31.76	9.411f	7.501p	3.872	3.1
NO1_FO.S	0.850	98.00	31.76	9.411f	7.499s	3.872	3.1
NO1_FO.U.P	0.850	98.00	43.11	9.387f	6.622p	0.972	3.1
NO1_FO.U.S	0.850	98.00	43.11	9.387f	6.620s	0.972	3.1
NO1_FO.U.C	0.850	98.00	42.86	9.425f	0.039p	0.417	120.7
NO2_FO.U.P	0.850	98.00	43.22	2.994f	4.641p	0.735	198.5
NO2_FO.U.S	0.850	98.00	38.33	2.943f	4.277s	0.579	198.5
NO3_FO.P	0.850	98.00	21.99	2.595a	7.501p	3.872	2.2
NO3_FO.S	0.850	98.00	21.99	2.595a	7.499s	3.872	2.2
NO3_FO.U.P	0.850	98.00	31.43	2.565a	6.466p	0.947	2.2
NO3_FO.U.S	0.850	98.00	31.43	2.565a	6.465s	0.947	2.2
NO4_FO.U.P	0.850	98.00	39.79	8.630a	6.401p	0.926	2.7
NO4_FO.U.S	0.850	98.00	39.79	8.630a	6.400s	0.926	2.7
FODAY.P	0.850	98.00	21.99	7.995a	7.501p	3.872	2.2
FODAY.S	0.850	98.00	26.88	8.592a	7.499s	3.872	2.6
Subtotals:		**98.00%**	**509.44**	**1.344f**	**0.004p**	**1.755**	**549.0**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	4.002f	7.501p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.701f	7.499s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.301f	7.501p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	1.001f	7.499s	3.872	0.8
DIRTYOIL.P	0.900	9.80	0.62	1.070a	1.567p	0.049	4.8
BILGEHOLD.S	1.000	9.80	0.69	1.070a	1.433s	0.049	5.3
SEWAGE	1.000	9.80	1.01	14.781f	0.262p	0.050	30.3
Subtotals:		**40.12%**	**51.30**	**2.970f**	**0.922p**	**3.699**	**45.3**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**53.75%**	**1,190.32**	**0.832f**	**0.288s**	**2.654**	**721.5**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	3,087.08	0.202f	0.061p	2.176	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**3,110.38**	**0.027f**	**0.060p**	**2.167**	

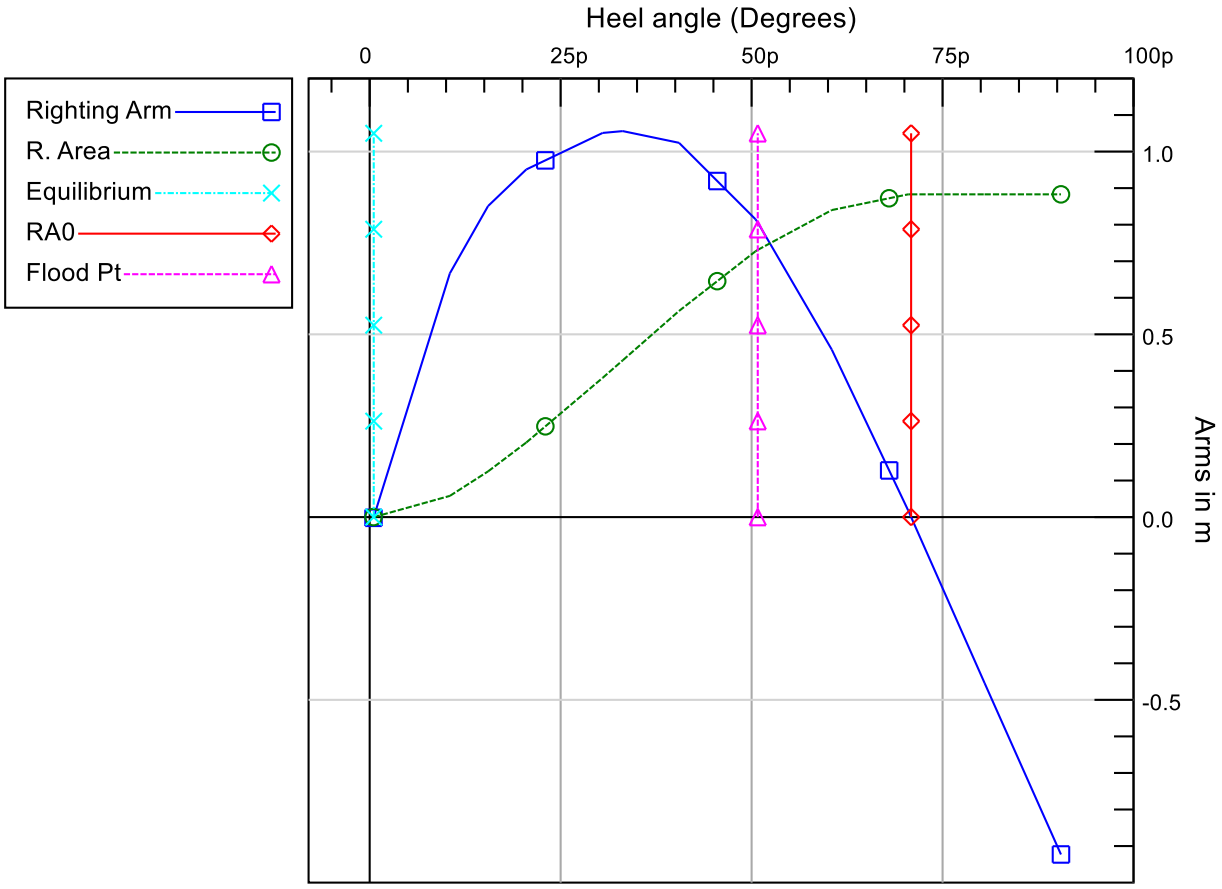
: OSV CHEMICAL CODE – CHAPTER 2 , 2.2

Free Surface Adjustment 0.232
Adjusted VCG 4.831

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Area (m-Rad)	Flood Pt Height (m)	Notes
0.50p	0.35f	4.037	-0.002	0.000	7.995 (4)	
0.53p	0.35f	4.037	0.000	0.000	7.992 (4)	Equil
10.50p	0.51f	3.949	0.667	0.058	6.757 (4)	
15.50p	0.55f	3.911	0.851	0.125	6.015 (4)	
20.50p	0.41f	3.893	0.951	0.204	5.208 (4)	
30.50p	0.28a	3.850	1.050	0.381	3.471 (4)	
33.13p	0.54a	3.819	**1.056**	0.429	3.009 (4)	MaxRa
40.50p	1.44a	3.656	1.024	0.564	1.748 (4)	
50.50p	2.80a	3.314	0.815	0.727	0.052 (4)	
50.80p	2.84a	3.303	0.806	0.731	0.000 (4)	FldPt
60.50p	4.09a	2.890	0.458	0.840	-1.658 (4)	
70.50p	5.22a	2.399	0.018	0.883	-3.337 (4)	
70.88p	5.26a	2.379	0.000	0.883	-3.399 (4)	RaZero
80.50p	6.24a	1.832	-0.454	0.883	-4.897 (4)	
90.50p	7.07a	1.204	-0.923	0.883	-6.292 (4)	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 721.5 MT-m was used to adjust the VCG.

: OSV CHEMICAL CODE – CHAPTER 2 , 2.2



Limit	Min/Max	Actual	Margin	Pass
(1) Area from 0.00 deg to MaxRA at 30.00	>0.0550 m-R	0.381	0.326	Yes
(3) Area from 0.00 deg to 40.00 or Flood	>0.0900 m-R	0.564	0.474	Yes
(4) Area from 30.00 deg to 40.00 or Flood	>0.0300 m-R	0.183	0.153	Yes
(5) Righting Arm at 30.00 deg	>0.200 m	1.050	0.850	Yes
(6) Absolute Angle at MaxRA	>15.00 deg	33.13	18.13	Yes
(7) GM Upright	>0.150 m	4.104	3.954	Yes

Heeling Moment Derivation

Part	LPA (m^2)	HCP (m)	Arm (m)	Pressure (MT/m^2)	Moment (m-MT)
HULL.C	203.3	2.731	4.624	0.051	48.31
DECKHOUSE	108.4	8.321	10.214	0.051	56.92
FWDMAST	13.4	15.833	17.726	0.051	12.23
ANTENNA	0.3	15.685	17.578	0.051	0.24
CRANE1	23.5	14.533	16.426	0.051	19.80
DECK	45.1	2.334	4.227	0.051	9.81
CRANE2	37.2	6.645	8.538	0.051	16.31
LIFEBOAT	10.0	6.135	8.028	0.051	4.12
CRANE3	2.9	6.496	8.389	0.051	1.24

Total wind heeling moment 168.99 to port

IMO SEVERE WIND & ROLLING

Free Surface Adjustment 0.202
Adjusted VCG 4.801

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
22.04s	0.33f	3.890	-1.056	0.081	-1.137	0.000	4.948 (3)	-2.186	Roll
12.04s	0.54f	3.933	-0.820	0.081	-0.902	-0.178	6.537 (3)	-0.594	
2.04s	0.35f	4.034	-0.186	0.081	-0.267	-0.286	7.820 (3)	0.904	
1.29p	0.35f	4.036	0.054	0.081	-0.027	-0.294	7.907 (4)	1.015	SSWEq
1.66p	0.35f	4.035	0.081	0.081	0.000	-0.294	7.864 (4)	0.960	Equil
7.96p	0.45f	3.983	0.514	0.081	0.433	-0.271	7.097 (4)	0.027	
8.14p	0.45f	3.981	0.526	0.081	0.445	-0.269	7.073 (4)	0.000	Deck Imm.
16.00p	0.54f	3.908	0.871	0.081	0.790	-0.182	5.937 (4)	-1.220	
17.96p	0.50f	3.901	0.915	0.081	0.834	-0.154	5.625 (4)	-1.533	
27.96p	0.06a	3.869	1.049	0.081	0.968	0.007	3.920 (4)	-3.156	
33.42p	0.57a	3.815	1.072	0.081	**0.991**	0.100	2.959 (4)	-4.315	MaxRa
37.96p	1.10a	3.725	1.060	0.081	0.979	0.178	2.177 (4)	-5.291	
47.96p	2.45a	3.410	0.907	0.081	0.826	0.338	0.485 (4)	-7.391	
50.00p	2.73a	3.334	0.852	0.081	0.771	0.367	0.137 (4)	-7.801	
50.80p	2.84a	3.303	0.829	0.081	0.747	0.377	0.000 (4)	-7.960	FldPt
57.96p	3.77a	3.004	0.584	0.081	0.502	0.455	-1.224 (4)	-9.315	
67.96p	4.94a	2.531	0.161	0.081	0.080	0.508	-2.919 (4)	-10.964	

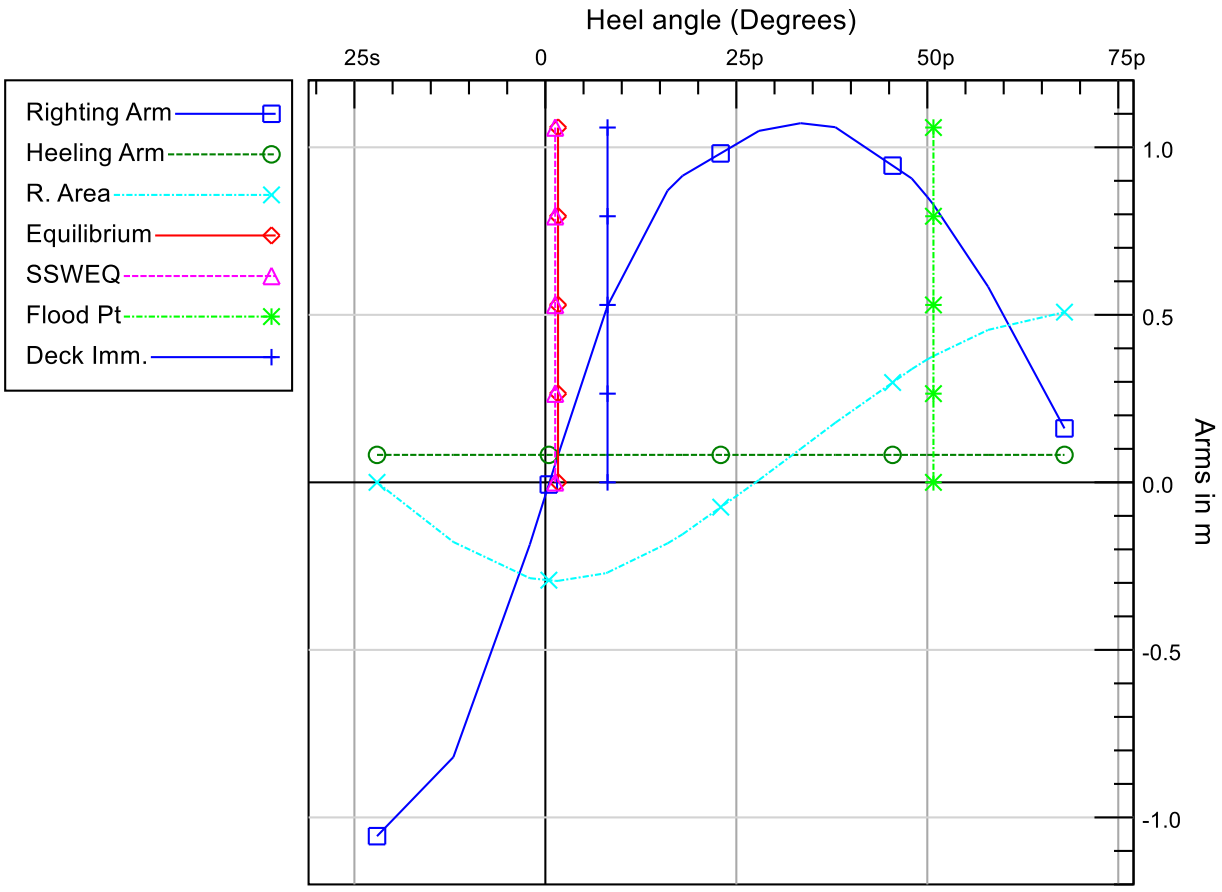
Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.

A Free Surface Moment of 628.3 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
wind heeling arms derived from this moment (in m-MT):
Port heeling moment = 253.48

Roll angle is 23.32
Equilibrium for load condition without gust is 1.29p

IMO SEVERE WIND & ROLLING



Limit	Min/Max	Actual	Margin	Pass
(1) Res. Ratio from Roll to Abs 50.00 deg or Flood	>1.000	2.245	1.245	Yes
(2) Angle from SSWEQ to 80% of Dk. Imm. or Abs 16.00 deg	>0.00 deg	5.23	5.23	Yes

Heeling Moment Report

FIFI HEELING MOMENT CALCULATION			
Reaction Force of the Fire Monitor	=	1.530	MT
Number of Fire Monitors	=	2	
Draft of the loading condition	=	4.038	m
D/2	=	2.019	m
Height of firefighting monitor above baseline	=	19.600	m
Moment due to Fire Fighting operation	=	53.798	MT-m
Reaction Force of the Bow Thruster	=	18	MT
D/2	=	2.019	m
Height of Bow Thruster above baseline	=	1.200	m
Moment due to Bow Thruster operation	=	14.742	MT-m
Total Heeling Moment	=	68.540	Tm

Heeling moment: 68.54 MT-m to port

FIRE FIGHTING CRITERIA

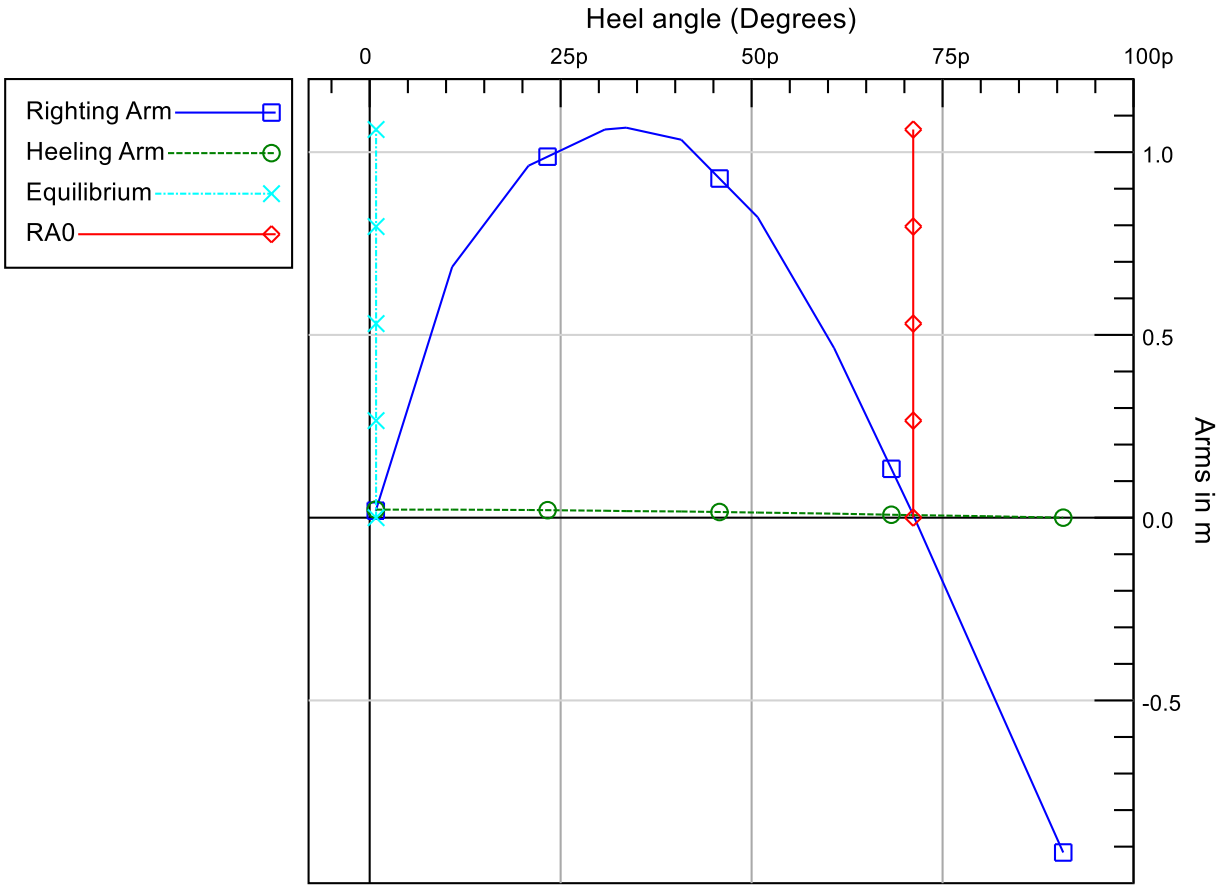
Free Surface Adjustment 0.211
Adjusted VCG 4.810

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Notes
0.80p	0.35f	4.037	0.019	0.022	-0.003	
0.84p	0.35f	4.037	0.022	0.022	0.000	Equil
10.80p	0.52f	3.946	0.686	0.022	0.664	
20.80p	0.39f	3.893	0.963	0.021	0.942	
30.80p	0.31a	3.847	1.062	0.019	1.043	
33.57p	0.59a	3.812	1.067	0.018	**1.049**	MaxRa
40.80p	1.48a	3.648	1.034	0.017	1.018	
50.80p	2.84a	3.303	0.822	0.014	0.808	
60.80p	4.12a	2.877	0.464	0.011	0.453	
70.80p	5.25a	2.383	0.024	0.007	0.016	
71.16p	5.29a	2.364	0.007	0.007	0.000	RaZero
80.80p	6.26a	1.815	-0.447	0.004	-0.451	
90.80p	7.10a	1.184	-0.915	0.000	-0.915	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 656.4 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling mmt = 68.54Cos(heel) + 0.00Sin(heel) <= 90 deg
Port heeling mmt = 0.00 > 90 deg

FIRE FIGHTING CRITERIA



Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	0.84	4.16	Yes

12.4.1 MAX TRANSVERSE CRANE OPERATION

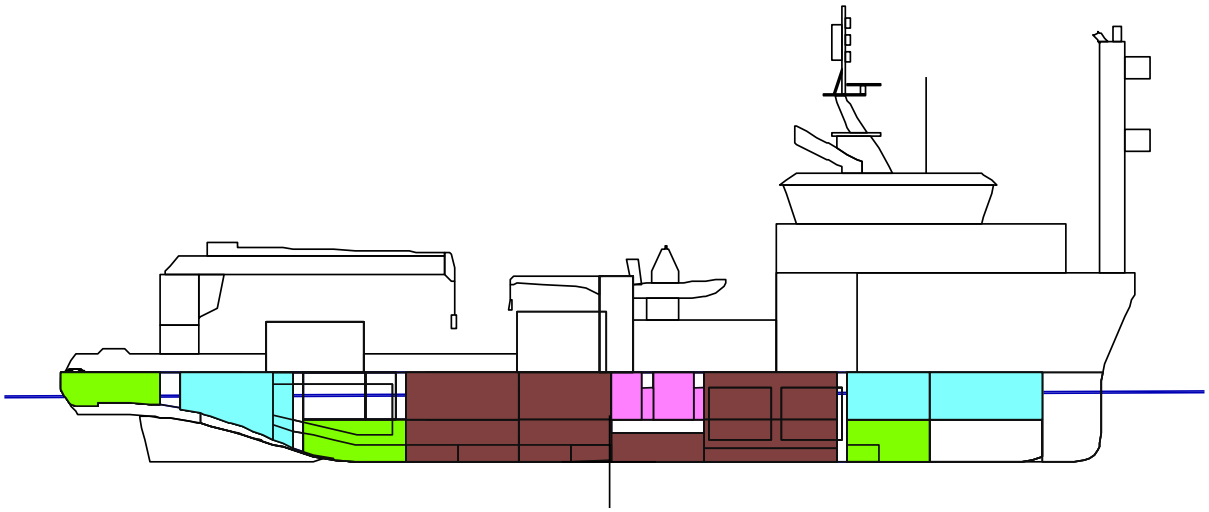
Max. Transverse moment – Lifting 3.500t @ 35.00 m radius (35.530 m boom length –Side Lifting to port side)

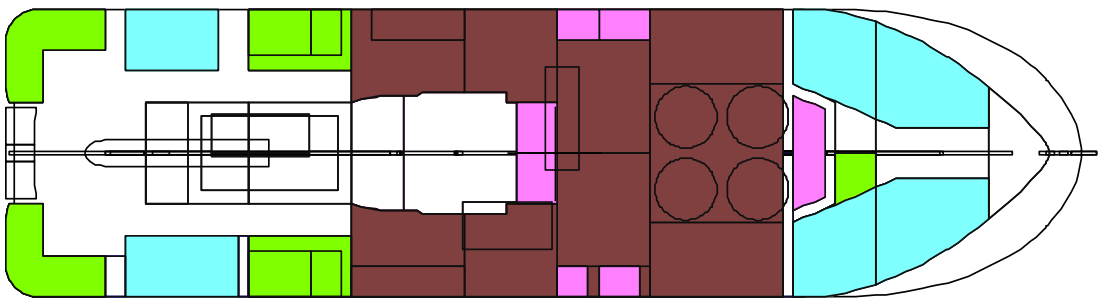
Floating Status

Draft FP	4.174 m	Heel	3.52p deg	GM(Solid)	4.233 m
Draft MS	4.034 m	Equil	Yes	F/S Corr.	0.194 m
Draft AP	3.894 m	Wind	Off	GM(Fluid)	4.039 m
Trim	0.28f m	Wave	No	KMT	8.828 m
LCG	0.096a m	VCG-Solid	4.603 m	TPcm	9.47
Displacement	3113.9 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	1,201.90	0.693f	0.305p	2.712
Displacement	3,113.88	0.096a	0.267p	4.603





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	223.93	53.85	FRESH WATER	405.66	100.00
FUEL OIL	509.44	98.00	MISC	51.30	40.12

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	4.00	24.100f	0.000	6.800
CRANE BOOM FINAL	32.55	25.100a	23.235p	11.190
CRANE BOOM INITIAL	-32.55	17.438a	6.000p	11.504
LIFTED WEIGHT FINAL	3.50	25.100a	41.000p	11.190
Total Fixed:	**1,923.55**	**0.669a**	**0.608p**	**5.809**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	0.0
NO3_DWSWB.P	1.025	100.00	49.20	14.761a	6.538p	1.351	0.0
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
DWSWBAP.P	1.025	100.00	33.58	29.636a	6.202p	4.439	0.0
DWSWBAP.S	1.025	100.00	33.58	29.636a	6.202s	4.439	0.0
Subtotals:		**53.85%**	**223.93**	**11.087a**	**1.247s**	**2.247**	**0.0**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	100.00	73.83	21.497f	3.747p	3.946	63.6
NO1_FW.S	1.000	100.00	73.83	21.497f	3.747s	3.946	63.6
NO2_FW.P	1.000	100.00	66.44	16.374f	5.508p	3.914	0.0
NO2_FW.S	1.000	100.00	66.44	16.374f	5.508s	3.914	0.0
NO3_FW.S	1.000	100.00	71.18	21.378a	6.556s	3.689	0.0
NO3_FW.P	1.000	100.00	53.94	22.110a	6.554p	3.828	0.0
Subtotals:		**100.00%**	**405.66**	**6.497f**	**0.279s**	**3.875**	**127.2**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.P	0.850	98.00	31.76	9.409f	7.506p	3.872	2.8
NO1_FO.S	0.850	98.00	31.76	9.409f	7.494s	3.872	2.8
NO1_FO.U.P	0.850	98.00	43.11	9.385f	6.625p	0.972	3.1

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.S	0.850	98.00	43.11	9.385f	6.616s	0.972	3.1
NO1_FO.U.C	0.850	98.00	42.86	9.408f	0.067p	0.417	5.2
NO2_FO.U.P	0.850	98.00	43.22	2.994f	4.643p	0.735	198.5
NO2_FO.U.S	0.850	98.00	38.33	2.943f	4.274s	0.579	198.5
NO3_FO.P	0.850	98.00	21.99	2.596a	7.506p	3.872	2.0
NO3_FO.S	0.850	98.00	21.99	2.596a	7.494s	3.872	2.0
NO3_FO.U.P	0.850	98.00	31.43	2.565a	6.469p	0.947	2.2
NO3_FO.U.S	0.850	98.00	31.43	2.565a	6.461s	0.947	2.2
NO4_FO.U.P	0.850	98.00	39.79	8.631a	6.405p	0.926	2.7
NO4_FO.U.S	0.850	98.00	39.79	8.631a	6.397s	0.926	2.7
FODAY.P	0.850	98.00	21.99	7.996a	7.506p	3.872	2.0
FODAY.S	0.850	98.00	26.88	8.594a	7.494s	3.872	2.4
Subtotals:		**98.00%**	**509.44**	**1.341f**	**0.010p**	**1.756**	**432.2**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	4.001f	7.506p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.701f	7.494s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.301f	7.506p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	1.000f	7.494s	3.872	0.8
DIRTYOIL.P	0.900	9.80	0.62	1.076a	1.973p	0.064	4.8
BILGEHOLD.S	1.000	9.80	0.69	1.076a	1.027s	0.064	5.3
SEWAGE	1.000	9.80	1.01	14.761f	1.516p	0.089	30.3
Subtotals:		**40.12%**	**51.30**	**2.969f**	**0.963p**	**3.700**	**45.3**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**53.75%**	**1,190.32**	**0.831f**	**0.284s**	**2.654**	**604.7**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	3,090.58	0.088f	0.419p	2.190	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**3,113.88**	**0.085a**	**0.416p**	**2.181**	

Heeling Moment Report

Draft	4.038	m					Resultant	1.0000
WIND	20	Knots					(Sin 90Deg)	
Part	**Wind area**	**Effective area**	**Centre of Area from sea-level**	**Ch**	**Lever**	**Cs**	**Wind Force**	**Moment**
	(m2)	**(m2)**	**m**		**(m)**		**(N)**	**(m-MT)**
HULL.C	203.300	203.300	2.524	1.000	4.624	1.000	13126.980	6.187
DECKHOUSE	108.400	108.400	8.114	1.000	10.214	1.000	6999.334	7.288
FWDMAST	13.400	13.400	15.626	1.100	17.726	1.000	951.754	1.720
ANTENNA	0.300	0.300	15.478	1.100	17.578	1.000	21.308	0.038
CRANE1	23.500	23.500	14.326	1.000	16.426	1.500	2276.075	3.811
DECK	45.100	45.100	2.127	1.000	4.227	1.000	2912.085	1.255
CRANE2	37.200	37.200	6.438	1.000	8.538	1.500	3602.978	3.136
LIFEBOAT	10.000	10.000	5.928	1.000	8.028	1.000	645.695	0.528
CRANE3	2.900	2.900	6.289	1.000	8.389	1.500	280.877	0.240
TOTAL WIND HMMT								**24.203**

Heeling moment: 24.20 MT-m to port

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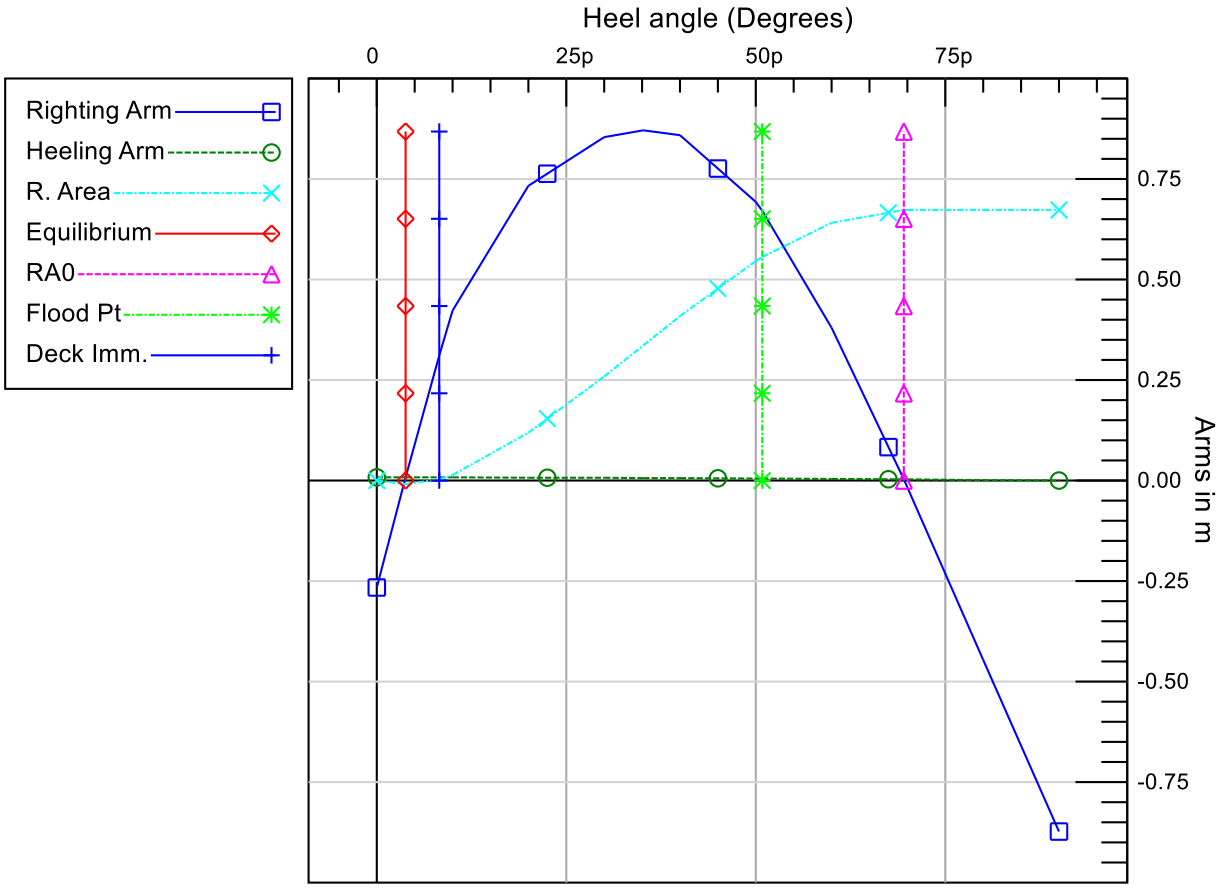
Free Surface Adjustment 0.194
Adjusted VCG 4.797

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
0.00	0.27f	4.036	-0.266	0.008	-0.274	0.000	8.065 (3,4)	1.218	
3.80p	0.28f	4.025	0.008	0.008	0.000	-0.009	7.629 (4)	0.658	Equil
8.24p	0.37f	3.979	0.311	0.008	0.303	0.003	7.076 (4)	0.000	Deck Imm.
10.00p	0.41f	3.955	0.423	0.008	0.415	0.014	6.842 (4)	-0.264	
20.00p	0.31f	3.900	0.733	0.007	0.726	0.119	5.306 (4)	-1.846	
30.00p	0.37a	3.866	0.854	0.007	0.848	0.259	3.573 (4)	-3.670	
35.15p	0.91a	3.800	0.871	0.006	**0.865**	0.336	2.669 (4)	-4.780	MaxRa
40.00p	1.53a	3.688	0.859	0.006	0.853	0.409	1.843 (4)	-5.828	
50.00p	2.90a	3.356	0.693	0.005	0.688	0.546	0.145 (4)	-7.915	
50.85p	3.01a	3.323	0.671	0.005	0.666	0.556	0.000 (4)	-8.084	FldPt
60.00p	4.20a	2.937	0.380	0.004	0.376	0.641	-1.565 (4)	-9.797	
69.53p	5.30a	2.475	0.003	0.003	0.000	0.673	-3.171 (4)	-11.319	RaZero
70.00p	5.35a	2.451	-0.017	0.003	-0.019	0.673	-3.248 (4)	-11.386	
80.00p	6.37a	1.888	-0.446	0.001	-0.447	0.673	-4.815 (4)	-12.649	
90.00p	7.21a	1.262	-0.873	0.000	-0.873	0.673	-6.219 (4)	-13.526	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 604.2 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling moment = 24.20Cos(heel) + 0.00Sin(heel)

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Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	3.80	1.20	Yes
(2) Abs. Ratio from 0.00 deg to Flood or RAzero	>1.400	93.287	91.887	Yes
(3) Area from Equilibrium to 20.00 deg or Flood	>0.0300 m-R	0.128	0.098	Yes
(4) Absolute Angle at Deck Immersion	>0.00 deg	8.24	8.24	Yes

12.4.2 MAX VERTICAL CRANE OPERATION

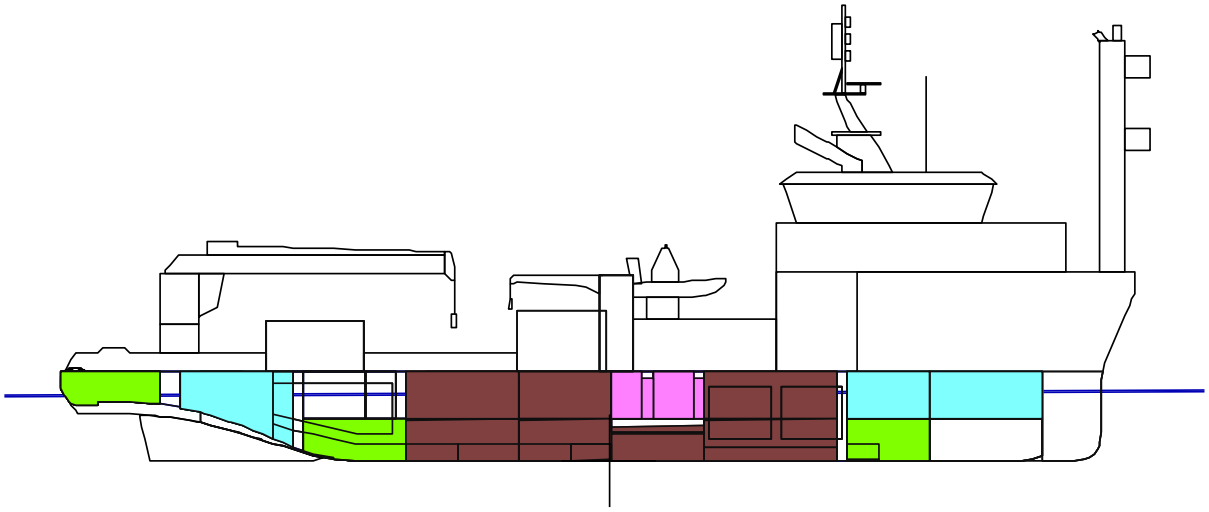
Max. Vertical moment – Lifting 3.500t @ 9.600 m radius (35.530 m boom length - Side Lifting to port side)

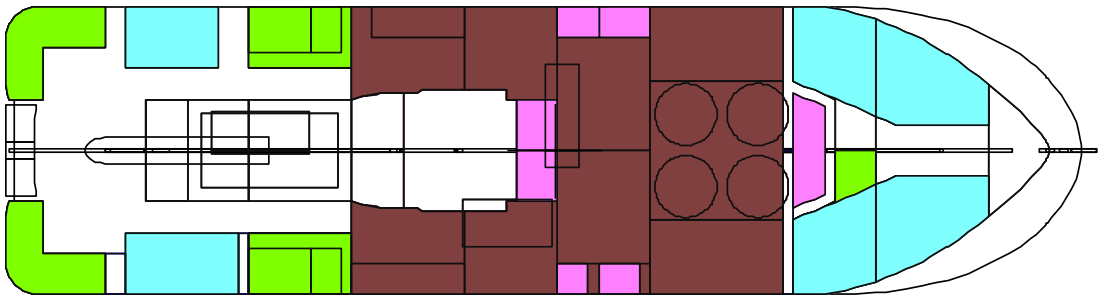
Floating Status

Draft FP	4.171 m	Heel	1.44p deg	GM(Solid)	4.116 m
Draft MS	4.036 m	Equil	Yes	F/S Corr.	0.200 m
Draft AP	3.901 m	Wind	Off	GM(Fluid)	3.916 m
Trim	0.27f m	Wave	No	KMT	8.934 m
LCG	0.096a m	VCG-Solid	4.819 m	TPcm	9.55
Displacement	3113.9 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	1,201.90	0.693f	0.116s	3.272
Displacement	3,113.88	0.096a	0.105p	4.819





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	223.93	53.85	FRESH WATER	405.66	100.00
FUEL OIL	509.44	98.00	MISC	51.30	40.12

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	4.00	24.100f	0.000	6.800
CRANE BOOM FINAL	32.55	25.100a	10.535p	28.218
CRANE BOOM INITIAL	-32.55	17.438a	6.000p	11.504
LIFTED WEIGHT FINAL	3.50	25.100a	15.600p	45.245
Total Fixed:	**1,923.55**	**0.669a**	**0.347p**	**6.159**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	0.0
NO3_DWSWB.P	1.025	100.00	49.20	14.761a	6.538p	1.351	0.0
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
DWSWBAP.P	1.025	100.00	33.58	29.636a	6.202p	4.439	0.0
DWSWBAP.S	1.025	100.00	33.58	29.636a	6.202s	4.439	0.0
Subtotals:		**53.85%**	**223.93**	**11.087a**	**1.247s**	**2.247**	**0.0**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.P	1.000	100.00	73.83	21.497f	3.747p	3.946	63.6
NO1_FW.S	1.000	100.00	73.83	21.497f	3.747s	3.946	63.6

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_FW.P	1.000	100.00	66.44	16.374f	5.508p	3.914	0.0
NO2_FW.S	1.000	100.00	66.44	16.374f	5.508s	3.914	0.0
NO3_FW.S	1.000	100.00	71.18	21.378a	6.556s	3.689	0.0
NO3_FW.P	1.000	100.00	53.94	22.110a	6.554p	3.828	0.0
Subtotals:		**100.00%**	**405.66**	**6.497f**	**0.279s**	**3.875**	**127.2**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.P	0.850	98.00	31.76	9.409f	7.502p	3.872	3.1
NO1_FO.S	0.850	98.00	31.76	9.409f	7.497s	3.872	3.1
NO1_FO.U.P	0.850	98.00	43.11	9.385f	6.623p	0.972	3.1
NO1_FO.U.S	0.850	98.00	43.11	9.385f	6.619s	0.972	3.1
NO1_FO.U.C	0.850	98.00	42.86	9.412f	0.059p	0.417	20.5
NO2_FO.U.P	0.850	98.00	43.22	2.994f	4.641p	0.735	198.5
NO2_FO.U.S	0.850	98.00	38.33	2.943f	4.276s	0.579	198.5
NO3_FO.P	0.850	98.00	21.99	2.596a	7.502p	3.872	2.2
NO3_FO.S	0.850	98.00	21.99	2.596a	7.497s	3.872	2.2
NO3_FO.U.P	0.850	98.00	31.43	2.565a	6.467p	0.947	2.2
NO3_FO.U.S	0.850	98.00	31.43	2.565a	6.464s	0.947	2.2
NO4_FO.U.P	0.850	98.00	39.79	8.631a	6.402p	0.926	2.7
NO4_FO.U.S	0.850	98.00	39.79	8.631a	6.399s	0.926	2.7
FODAY.P	0.850	98.00	21.99	7.996a	7.502p	3.872	2.2
FODAY.S	0.850	98.00	26.88	8.594a	7.497s	3.872	2.6
Subtotals:		**98.00%**	**509.44**	**1.342f**	**0.007p**	**1.756**	**448.8**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	4.001f	7.502p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.701f	7.497s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.301f	7.502p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	1.000f	7.497s	3.872	0.8
DIRTYOIL.P	0.900	9.80	0.62	1.077a	1.693p	0.051	4.8
BILGEHOLD.S	1.000	9.80	0.69	1.077a	1.307s	0.051	5.3
SEWAGE	1.000	9.80	1.01	14.777f	0.754p	0.059	30.3
Subtotals:		**40.12%**	**51.30**	**2.970f**	**0.937p**	**3.699**	**45.3**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**53.75%**	**1,190.32**	**0.831f**	**0.287s**	**2.654**	**621.3**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	3,090.58	0.089f	0.172p	2.179	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**3,113.88**	**0.084a**	**0.171p**	**2.170**	

Heeling Moment Report

Draft	4.038	m					Resultant	1.0000
WIND	20	Knots					(Sin 90Deg)	
Part	**Wind area**	**Effective area**	**Centre of Area from sea-level**	**Ch**	**Lever**	**Cs**	**Wind Force**	**Moment**
	(m2)	**(m2)**	**m**		**(m)**		**(N)**	**(m-MT)**
HULL.C	203.300	203.300	2.524	1.000	4.624	1.000	13126.980	6.187
DECKHOUSE	108.400	108.400	8.114	1.000	10.214	1.000	6999.334	7.288
FWDMAST	13.400	13.400	15.626	1.100	17.726	1.000	951.754	1.720
ANTENNA	0.300	0.300	15.478	1.100	17.578	1.000	21.308	0.038
CRANE1	23.500	23.500	14.326	1.000	16.426	1.500	2276.075	3.811
DECK	45.100	45.100	2.127	1.000	4.227	1.000	2912.085	1.255
CRANE2	37.200	37.200	6.438	1.000	8.538	1.500	3602.978	3.136
LIFEBOAT	10.000	10.000	5.928	1.000	8.028	1.000	645.695	0.528
CRANE3	2.900	2.900	6.289	1.000	8.389	1.500	280.877	0.240
TOTAL WIND HMMT								**24.203**

Heeling moment: 24.20 MT-m to port

IS CODE CRANE OPERATION CRITERIA

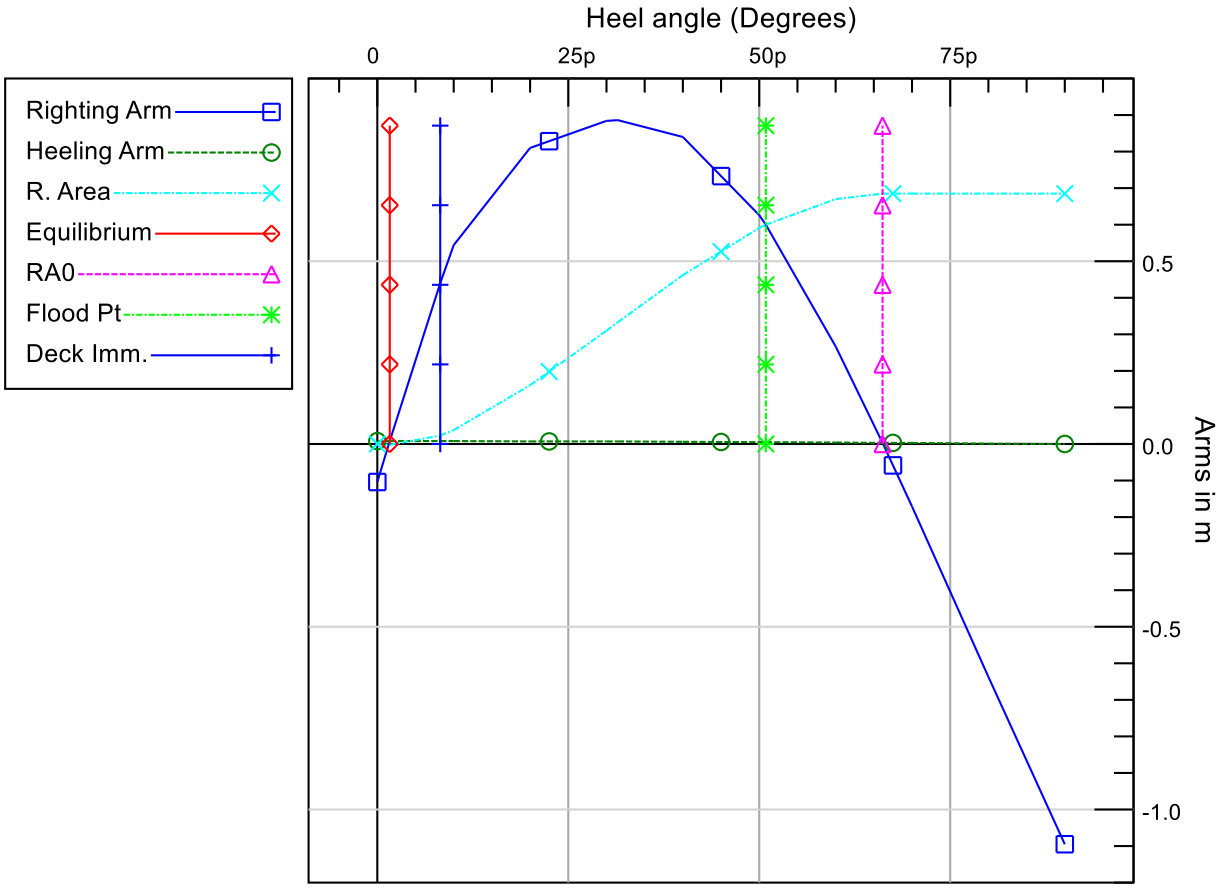
Free Surface Adjustment 0.199
Adjusted VCG 5.018

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
0.00	0.27f	4.036	-0.104	0.008	-0.112	0.000	8.065 (3,4)	1.218	
1.63p	0.27f	4.034	0.008	0.008	0.000	-0.002	7.883 (4)	0.979	Equil
8.24p	0.37f	3.979	0.439	0.008	0.432	0.023	7.076 (4)	0.000	Deck Imm.
10.00p	0.41f	3.955	0.544	0.008	0.537	0.038	6.841 (4)	-0.265	
20.00p	0.31f	3.900	0.810	0.007	0.803	0.161	5.306 (4)	-1.846	
30.00p	0.37a	3.866	0.884	0.007	0.878	0.310	3.573 (4)	-3.671	
31.48p	0.52a	3.852	0.886	0.007	**0.879**	0.333	3.311 (4)	-3.989	MaxRa
40.00p	1.54a	3.689	0.840	0.006	0.834	0.462	1.844 (4)	-5.834	
50.00p	2.92a	3.357	0.626	0.005	0.621	0.591	0.147 (4)	-7.929	
50.86p	3.04a	3.324	0.600	0.005	0.595	0.600	0.000 (4)	-8.101	FldPt
60.00p	4.24a	2.940	0.267	0.004	0.263	0.670	-1.562 (4)	-9.818	
66.14p	4.96a	2.650	0.003	0.003	0.000	0.685	-2.605 (4)	-10.833	RaZero
70.00p	5.39a	2.454	-0.171	0.003	-0.174	0.685	-3.244 (4)	-11.412	
80.00p	6.41a	1.892	-0.637	0.001	-0.639	0.685	-4.810 (4)	-12.677	
90.00p	7.25a	1.265	-1.095	0.000	-1.095	0.685	-6.214 (4)	-13.552	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 619.1 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling moment = 24.20Cos(heel) + 0.00Sin(heel)

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Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	1.63	3.37	Yes
(2) Abs. Ratio from 0.00 deg to Flood or RAzero	>1.400	100.579	99.179	Yes
(3) Area from Equilibrium to 20.00 deg or Flood	>0.0300 m-R	0.163	0.133	Yes
(4) Absolute Angle at Deck Immersion	>0.00 deg	8.24	8.24	Yes

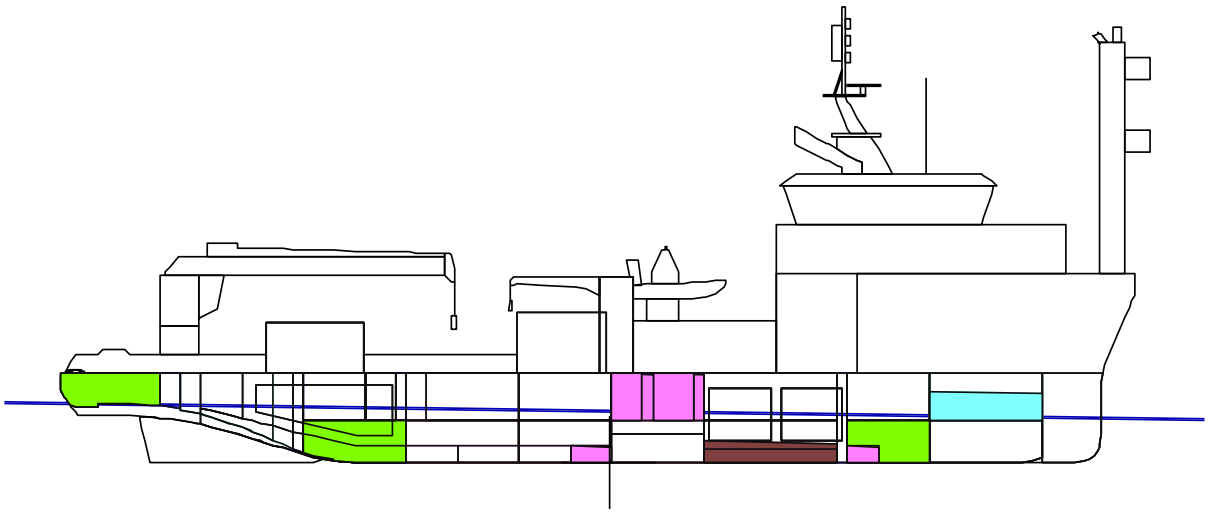
12.5 CONDITION NO.5: BALLAST ARRIVAL (WITHOUT CARGO)

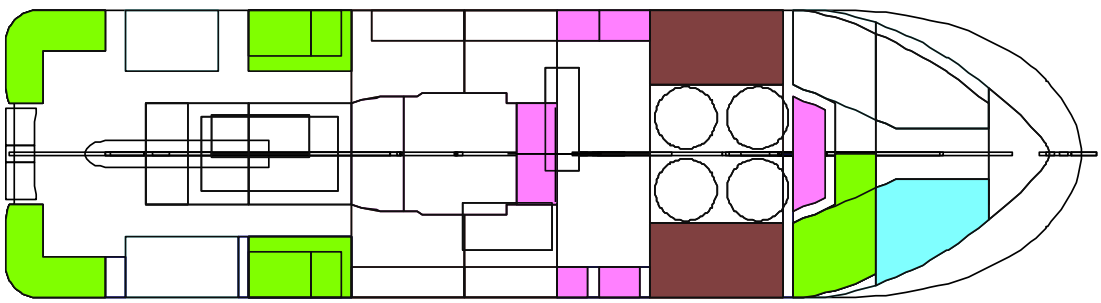
Floating Status

Draft FP	2.691 m	Heel	0.14p deg	GM(Solid)	5.151 m
Draft MS	3.109 m	Equil	Yes	F/S Corr.	0.268 m
Draft AP	3.527 m	Wind	Off	GM(Fluid)	4.883 m
Trim	0.84a m	Wave	No	KMT	10.352 m
LCG	0.826a m	VCG-Solid	5.202 m	TPcm	9.26
Displacement	2304.0 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	392.02	1.969a	1.106s	2.325
Displacement	2,303.99	0.826a	0.014p	5.202





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	223.93	53.85	FRESH WATER	40.53	9.99
FUEL OIL	50.86	9.78	MISC	72.22	56.48

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	0.40	24.100f	0.000	6.800
Total Fixed:	**1,916.45**	**0.541a**	**0.242p**	**5.803**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	0.0
NO3_DWSWB.P	1.025	100.00	49.20	14.761a	6.538p	1.351	0.0
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
DWSWBAP.P	1.025	100.00	33.58	29.636a	6.202p	4.439	0.0
DWSWBAP.S	1.025	100.00	33.58	29.636a	6.202s	4.439	0.0
Subtotals:		**53.85%**	**223.93**	**11.087a**	**1.247s**	**2.247**	**0.0**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.S	1.000	54.90	40.53	21.438f	3.677s	3.319	63.6
Subtotals:		**9.99%**	**40.53**	**21.438f**	**3.677s**	**3.319**	**127.2**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.P	0.850	49.00	21.56	9.273f	6.061p	0.422	44.2
NO1_FO.U.S	0.850	66.60	29.30	9.342f	6.207s	0.569	3.1
Subtotals:		**9.78%**	**50.86**	**9.313f**	**1.007s**	**0.507**	**444.3**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.996f	7.500p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.697f	7.500s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.297f	7.500p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	0.999f	7.500s	3.872	0.8
DIRTYOIL.P	0.900	98.00	6.22	1.107a	1.502p	0.490	4.8
BILGEHOLD.S	1.000	98.00	6.91	1.107a	1.498s	0.490	5.3
SEWAGE	1.000	98.00	10.11	14.760f	0.007p	0.490	30.3
Subtotals:		**56.48%**	**72.22**	**3.784f**	**0.638p**	**2.783**	**45.3**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**17.50%**	**387.54**	**2.237a**	**1.119s**	**2.231**	**616.8**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	2,280.68	0.650a	0.022p	1.684	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**2,303.99**	**0.876a**	**0.022p**	**1.677**	

Unprotected Flood Points

Name	L (m)	T (m)	V (m)	Height (m)
(1) POSITION 1(S)	9.700f	5.400s	12.150	9.194
(2) POSITION 1(P)	9.700f	5.400p	12.150	9.168
(3) POSITION 2(S)	10.300f	6.300s	12.150	9.205
(4) POSITION 2(P)	10.300f	6.300p	12.150	9.175

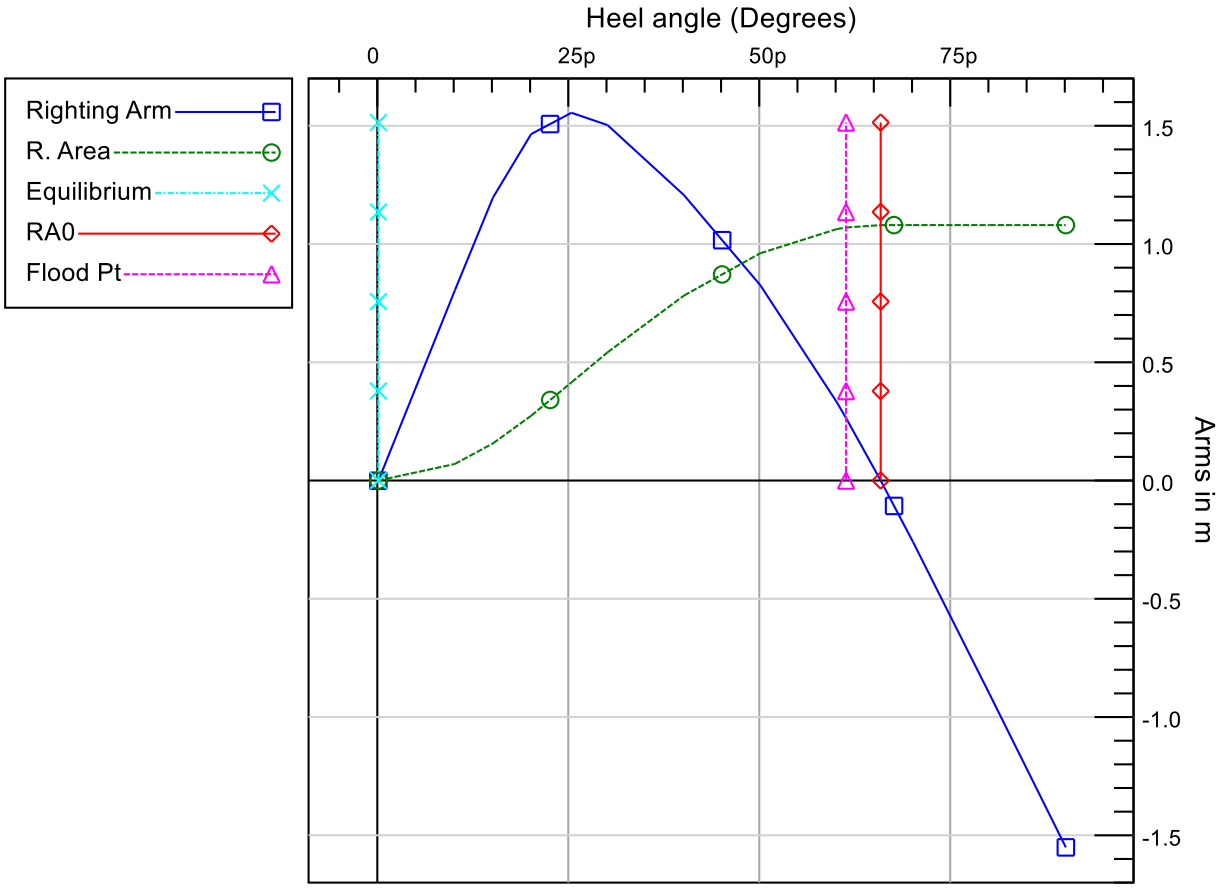
: OSV CHEMICAL CODE – CHAPTER 2 , 2.2

Free Surface Adjustment 0.268
Adjusted VCG 5.470

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Area (m-Rad)	Flood Pt Height (m)	Notes
0.14p	0.84a	3.108	-0.001	0.000	9.169 (2)	Equil
10.14p	0.58a	3.021	0.803	0.070	7.934 (4)	
15.14p	0.38a	2.925	1.196	0.157	7.226 (4)	
20.14p	0.27a	2.830	1.465	0.274	6.457 (4)	
25.41p	0.29a	2.718	**1.555**	0.415	5.605 (4)	MaxRa
30.14p	0.41a	2.576	1.503	0.542	4.842 (4)	
40.14p	0.94a	2.151	1.207	0.782	3.245 (4)	
50.14p	1.82a	1.577	0.826	0.961	1.700 (4)	
60.14p	2.89a	0.920	0.332	1.063	0.185 (4)	
61.36p	3.02a	0.836	0.264	1.070	0.000 (4)	FldPt
65.90p	3.52a	0.522	0.000	1.080	-0.677 (4)	RaZero
70.14p	4.00a	0.227	-0.260	1.080	-1.301 (4)	
80.14p	5.02a	-0.467	-0.902	1.080	-2.741 (4)	
90.14p	5.87a	-1.132	-1.551	1.080	-4.112 (4)	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 616.8 MT-m was used to adjust the VCG.

: OSV CHEMICAL CODE – CHAPTER 2 , 2.2



Limit	Min/Max	Actual	Margin	Pass
(1) Area from 0.00 deg to MaxRA at 25.41	>0.0596 m-R	0.415	0.355	Yes
(2) Area from 0.00 deg to 40.00 or Flood	>0.0900 m-R	0.782	0.692	Yes
(3) Area from 30.00 deg to 40.00 or Flood	>0.0300 m-R	0.240	0.210	Yes
(4) Righting Arm at 30.00 deg	>0.200 m	1.503	1.303	Yes
(5) Absolute Angle at MaxRA	>15.00 deg	25.41	10.41	Yes
(6) GM Upright	>0.150 m	4.912	4.762	Yes

Heeling Moment Derivation

Part	LPA (m^2)	HCP (m)	Arm (m)	Pressure (MT/m^2)	Moment (m-MT)
HULL.C	256.9	2.940	4.444	0.051	58.70
DECKHOUSE	108.4	9.191	10.695	0.051	59.60
FWDMAST	13.4	16.703	18.207	0.051	12.56
ANTENNA	0.3	16.554	18.058	0.051	0.25
CRANE1	23.5	15.402	16.906	0.051	20.38
DECK	45.1	3.203	4.707	0.051	10.92
CRANE2	37.2	7.514	9.018	0.051	17.23
LIFEBOAT	10.0	7.005	8.509	0.051	4.37
CRANE3	2.9	7.365	8.869	0.051	1.32

Total wind heeling moment 185.32 to port

IMO SEVERE WIND & ROLLING

Free Surface Adjustment 0.268
Adjusted VCG 5.470

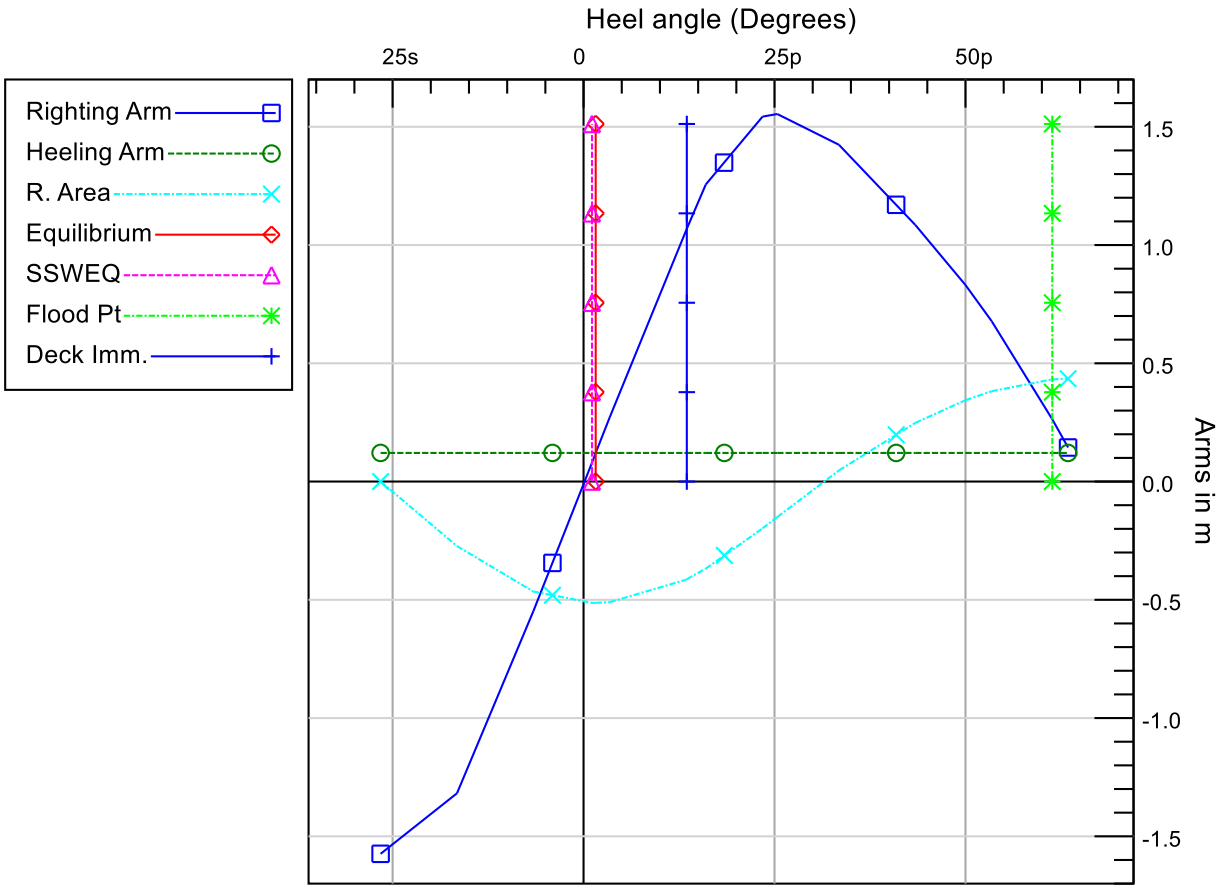
Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
26.57s	0.31a	2.687	-1.574	0.121	-1.694	0.000	5.418 (3)	-1.867	Roll
16.57s	0.33a	2.896	-1.318	0.121	-1.439	-0.273	7.012 (3)	-0.386	
6.57s	0.70a	3.069	-0.548	0.121	-0.669	-0.465	8.405 (3)	0.866	
1.11p	0.83a	3.107	0.080	0.121	-0.040	-0.513	9.067 (4)	1.583	SSWEq
1.59p	0.82a	3.106	0.121	0.121	0.000	-0.513	9.012 (4)	1.519	Equil
3.43p	0.79a	3.097	0.271	0.121	0.151	-0.510	8.795 (4)	1.274	
13.43p	0.44a	2.961	1.062	0.121	0.942	-0.415	7.473 (4)	0.011	
13.52p	0.44a	2.960	1.069	0.121	0.949	-0.413	7.460 (4)	0.000	Deck Imm.
16.00p	0.35a	2.907	1.256	0.121	1.136	-0.368	7.098 (4)	-0.310	
23.43p	0.27a	2.765	1.543	0.121	1.422	-0.198	5.927 (4)	-1.382	
25.36p	0.29a	2.719	1.554	0.121	**1.433**	-0.150	5.614 (4)	-1.679	MaxRa
33.43p	0.55a	2.454	1.424	0.121	1.303	0.047	4.314 (4)	-2.944	
43.43p	1.19a	1.978	1.085	0.121	0.965	0.248	2.728 (4)	-4.527	
50.00p	1.81a	1.586	0.831	0.121	0.710	0.344	1.721 (4)	-5.560	
53.43p	2.16a	1.366	0.678	0.121	0.557	0.382	1.200 (4)	-6.088	
61.36p	3.02a	0.836	0.263	0.121	0.142	0.432	0.000 (4)	-7.262	FldPt
63.43p	3.25a	0.694	0.144	0.121	0.024	0.435	-0.309 (4)	-7.554	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 617.0 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the wind heeling arms derived from this moment (in m-MT):
Port heeling moment = 277.98

Roll angle is 27.67
Equilibrium for load condition without gust is 1.11p

IMO SEVERE WIND & ROLLING



Limit	Min/Max	Actual	Margin	Pass
(1) Res. Ratio from Roll to Abs 50.00 deg or Flood	>1.000	1.671	0.671	Yes
(2) Angle from SSWEQ to 80% of Dk. Imm. or Abs 16.00 deg	>0.00 deg	9.71	9.71	Yes

Heeling Moment Report

FIFI HEELING MOMENT CALCULATION			
Reaction Force of the Fire Monitor	=	1.530	MT
Number of Fire Monitors	=	2	
Draft of the loading condition	=	3.109	m
D/2	=	1.5545	m
Height of firefighting monitor above baseline	=	19.600	m
Moment due to Fire Fighting operation	=	55.219	MT-m
Reaction Force of the Bow Thruster	=	18	MT
D/2	=	1.5545	m
Height of Bow Thruster above baseline	=	1.200	m
Moment due to Bow Thruster operation	=	6.381	MT-m
Total Heeling Moment	=	61.600	Tm

Heeling moment: 61.60 MT-m to port

FIRE FIGHTING CRITERIA

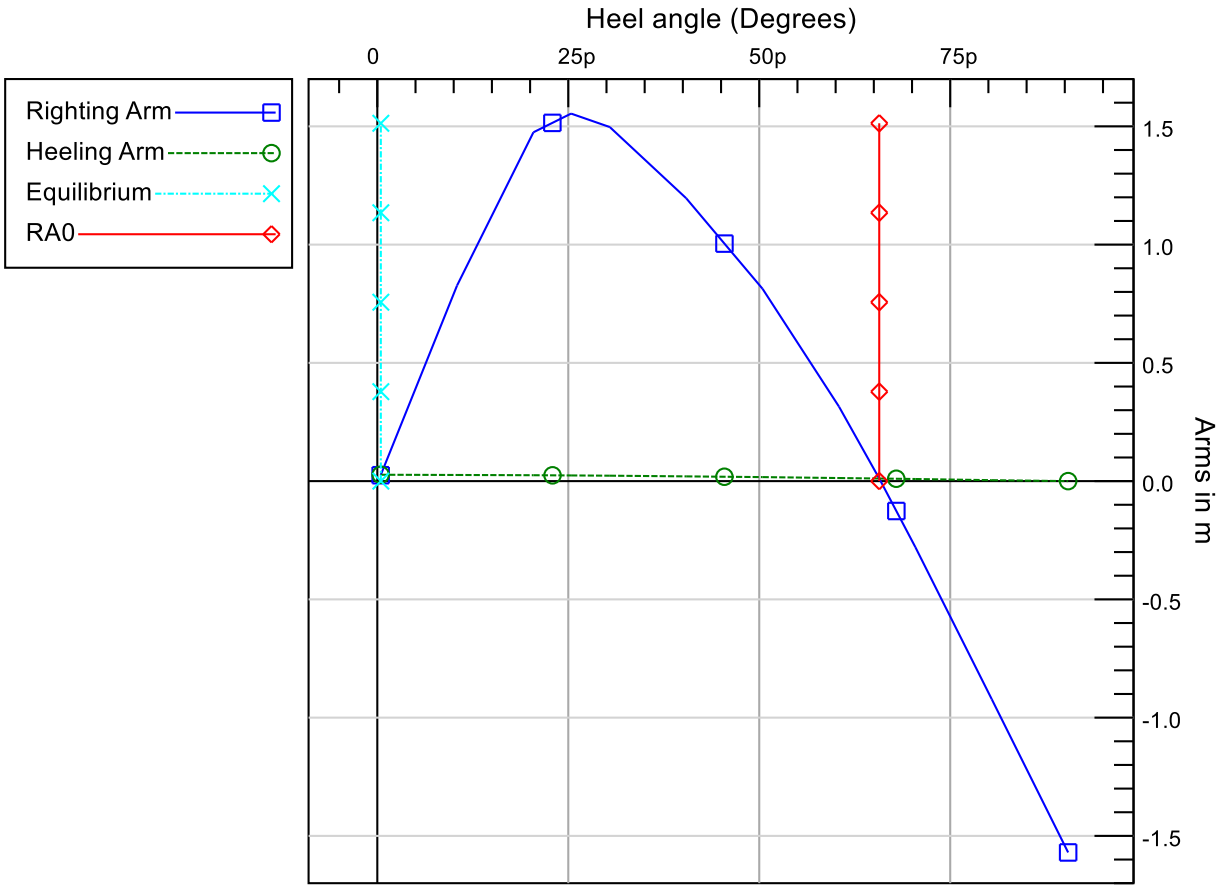
Free Surface Adjustment 0.268
Adjusted VCG 5.470

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Notes
0.44p	0.84a	3.108	0.025	0.027	-0.002	
0.46p	0.84a	3.108	0.027	0.027	0.000	Equil
10.44p	0.56a	3.016	0.827	0.026	0.800	
20.44p	0.27a	2.824	1.475	0.025	1.450	
25.37p	0.29a	2.719	1.554	0.024	**1.530**	MaxRa
30.44p	0.42a	2.566	1.497	0.023	1.474	
40.44p	0.96a	2.136	1.196	0.020	1.176	
50.44p	1.85a	1.558	0.813	0.017	0.796	
60.44p	2.92a	0.899	0.315	0.013	0.302	
65.71p	3.51a	0.536	0.011	0.011	0.000	RaZero
70.44p	4.03a	0.206	-0.279	0.009	-0.288	
80.44p	5.05a	-0.487	-0.921	0.004	-0.926	
90.44p	5.89a	-1.151	-1.570	0.000	-1.570	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 616.9 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling mmt = 61.60Cos(heel) + 0.00Sin(heel) <= 90 deg
Port heeling mmt = 0.00 > 90 deg

FIRE FIGHTING CRITERIA



Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	0.46	4.54	Yes

12.5.1 MAX TRANSVERSE CRANE OPERATION

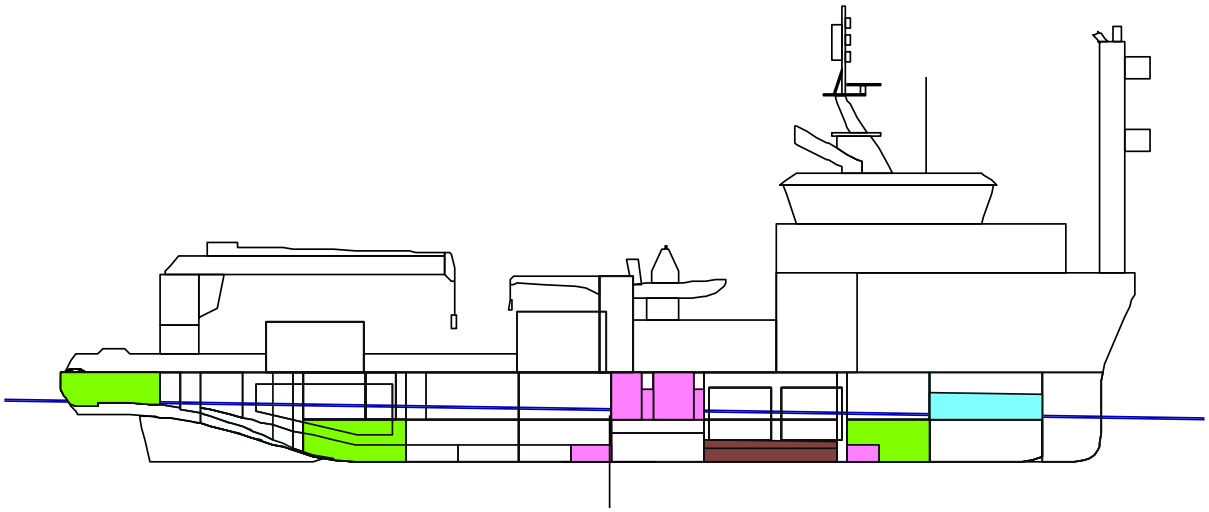
Max. Transverse moment – Lifting 3.500t @ 35.00 m radius (35.530 m boom length –Side Lifting to port side)

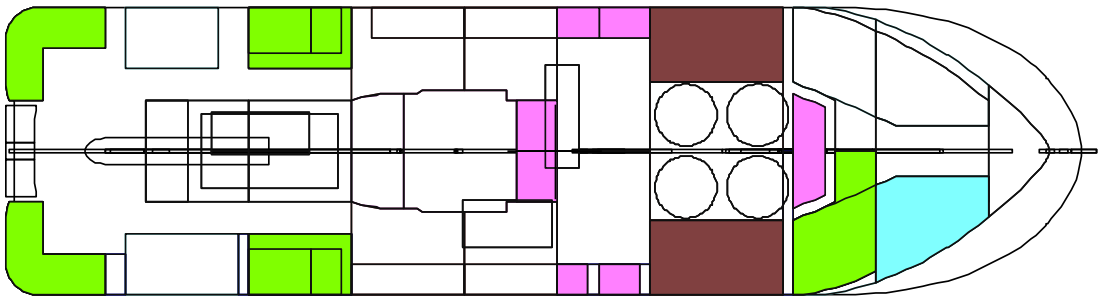
Floating Status

Draft FP	2.666 m	Heel	3.62p deg	GM(Solid)	4.989 m
Draft MS	3.101 m	Equil	Yes	F/S Corr.	0.268 m
Draft AP	3.537 m	Wind	Off	GM(Fluid)	4.722 m
Trim	0.87a m	Wave	No	KMT	10.186 m
LCG	0.971a m	VCG-Solid	5.207 m	TPcm	9.13
Displacement	2307.5 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	395.52	2.801a	0.703p	2.378
Displacement	2,307.49	0.971a	0.322p	5.207





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	223.93	53.85	FRESH WATER	40.53	9.99
FUEL OIL	50.86	9.78	MISC	72.22	56.48

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	0.40	24.100f	0.000	6.800
CRANE BOOM FINAL	32.55	25.100a	23.235p	11.190
CRANE BOOM INITIAL	-32.55	17.438a	6.000p	11.504
LIFTED WEIGHT FINAL	3.50	25.100a	41.000p	11.190
Total Fixed:	**1,919.95**	**0.716a**	**0.609p**	**5.807**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	0.0
NO3_DWSWB.P	1.025	100.00	49.20	14.761a	6.538p	1.351	0.0
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
DWSWBAP.P	1.025	100.00	33.58	29.636a	6.202p	4.439	0.0
DWSWBAP.S	1.025	100.00	33.58	29.636a	6.202s	4.439	0.0
Subtotals:		**53.85%**	**223.93**	**11.087a**	**1.247s**	**2.247**	**0.0**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.S	1.000	54.90	40.53	21.478f	3.597s	3.322	63.6
Subtotals:		**9.99%**	**40.53**	**21.478f**	**3.597s**	**3.322**	**127.2**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.P	0.850	49.00	21.56	9.265f	6.187p	0.426	45.1
NO1_FO.U.S	0.850	66.60	29.30	9.342f	6.200s	0.569	3.1
Subtotals:		**9.78%**	**50.86**	**9.309f**	**0.950s**	**0.509**	**445.2**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.996f	7.506p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.697f	7.494s	3.872	1.3
HYDOIL.P	0.924	98.00	10.73	1.297f	7.506p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	0.999f	7.494s	3.872	0.8
DIRTYOIL.P	0.900	98.00	6.22	1.103a	1.521p	0.490	4.8
BILGEHOLD.S	1.000	98.00	6.91	1.103a	1.479s	0.490	5.3
SEWAGE	1.000	98.00	10.11	14.765f	0.047p	0.491	30.3
Subtotals:		**56.48%**	**72.22**	**3.785f**	**0.651p**	**2.784**	**45.3**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**17.50%**	**387.54**	**2.233a**	**1.100s**	**2.231**	**617.7**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	2,284.19	0.800a	0.550p	1.706	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**2,307.49**	**1.024a**	**0.544p**	**1.699**	

Heeling Moment Report

Draft	3.109	m					Resultant	1.0000
WIND	20	Knots					(Sin 90Deg)	
Part	**Wind area**	**Effective area**	**Centre of Area from sea-level**	**Ch**	**Lever**	**Cs**	**Wind Force**	**Moment**
	(m2)	**(m2)**	**m**		**(m)**		**(N)**	**(m-MT)**
HULL.C	256.900	256.900	2.344	1.000	4.444	1.000	16587.905	7.514
DECKHOUSE	108.400	108.400	8.595	1.000	10.695	1.000	6999.334	7.631
FWDMAST	13.400	13.400	16.107	1.100	18.207	1.000	951.754	1.766
ANTENNA	0.300	0.300	15.958	1.100	18.058	1.000	21.308	0.039
CRANE1	23.500	23.500	14.806	1.000	16.906	1.500	2276.075	3.922
DECK	45.100	45.100	2.607	1.000	4.707	1.000	2912.085	1.397
CRANE2	37.200	37.200	6.918	1.000	9.018	1.500	3602.978	3.312
LIFEBOAT	10.000	10.000	6.409	1.000	8.509	1.000	645.695	0.560
CRANE3	2.900	2.900	6.769	1.000	8.869	1.500	280.877	0.254
TOTAL WIND HMMT								**26.397**

Heeling moment: 26.39 MT-m to port

IS CODE CRANE OPERATION CRITERIA

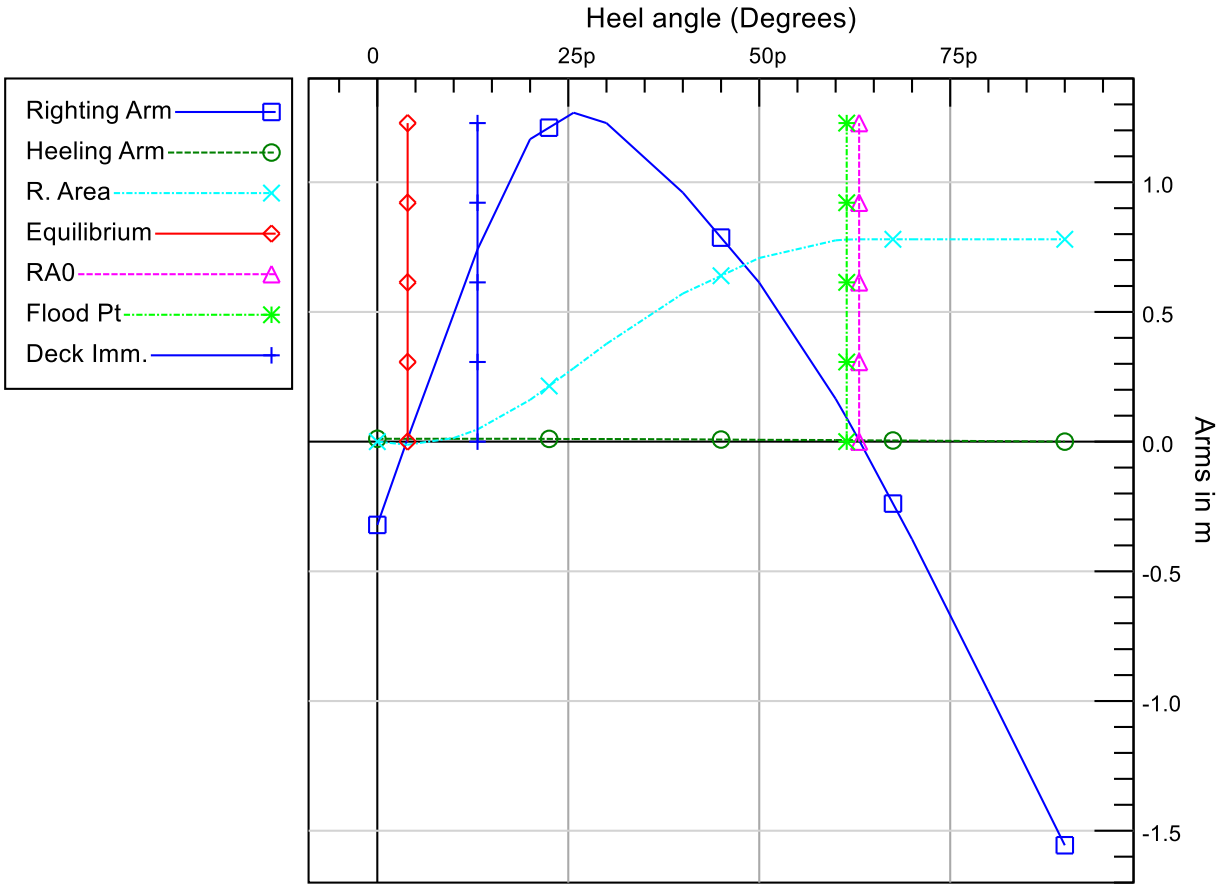
Free Surface Adjustment 0.268
Adjusted VCG 5.475

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
0.00	0.92a	3.107	-0.321	0.011	-0.332	0.000	9.197 (1,2)	1.675	
3.97p	0.86a	3.092	0.011	0.011	0.000	-0.012	8.746 (4)	1.156	Equil
10.00p	0.67a	3.023	0.492	0.011	0.481	0.014	7.969 (4)	0.385	
13.12p	0.55a	2.968	0.741	0.011	0.730	0.047	7.534 (4)	0.000	Deck Imm.
20.00p	0.38a	2.835	1.166	0.011	1.155	0.161	6.496 (4)	-0.928	
25.70p	0.42a	2.716	1.268	0.010	**1.258**	0.283	5.576 (4)	-1.805	MaxRa
30.00p	0.55a	2.589	1.228	0.010	1.218	0.377	4.881 (4)	-2.485	
40.00p	1.09a	2.172	0.960	0.009	0.951	0.571	3.282 (4)	-4.085	
50.00p	1.98a	1.605	0.613	0.007	0.606	0.708	1.733 (4)	-5.674	
60.00p	3.06a	0.951	0.165	0.006	0.159	0.776	0.218 (4)	-7.193	
61.45p	3.22a	0.852	0.091	0.005	0.085	0.779	0.000 (4)	-7.402	FldPt
63.08p	3.41a	0.741	0.005	0.005	0.000	0.780	-0.244 (4)	-7.633	RaZero
70.00p	4.18a	0.259	-0.376	0.004	-0.380	0.780	-1.268 (4)	-8.561	
80.00p	5.21a	-0.434	-0.963	0.002	-0.965	0.780	-2.707 (4)	-9.695	
90.00p	6.05a	-1.100	-1.556	0.000	-1.556	0.780	-4.080 (4)	-10.530	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 617.7 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling moment = 26.40Cos(heel) + 0.00Sin(heel)

IS CODE CRANE OPERATION CRITERIA



Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	3.97	1.03	Yes
(2) Abs. Ratio from 0.00 deg to Flood or RAzero	>1.400	78.535	77.135	Yes
(3) Area from Equilibrium to 20.00 deg or Flood	>0.0300 m-R	0.173	0.143	Yes
(4) Absolute Angle at Deck Immersion	>0.00 deg	13.12	13.12	Yes

12.5.2 MAX VERTICAL CRANE OPERATION

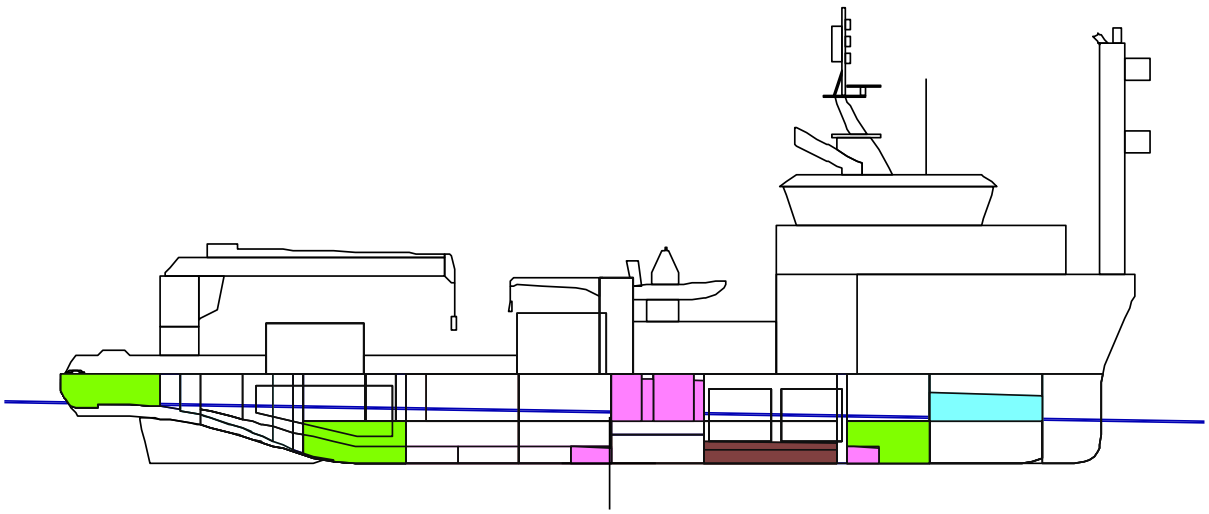
Max. Vertical moment – Lifting 3.500t @ 9.600 m radius (35.530 m boom length - Side Lifting to port side)

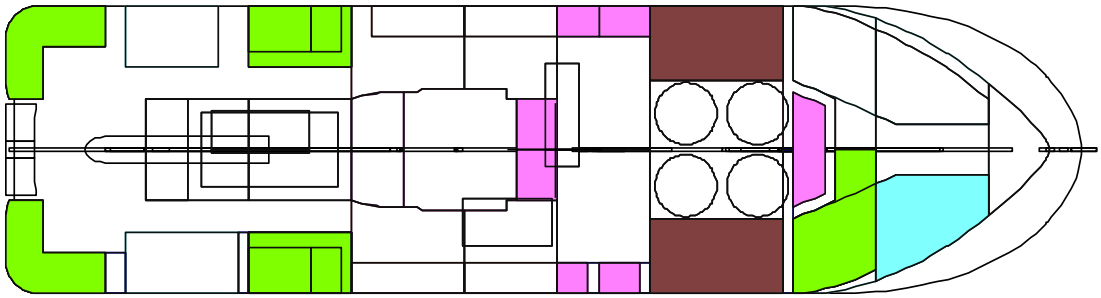
Floating Status

Draft FP	2.648 m	Heel	1.18p deg	GM(Solid)	4.798 m
Draft MS	3.107 m	Equil	Yes	F/S Corr.	0.267 m
Draft AP	3.565 m	Wind	Off	GM(Fluid)	4.530 m
Trim	0.92a m	Wave	No	KMT	10.294 m
LCG	0.971a m	VCG-Solid	5.499 m	TPcm	9.20
Displacement	2307.5 MT	WaterSpgr	1.025		

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1,911.97	0.592a	0.243p	5.792
Deadweight	395.52	2.804a	0.579s	4.080
Displacement	2,307.49	0.971a	0.102p	5.499





Tank Group	Weight (MT)	Load%	Tank Group	Weight (MT)	Load%
WATER BALLAST	223.93	53.85	FRESH WATER	40.53	9.99
FUEL OIL	50.86	9.78	MISC	72.22	56.48

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1,911.97	0.592a	0.243p	5.792
01_CREW & EFFECT	4.08	20.924f	0.000	10.853
02_PROVISION STORE	0.40	24.100f	0.000	6.800
CRANE BOOM FINAL	32.55	25.100a	10.535p	28.218
CRANE BOOM INITIAL	-32.55	17.438a	6.000p	11.504
LIFTED WEIGHT FINAL	3.50	25.100a	15.600p	45.245
Total Fixed:	**1,919.95**	**0.716a**	**0.347p**	**6.158**

Tank Status

WATER BALLAST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO2_DWSWB.S	1.025	100.00	58.39	16.438f	4.784s	1.234	0.0
NO3_DWSWB.P	1.025	100.00	49.20	14.761a	6.538p	1.351	0.0
NO3_DWSWB.S	1.025	100.00	49.20	14.761a	6.538s	1.351	0.0
DWSWBAP.P	1.025	100.00	33.58	29.636a	6.202p	4.439	0.0
DWSWBAP.S	1.025	100.00	33.58	29.636a	6.202s	4.439	0.0
Subtotals:		**53.85%**	**223.93**	**11.087a**	**1.247s**	**2.247**	**0.0**

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FW.S	1.000	54.90	40.53	21.447f	3.654s	3.320	63.6
Subtotals:		**9.99%**	**40.53**	**21.447f**	**3.654s**	**3.320**	**127.2**

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
NO1_FO.U.P	0.850	49.00	21.56	9.263f	6.099p	0.423	44.5
NO1_FO.U.S	0.850	66.60	29.30	9.340f	6.205s	0.569	3.1
Subtotals:		**9.78%**	**50.86**	**9.307f**	**0.990s**	**0.507**	**444.5**

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
DISPERSANT.P	1.163	98.00	16.89	3.996f	7.502p	3.872	1.7
FOAM.S	1.145	98.00	13.30	3.697f	7.498s	3.872	1.3

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
HYDOIL.P	0.924	98.00	10.73	1.297f	7.502p	3.872	1.1
LUBOIL.S	0.924	98.00	8.05	0.998f	7.498s	3.872	0.8
DIRTYOIL.P	0.900	98.00	6.22	1.106a	1.511p	0.490	4.8
BILGEHOLD.S	1.000	98.00	6.91	1.106a	1.489s	0.490	5.3
SEWAGE	1.000	98.00	10.11	14.762f	0.036p	0.490	30.3
Subtotals:		**56.48%**	**72.22**	**3.784f**	**0.645p**	**2.784**	**45.3**

Totals

		Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	FSMT (MT-m)
Totals:		**17.50%**	**387.54**	**2.237a**	**1.113s**	**2.231**	**617.0**

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Perm
HULL.C	Intact	1.025	2,284.19	0.808a	0.183p	1.690	1.000
SKEG.C	Intact	1.025	23.30	23.061a	0.000	1.014	1.000
SubTotals:			**2,307.49**	**1.033a**	**0.181p**	**1.684**	

Heeling Moment Report

Draft	3.109	m					Resultant	1.0000
WIND	20	Knots					(Sin 90Deg)	
Part	**Wind area**	**Effective area**	**Centre of Area from sea-level**	**Ch**	**Lever**	**Cs**	**Wind Force**	**Moment**
	(m2)	**(m2)**	**m**		**(m)**		**(N)**	**(m-MT)**
HULL.C	256.900	256.900	2.344	1.000	4.444	1.000	16587.905	7.514
DECKHOUSE	108.400	108.400	8.595	1.000	10.695	1.000	6999.334	7.631
FWDMAST	13.400	13.400	16.107	1.100	18.207	1.000	951.754	1.766
ANTENNA	0.300	0.300	15.958	1.100	18.058	1.000	21.308	0.039
CRANE1	23.500	23.500	14.806	1.000	16.906	1.500	2276.075	3.922
DECK	45.100	45.100	2.607	1.000	4.707	1.000	2912.085	1.397
CRANE2	37.200	37.200	6.918	1.000	9.018	1.500	3602.978	3.312
LIFEBOAT	10.000	10.000	6.409	1.000	8.509	1.000	645.695	0.560
CRANE3	2.900	2.900	6.769	1.000	8.869	1.500	280.877	0.254
TOTAL WIND HMMT								**26.397**

Heeling moment: 26.39 MT-m to port

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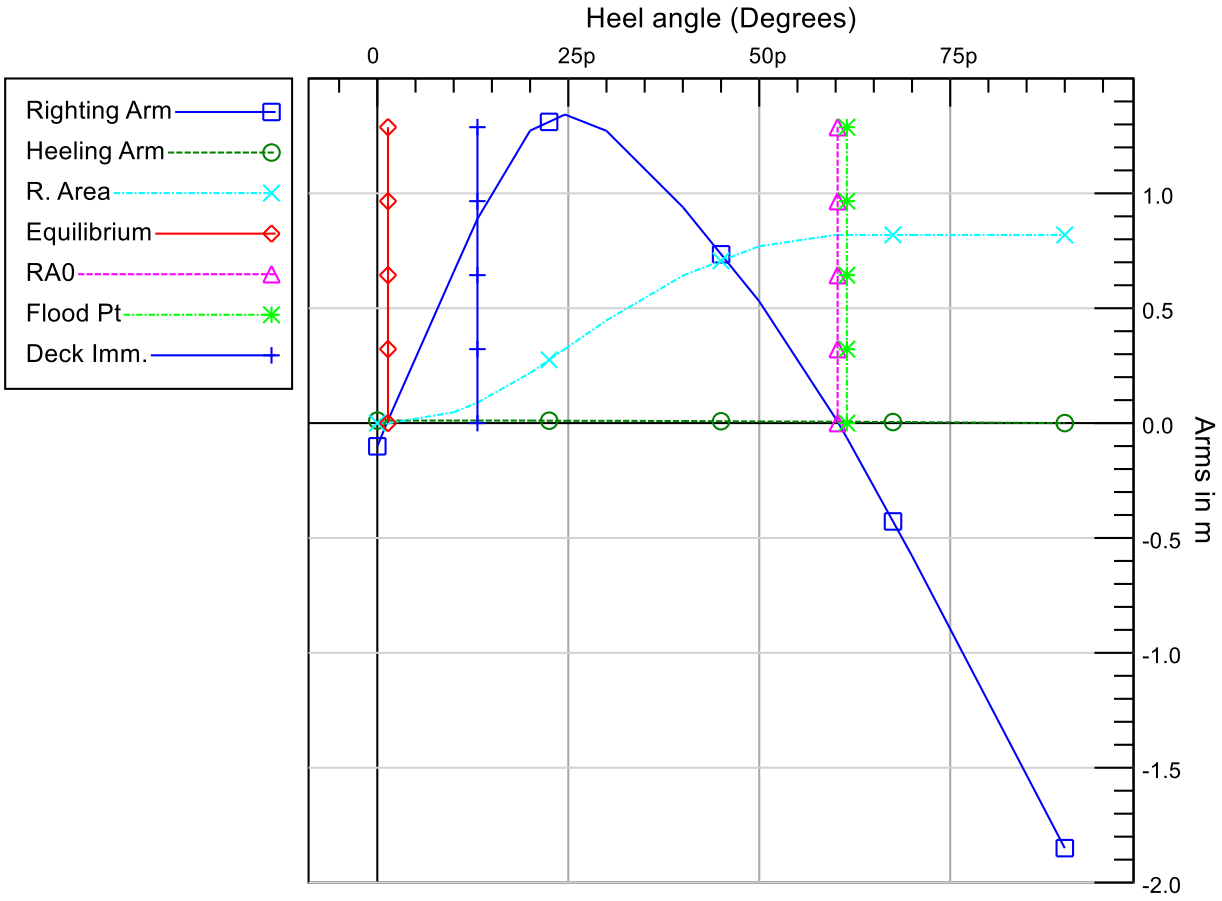
Free Surface Adjustment 0.267
Adjusted VCG 5.766

Heel Angle (deg)	Trim Angle (deg)	Origin Depth (m)	Righting Arm (m)	Heeling Arm (m)	Residual Arm (m)	Residual Area (m-Rad)	Flood Pt Height (m)	Freeboard (m)	Notes
0.00	0.92a	3.107	-0.101	0.011	-0.112	0.000	9.198 (1,2)	1.674	
1.40p	0.92a	3.105	0.011	0.011	0.000	-0.001	9.051 (4)	1.497	Equil
10.00p	0.68a	3.023	0.658	0.011	0.647	0.047	7.969 (4)	0.384	
13.11p	0.55a	2.968	0.889	0.011	0.877	0.089	7.536 (4)	0.000	Deck Imm.
20.00p	0.39a	2.835	1.273	0.011	1.262	0.218	6.497 (4)	-0.929	
24.57p	0.41a	2.744	1.343	0.010	**1.333**	0.323	5.759 (4)	-1.629	MaxRa
30.00p	0.55a	2.589	1.272	0.010	1.262	0.447	4.882 (4)	-2.487	
40.00p	1.10a	2.173	0.940	0.009	0.932	0.642	3.283 (4)	-4.089	
50.00p	2.00a	1.606	0.530	0.007	0.523	0.770	1.735 (4)	-5.683	
60.00p	3.09a	0.953	0.020	0.006	0.015	0.819	0.220 (4)	-7.209	
60.26p	3.12a	0.935	0.006	0.006	0.000	0.819	0.181 (4)	-7.246	RaZero
61.47p	3.26a	0.853	-0.064	0.005	-0.069	0.819	0.000 (4)	-7.422	FldPt
70.00p	4.22a	0.262	-0.578	0.004	-0.581	0.819	-1.264 (4)	-8.583	
80.00p	5.25a	-0.432	-1.214	0.002	-1.216	0.819	-2.702 (4)	-9.721	
90.00p	6.09a	-1.097	-1.850	0.000	-1.850	0.819	-4.075 (4)	-10.553	

Weight and C.G. used above include tank loads.
The tank load centers were not allowed to shift with heel and trim changes.
A Free Surface Moment of 617.1 MT-m was used to adjust the VCG.

Note:
Residual Righting Arms shown above are in excess of the
overturning arms derived from these moments (in m-MT):
Port heeling moment = 26.40Cos(heel) + 0.00Sin(heel)

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Limit	Min/Max	Actual	Margin	Pass
(1) Absolute Angle at Equilibrium	<5.00 deg	1.40	3.60	Yes
(2) Abs. Ratio from 0.00 deg to Flood or RAzero	>1.400	83.444	82.044	Yes
(3) Area from Equilibrium to 20.00 deg or Flood	>0.0300 m-R	0.220	0.190	Yes
(4) Absolute Angle at Deck Immersion	>0.00 deg	13.11	13.11	Yes

13 SUMMARY OF LOADING CONDITIONS

Loading Conditions		COND.1	COND.2	COND.3	COND.4	COND.5
Lightship weight (MT)		1,911.97	1,911.97	1,911.97	1,911.97	1,911.97
Deadweight (MT)		-	1,393.70	977.75	1,198.40	392.02
Displacement (MT)		1,911.97	3,305.67	2,889.73	3,110.38	2,303.99
GM (m)		5.031	3.632	3.897	4.103	4.883
Draft (m)	Forward	2.220	4.061	3.485	4.211	2.691
	Amidships	2.663	4.200	3.750	4.038	3.109
	Aft	3.106	4.339	4.015	3.864	3.527
Trim (m)		0.880a	0.280a	0.53a	0.35f	0.84a

- CONDITION NO.1: LIGHTSHIP (NOT A SAILING CONDITION)
- CONDITION NO.2: FULLY LOADED DEPARTURE
- CONDITION NO.3: FULLY LOADED ARRIVAL
- CONDITION NO.4: BALLAST DEPARTURE (WITHOUT CARGO)
- CONDITION NO.5: BALLAST ARRIVAL (WITHOUT CARGO)

14 VESSEL DATA

14.1 HYDROSTATICS

Hydrostatic Properties
Draft is from Baseline.
No Trim, No heel, VCG = 0.000

Draft at Origin (m)	Displ (MT)	LCB (m)	VCB (m)	LCF (m)	TPcm (MT/cm)	MTcm (MT-m /cm)	KML (m)	KMT (m)
2.250	1564.894	0.927f	1.176	0.310f	7.86	25.33	92.924	11.733
2.350	1643.812	0.894f	1.230	0.167f	7.93	25.99	90.738	11.374
2.450	1723.523	0.856f	1.284	0.038f	7.99	26.60	88.581	11.061
2.550	1803.908	0.818f	1.339	0.039a	8.08	27.06	86.118	10.867
2.650	1885.019	0.779f	1.393	0.153a	8.14	27.65	84.193	10.562
2.750	1966.820	0.736f	1.447	0.375a	8.23	28.62	83.530	10.314
2.850	2050.140	0.679f	1.502	0.872a	8.41	30.72	86.019	10.135
2.950	2134.755	0.612f	1.558	1.198a	8.53	32.15	86.450	9.954
3.050	2220.581	0.537f	1.613	1.465a	8.64	33.39	86.300	9.758
3.150	2307.610	0.455f	1.670	1.788a	8.76	34.84	86.661	9.619
3.250	2395.888	0.366f	1.726	2.119a	8.89	36.34	87.062	9.504
3.350	2485.489	0.271f	1.783	2.451a	9.03	37.85	87.421	9.416
3.450	2576.606	0.167f	1.840	2.915a	9.21	39.95	88.993	9.408
3.550	2669.641	0.050f	1.898	3.532a	9.44	42.77	91.969	9.542
3.650	2763.349	0.063a	1.956	3.351a	9.39	42.16	87.576	9.429
3.750	2857.523	0.173a	2.013	3.458a	9.44	42.85	86.066	9.305
3.850	2952.362	0.282a	2.070	3.625a	9.52	43.82	85.198	9.178
3.950	3047.738	0.388a	2.128	3.705a	9.56	44.40	83.619	9.050
4.050	3143.497	0.490a	2.185	3.770a	9.59	44.91	82.008	8.930
4.150	3239.563	0.588a	2.242	3.821a	9.62	45.36	80.379	8.814
4.250	3335.888	0.682a	2.298	3.842a	9.64	45.68	78.602	8.700
4.350	3432.390	0.771a	2.354	3.855a	9.66	45.96	76.856	8.591
4.450	3529.015	0.855a	2.410	3.850a	9.67	46.16	75.081	8.484
4.550	3625.726	0.935a	2.466	3.838a	9.67	46.33	73.345	8.383
4.650	3722.503	1.010a	2.522	3.821a	9.68	46.48	71.672	8.289
4.750	3819.332	1.081a	2.577	3.801a	9.69	46.62	70.060	8.203

Hydrostatic Properties
Draft is from Baseline.
Trim: aft 0.500/57.400, No heel, VCG = 0.000

Draft at Origin (m)	Displ (MT)	LCB (m)	VCB (m)	LCF (m)	TPcm (MT/cm)	MTcm (MT-m/cm)	KML (m)	KMT (m)
2.250	1564.164	0.113f	1.179	0.078a	7.95	26.20	96.125	11.877
2.350	1643.948	0.102f	1.234	0.160a	8.00	26.69	93.177	11.468
2.450	1724.276	0.087f	1.288	0.282a	8.06	27.32	90.944	11.105
2.550	1805.711	0.063f	1.343	0.727a	8.27	29.25	92.975	10.951
2.650	1889.152	0.020f	1.399	1.069a	8.40	30.73	93.356	10.720
2.750	1973.754	0.033a	1.455	1.393a	8.53	32.15	93.501	10.498
2.850	2059.596	0.096a	1.512	1.675a	8.64	33.46	93.261	10.264
2.950	2146.709	0.167a	1.569	2.003a	8.77	34.95	93.437	10.094
3.050	2235.134	0.246a	1.626	2.332a	8.91	36.44	93.578	9.956
3.150	2324.978	0.334a	1.684	2.752a	9.07	38.34	94.646	9.884
3.250	2416.623	0.435a	1.743	3.235a	9.26	40.55	96.301	9.879
3.350	2509.836	0.544a	1.802	3.405a	9.34	41.42	94.734	9.953
3.450	2603.500	0.648a	1.860	3.491a	9.39	42.03	92.655	9.798
3.550	2697.790	0.751a	1.919	3.673a	9.47	43.07	91.637	9.642
3.650	2792.714	0.852a	1.977	3.764a	9.51	43.70	89.825	9.483
3.750	2888.050	0.950a	2.035	3.836a	9.55	44.25	87.953	9.334
3.850	2983.730	1.043a	2.092	3.889a	9.58	44.72	86.025	9.190
3.950	3079.698	1.132a	2.150	3.917a	9.61	45.07	84.003	9.050
4.050	3175.863	1.217a	2.207	3.933a	9.63	45.37	82.000	8.917
4.150	3272.172	1.297a	2.264	3.930a	9.64	45.58	79.960	8.787
4.250	3368.580	1.372a	2.320	3.919a	9.65	45.76	77.971	8.663
4.350	3465.059	1.443a	2.376	3.904a	9.65	45.91	76.053	8.548
4.450	3561.597	1.509a	2.432	3.885a	9.66	46.05	74.213	8.441
4.550	3658.177	1.572a	2.487	3.862a	9.66	46.17	72.443	8.343
4.650	3754.788	1.630a	2.543	3.835a	9.66	46.27	70.736	8.252
4.750	3851.412	1.685a	2.598	3.803a	9.66	46.35	69.080	8.168

Hydrostatic Properties
Draft is from Baseline.
Trim: aft 1.000/57.400, No heel, VCG = 0.000

Draft at Origin (m)	Displ (MT)	LCB (m)	VCB (m)	LCF (m)	TPcm (MT/cm)	MTcm (MT-m/cm)	KML (m)	KMT (m)
2.250	1565.968	0.721a	1.191	0.636a	7.97	26.40	96.763	11.898
2.350	1647.166	0.725a	1.247	0.955a	8.21	29.20	101.747	11.563
2.450	1730.061	0.745a	1.303	1.296a	8.35	30.69	101.808	11.278
2.550	1814.335	0.777a	1.360	1.612a	8.53	32.16	101.721	11.150
2.650	1900.168	0.821a	1.417	1.907a	8.65	33.54	101.297	10.874
2.750	1987.344	0.876a	1.475	2.240a	8.78	35.06	101.255	10.664
2.850	2075.898	0.942a	1.533	2.599a	8.93	36.70	101.450	10.509
2.950	2166.120	1.021a	1.592	3.065a	9.12	38.80	102.806	10.438
3.050	2258.276	1.114a	1.652	3.456a	9.28	40.59	103.144	10.604
3.150	2351.363	1.209a	1.712	3.550a	9.33	41.23	100.631	10.417
3.250	2445.038	1.302a	1.771	3.725a	9.41	42.25	99.179	10.224
3.350	2539.438	1.394a	1.831	3.843a	9.47	43.03	97.242	10.027
3.450	2634.303	1.483a	1.890	3.909a	9.51	43.57	94.914	9.839
3.550	2729.551	1.569a	1.948	3.963a	9.54	44.05	92.615	9.660
3.650	2825.114	1.651a	2.006	4.006a	9.57	44.49	90.374	9.489
3.750	2920.911	1.728a	2.064	4.017a	9.59	44.78	87.978	9.324
3.850	3016.875	1.801a	2.122	4.016a	9.60	45.01	85.615	9.164
3.950	3112.954	1.869a	2.179	4.006a	9.61	45.19	83.320	9.013
4.050	3209.118	1.933a	2.235	3.990a	9.62	45.35	81.109	8.871
4.150	3305.349	1.993a	2.292	3.970a	9.63	45.50	78.996	8.740
4.250	3401.632	2.048a	2.348	3.947a	9.63	45.62	76.971	8.619
4.350	3497.953	2.100a	2.403	3.921a	9.63	45.73	75.027	8.507
4.450	3594.294	2.149a	2.459	3.890a	9.64	45.82	73.154	8.404
4.550	3690.636	2.194a	2.514	3.854a	9.64	45.88	71.343	8.308
4.650	3786.961	2.235a	2.569	3.812a	9.63	45.91	69.575	8.220
4.750	3883.210	2.274a	2.623	3.587a	9.57	45.09	66.634	8.098

Hydrostatic Properties
Draft is from Baseline.
Trim: fwd 0.500/57.400, No heel, VCG = 0.000

Draft at Origin (m)	Displ (MT)	LCB (m)	VCB (m)	LCF (m)	TPcm (MT/cm)	MTcm (MT-m/cm)	KML (m)	KMT (m)
2.250	1567.855	1.717f	1.182	0.647f	7.78	24.63	90.167	11.610
2.350	1646.172	1.660f	1.235	0.509f	7.85	25.24	87.991	11.285
2.450	1724.970	1.604f	1.289	0.408f	7.90	25.74	85.648	10.933
2.550	1804.539	1.548f	1.342	0.249f	8.02	26.43	84.078	10.796
2.650	1885.063	1.489f	1.396	0.141f	8.07	26.97	82.107	10.524
2.750	1965.996	1.432f	1.450	0.062f	8.11	27.41	80.030	10.243
2.850	2047.401	1.376f	1.503	0.046a	8.17	27.96	78.393	9.991
2.950	2129.428	1.318f	1.557	0.226a	8.24	28.77	77.539	9.787
3.050	2212.598	1.252f	1.611	0.600a	8.38	30.33	78.672	9.647
3.150	2297.165	1.176f	1.666	0.990a	8.52	32.04	80.061	9.481
3.250	2382.958	1.093f	1.721	1.278a	8.63	33.34	80.303	9.335
3.350	2469.887	1.004f	1.776	1.592a	8.75	34.74	80.739	9.221
3.450	2558.008	0.909f	1.831	1.917a	8.88	36.21	81.247	9.127
3.550	2647.395	0.808f	1.887	2.247a	9.01	37.72	81.774	9.051
3.650	2738.120	0.702f	1.943	2.590a	9.14	39.30	82.376	9.004
3.750	2830.422	0.587f	2.000	3.044a	9.32	41.39	83.938	9.020
3.850	2924.371	0.464f	2.057	3.221a	9.40	42.30	83.034	9.109
3.950	3018.574	0.348f	2.113	3.301a	9.44	42.86	81.502	9.011
4.050	3113.232	0.235f	2.170	3.448a	9.50	43.74	80.637	8.912
4.150	3208.564	0.124f	2.226	3.578a	9.56	44.54	79.683	8.807
4.250	3304.353	0.015f	2.283	3.652a	9.60	45.09	78.330	8.705
4.350	3400.497	0.089a	2.339	3.707a	9.63	45.57	76.915	8.608
4.450	3496.926	0.190a	2.395	3.755a	9.66	46.01	75.521	8.514
4.550	3593.586	0.286a	2.450	3.773a	9.68	46.32	73.980	8.421
4.650	3690.412	0.378a	2.506	3.776a	9.69	46.57	72.432	8.331
4.750	3787.349	0.465a	2.561	3.768a	9.70	46.77	70.883	8.243

Hydrostatic Properties
Draft is from Baseline.
Trim: fwd 1.000/57.400, No heel, VCG = 0.000

Draft at Origin (m)	Displ (MT)	LCB (m)	VCB (m)	LCF (m)	TPcm (MT/cm)	MTcm (MT-m/cm)	KML (m)	KMT (m)
2.250	1572.917	2.481f	1.195	0.886f	7.73	24.20	88.315	11.551
2.350	1650.481	2.404f	1.248	0.806f	7.78	24.63	85.656	11.167
2.450	1728.551	2.329f	1.300	0.730f	7.82	25.04	83.127	10.826
2.550	1807.554	2.253f	1.353	0.597f	7.93	25.62	81.339	10.705
2.650	1887.158	2.180f	1.406	0.494f	7.98	26.12	79.435	10.410
2.750	1967.287	2.109f	1.459	0.296f	8.07	26.93	78.552	10.210
2.850	2048.130	2.036f	1.512	0.230f	8.10	27.32	76.542	9.960
2.950	2129.353	1.965f	1.565	0.152f	8.14	27.74	74.778	9.726
3.050	2211.021	1.896f	1.619	0.047f	8.19	28.27	73.391	9.517
3.150	2293.270	1.827f	1.672	0.117a	8.26	29.01	72.593	9.347
3.250	2376.497	1.754f	1.725	0.400a	8.37	30.19	72.910	9.237
3.350	2460.847	1.674f	1.779	0.806a	8.51	31.97	74.562	9.094
3.450	2546.583	1.585f	1.833	1.100a	8.63	33.28	75.008	8.979
3.550	2633.408	1.491f	1.887	1.410a	8.74	34.66	75.539	8.887
3.650	2721.371	1.393f	1.942	1.727a	8.87	36.09	76.118	8.809
3.750	2810.548	1.290f	1.996	2.034a	8.99	37.51	76.604	8.738
3.850	2901.004	1.181f	2.051	2.379a	9.12	39.10	77.353	8.702
3.950	2992.842	1.067f	2.107	2.725a	9.26	40.73	78.096	8.686
4.050	3086.268	0.945f	2.163	3.207a	9.44	42.99	79.935	8.750
4.150	3180.740	0.821f	2.218	3.184a	9.45	43.06	77.691	8.760
4.250	3275.380	0.705f	2.274	3.251a	9.48	43.57	76.335	8.678
4.350	3370.527	0.591f	2.330	3.424a	9.55	44.57	75.899	8.598
4.450	3466.311	0.478f	2.385	3.529a	9.60	45.29	74.982	8.512
4.550	3562.502	0.369f	2.441	3.592a	9.64	45.81	73.802	8.431
4.650	3659.026	0.264f	2.496	3.639a	9.67	46.27	72.578	8.351
4.750	3755.829	0.163f	2.551	3.679a	9.70	46.72	71.385	8.275

14.2 CROSS CURVES

Cross Curves of Stability
Righting Arms(heel) for VCG = 0.00
Trim zero at heel = 0 (RA Trim = 0)

Displ (MT)	5.000s	10.000s	20.000s	30.000s	40.000s	50.000s	60.000s
1564.894	1.030s	2.071s	3.948s	4.935s	5.383s	5.572s	5.589s
1643.812	0.999s	2.011s	3.881s	4.866s	5.322s	5.534s	5.562s
1723.523	0.971s	1.957s	3.814s	4.797s	5.259s	5.492s	5.531s
1803.908	0.946s	1.908s	3.747s	4.727s	5.197s	5.450s	5.496s
1885.019	0.923s	1.865s	3.679s	4.655s	5.135s	5.406s	5.456s
1966.820	0.903s	1.826s	3.609s	4.583s	5.074s	5.359s	5.413s
2050.140	0.885s	1.791s	3.538s	4.509s	5.011s	5.310s	5.367s
2134.755	0.869s	1.759s	3.464s	4.433s	4.947s	5.259s	5.318s
2220.581	0.857s	1.730s	3.389s	4.356s	4.882s	5.205s	5.268s
2307.610	0.847s	1.703s	3.313s	4.276s	4.814s	5.149s	5.216s
2395.888	0.838s	1.678s	3.236s	4.194s	4.744s	5.090s	5.162s
2485.489	0.830s	1.655s	3.157s	4.108s	4.672s	5.027s	5.104s
2576.606	0.822s	1.633s	3.078s	4.019s	4.597s	4.959s	5.044s
2669.641	0.814s	1.613s	2.997s	3.925s	4.517s	4.887s	4.979s
2763.349	0.807s	1.595s	2.915s	3.829s	4.436s	4.812s	4.911s
2857.523	0.799s	1.577s	2.832s	3.730s	4.351s	4.734s	4.842s
2952.362	0.791s	1.560s	2.749s	3.629s	4.265s	4.654s	4.770s
3047.738	0.784s	1.542s	2.665s	3.525s	4.175s	4.571s	4.696s
3143.497	0.775s	1.517s	2.580s	3.421s	4.083s	4.486s	4.622s
3239.563	0.766s	1.486s	2.495s	3.316s	3.988s	4.398s	4.545s
3335.888	0.757s	1.449s	2.409s	3.212s	3.891s	4.309s	4.468s
3432.390	0.748s	1.407s	2.323s	3.108s	3.792s	4.219s	4.389s
3529.015	0.739s	1.358s	2.236s	3.006s	3.693s	4.127s	4.310s
3625.726	0.731s	1.305s	2.149s	2.905s	3.594s	4.034s	4.230s
3722.503	0.720s	1.246s	2.062s	2.807s	3.494s	3.940s	4.151s
3819.332	0.697s	1.182s	1.973s	2.710s	3.395s	3.847s	4.071s

Cross Curves of Stability
Righting Arms(heel) for VCG = 0.00
Trim aft 0.500/57.400 at heel = 0 (RA Trim = 0)

Displ (MT)	5.000s	10.000s	20.000s	30.000s	40.000s	50.000s	60.000s
1564.164	1.039s	2.096s	3.996s	4.963s	5.380s	5.543s	5.533s
1643.948	1.009s	2.037s	3.926s	4.889s	5.313s	5.497s	5.504s
1724.276	0.981s	1.985s	3.857s	4.815s	5.247s	5.450s	5.471s
1805.711	0.957s	1.938s	3.788s	4.740s	5.181s	5.402s	5.432s
1889.152	0.935s	1.895s	3.716s	4.662s	5.114s	5.351s	5.388s
1973.754	0.916s	1.855s	3.641s	4.583s	5.045s	5.299s	5.341s
2059.596	0.902s	1.819s	3.563s	4.502s	4.976s	5.246s	5.293s
2146.709	0.890s	1.786s	3.482s	4.419s	4.905s	5.191s	5.242s
2235.134	0.879s	1.755s	3.399s	4.333s	4.831s	5.133s	5.190s
2324.978	0.869s	1.727s	3.314s	4.244s	4.755s	5.071s	5.134s
2416.623	0.858s	1.701s	3.228s	4.152s	4.676s	5.004s	5.075s
2509.836	0.848s	1.676s	3.140s	4.055s	4.593s	4.933s	5.011s
2603.500	0.838s	1.653s	3.052s	3.956s	4.509s	4.858s	4.945s
2697.790	0.829s	1.631s	2.963s	3.854s	4.422s	4.781s	4.877s
2792.714	0.819s	1.611s	2.873s	3.749s	4.332s	4.701s	4.806s
2888.050	0.809s	1.591s	2.782s	3.642s	4.240s	4.619s	4.733s
2983.730	0.798s	1.569s	2.691s	3.533s	4.147s	4.534s	4.659s
3079.698	0.787s	1.540s	2.601s	3.423s	4.051s	4.448s	4.584s
3175.863	0.776s	1.506s	2.510s	3.314s	3.953s	4.360s	4.508s
3272.172	0.765s	1.465s	2.419s	3.205s	3.854s	4.270s	4.431s
3368.580	0.755s	1.418s	2.328s	3.097s	3.754s	4.179s	4.352s
3465.059	0.746s	1.365s	2.236s	2.991s	3.654s	4.087s	4.273s
3561.597	0.735s	1.307s	2.145s	2.886s	3.554s	3.994s	4.193s
3658.177	0.719s	1.245s	2.052s	2.784s	3.454s	3.901s	4.113s
3754.788	0.695s	1.178s	1.960s	2.683s	3.355s	3.807s	4.032s
3851.412	0.660s	1.106s	1.867s	2.585s	3.257s	3.714s	3.950s

Cross Curves of Stability
Righting Arms(heel) for VCG = 0.00
Trim aft 1.000/57.400 at heel = 0 (RA Trim = 0)

Displ (MT)	5.000s	10.000s	20.000s	30.000s	40.000s	50.000s	60.000s
1565.968	1.049s	2.128s	4.039s	4.982s	5.369s	5.503s	5.466s
1647.166	1.020s	2.070s	3.967s	4.903s	5.297s	5.450s	5.432s
1730.061	0.994s	2.018s	3.895s	4.822s	5.225s	5.395s	5.393s
1814.335	0.973s	1.969s	3.821s	4.740s	5.153s	5.339s	5.349s
1900.168	0.956s	1.925s	3.742s	4.655s	5.078s	5.283s	5.303s
1987.344	0.941s	1.884s	3.659s	4.568s	5.003s	5.227s	5.254s
2075.898	0.926s	1.847s	3.572s	4.478s	4.925s	5.169s	5.203s
2166.120	0.913s	1.812s	3.482s	4.385s	4.845s	5.108s	5.149s
2258.276	0.900s	1.779s	3.389s	4.288s	4.761s	5.041s	5.091s
2351.363	0.888s	1.749s	3.295s	4.188s	4.675s	4.971s	5.029s
2445.038	0.876s	1.721s	3.200s	4.086s	4.587s	4.898s	4.965s
2539.438	0.863s	1.695s	3.103s	3.980s	4.496s	4.821s	4.899s
2634.303	0.851s	1.670s	3.007s	3.872s	4.404s	4.742s	4.830s
2729.551	0.838s	1.645s	2.910s	3.762s	4.309s	4.660s	4.759s
2825.114	0.825s	1.619s	2.813s	3.649s	4.213s	4.576s	4.686s
2920.911	0.811s	1.590s	2.716s	3.536s	4.115s	4.490s	4.613s
3016.875	0.798s	1.556s	2.619s	3.422s	4.015s	4.403s	4.538s
3112.954	0.786s	1.517s	2.522s	3.308s	3.915s	4.314s	4.462s
3209.118	0.774s	1.472s	2.426s	3.196s	3.813s	4.224s	4.386s
3305.349	0.762s	1.421s	2.330s	3.084s	3.712s	4.133s	4.308s
3401.632	0.749s	1.365s	2.234s	2.974s	3.611s	4.041s	4.230s
3497.953	0.732s	1.304s	2.138s	2.866s	3.510s	3.948s	4.150s
3594.294	0.710s	1.238s	2.041s	2.761s	3.410s	3.855s	4.069s
3690.636	0.682s	1.168s	1.945s	2.657s	3.311s	3.762s	3.987s
3786.961	0.645s	1.093s	1.848s	2.555s	3.212s	3.668s	3.902s
3883.210	0.597s	1.013s	1.750s	2.454s	3.113s	3.572s	3.814s

Cross Curves of Stability
Righting Arms(heel) for VCG = 0.00
Trim fwd 0.500/57.400 at heel = 0 (RA Trim = 0)

Displ (MT)	5.000s	10.000s	20.000s	30.000s	40.000s	50.000s	60.000s
1567.855	1.021s	2.051s	3.899s	4.900s	5.378s	5.595s	5.638s
1646.172	0.989s	1.990s	3.833s	4.836s	5.323s	5.562s	5.611s
1724.970	0.962s	1.935s	3.768s	4.772s	5.266s	5.528s	5.581s
1804.539	0.937s	1.885s	3.702s	4.706s	5.208s	5.491s	5.548s
1885.063	0.914s	1.840s	3.636s	4.639s	5.150s	5.450s	5.511s
1965.996	0.893s	1.800s	3.569s	4.572s	5.092s	5.407s	5.471s
2047.401	0.875s	1.764s	3.502s	4.504s	5.035s	5.361s	5.426s
2129.428	0.858s	1.732s	3.434s	4.434s	4.977s	5.313s	5.380s
2212.598	0.844s	1.704s	3.366s	4.364s	4.918s	5.264s	5.332s
2297.165	0.831s	1.678s	3.297s	4.291s	4.858s	5.211s	5.282s
2382.958	0.819s	1.654s	3.227s	4.217s	4.795s	5.157s	5.231s
2469.887	0.810s	1.633s	3.156s	4.140s	4.731s	5.099s	5.178s
2558.008	0.803s	1.612s	3.085s	4.060s	4.664s	5.039s	5.122s
2647.395	0.797s	1.594s	3.013s	3.977s	4.595s	4.975s	5.064s
2738.120	0.791s	1.576s	2.940s	3.891s	4.523s	4.907s	5.002s
2830.422	0.785s	1.560s	2.865s	3.801s	4.448s	4.835s	4.937s
2924.371	0.779s	1.544s	2.789s	3.707s	4.368s	4.759s	4.868s
3018.574	0.773s	1.528s	2.713s	3.612s	4.285s	4.680s	4.798s
3113.232	0.767s	1.509s	2.635s	3.514s	4.199s	4.600s	4.726s
3208.564	0.762s	1.487s	2.557s	3.415s	4.110s	4.516s	4.651s
3304.353	0.755s	1.460s	2.477s	3.316s	4.018s	4.430s	4.575s
3400.497	0.748s	1.427s	2.397s	3.217s	3.923s	4.342s	4.498s
3496.926	0.741s	1.389s	2.316s	3.118s	3.826s	4.252s	4.421s
3593.586	0.733s	1.345s	2.235s	3.021s	3.727s	4.161s	4.343s
3690.412	0.723s	1.296s	2.152s	2.924s	3.628s	4.069s	4.264s
3787.349	0.709s	1.241s	2.069s	2.829s	3.529s	3.975s	4.185s

Cross Curves of Stability
Righting Arms(heel) for VCG = 0.00
Trim fwd 1.000/57.400 at heel = 0 (RA Trim = 0)

Displ (MT)	5.000s	10.000s	20.000s	30.000s	40.000s	50.000s	60.000s
1572.917	1.012s	2.032s	3.848s	4.861s	5.365s	5.613s	5.679s
1650.481	0.981s	1.972s	3.783s	4.801s	5.317s	5.585s	5.653s
1728.551	0.954s	1.917s	3.719s	4.741s	5.267s	5.556s	5.623s
1807.554	0.928s	1.867s	3.655s	4.680s	5.214s	5.524s	5.590s
1887.158	0.906s	1.821s	3.591s	4.618s	5.161s	5.488s	5.555s
1967.287	0.886s	1.780s	3.527s	4.555s	5.107s	5.448s	5.518s
2048.130	0.867s	1.743s	3.463s	4.491s	5.052s	5.405s	5.476s
2129.353	0.850s	1.710s	3.399s	4.426s	4.998s	5.359s	5.431s
2211.021	0.835s	1.681s	3.335s	4.360s	4.944s	5.311s	5.385s
2293.270	0.821s	1.654s	3.270s	4.294s	4.889s	5.261s	5.337s
2376.497	0.808s	1.631s	3.206s	4.226s	4.833s	5.210s	5.287s
2460.847	0.797s	1.610s	3.142s	4.155s	4.776s	5.157s	5.237s
2546.583	0.787s	1.591s	3.077s	4.083s	4.716s	5.101s	5.185s
2633.408	0.779s	1.573s	3.013s	4.008s	4.655s	5.043s	5.131s
2721.371	0.774s	1.557s	2.947s	3.931s	4.591s	4.981s	5.075s
2810.548	0.769s	1.542s	2.881s	3.851s	4.525s	4.916s	5.015s
2901.004	0.764s	1.526s	2.814s	3.768s	4.454s	4.848s	4.953s
2992.842	0.760s	1.510s	2.745s	3.682s	4.380s	4.776s	4.888s
3086.268	0.756s	1.493s	2.675s	3.592s	4.300s	4.700s	4.819s
3180.740	0.751s	1.474s	2.603s	3.500s	4.216s	4.620s	4.747s
3275.380	0.747s	1.452s	2.531s	3.408s	4.130s	4.539s	4.674s
3370.527	0.742s	1.428s	2.457s	3.315s	4.040s	4.455s	4.599s
3466.311	0.738s	1.399s	2.383s	3.221s	3.948s	4.368s	4.524s
3562.502	0.731s	1.365s	2.307s	3.128s	3.853s	4.280s	4.448s
3659.026	0.722s	1.325s	2.231s	3.035s	3.756s	4.189s	4.371s
3755.829	0.709s	1.280s	2.154s	2.944s	3.658s	4.098s	4.294s

14.3 **DOWNFLOODING POINTS**

The following are the unprotected flood points

Name	L, T, V (m)
(1) POSITION 1(S)	9.700f, 5.400s, 12.150
(2) POSITION 1(P)	9.700f, 5.400p, 12.150
(3) POSITION 2(S)	10.300f, 6.300s, 12.150
(4) POSITION 2(P)	10.300f, 6.300s, 12.150

14.4 MAXIMUM ALLOWABLE VCG TABLE

MAXIMUM VCG FOR INTACT

OSV CHEMICAL CODE – CHAPTER 2 , 2.2

IS CODE 2008 INTACT STABILITY CRITERIA.

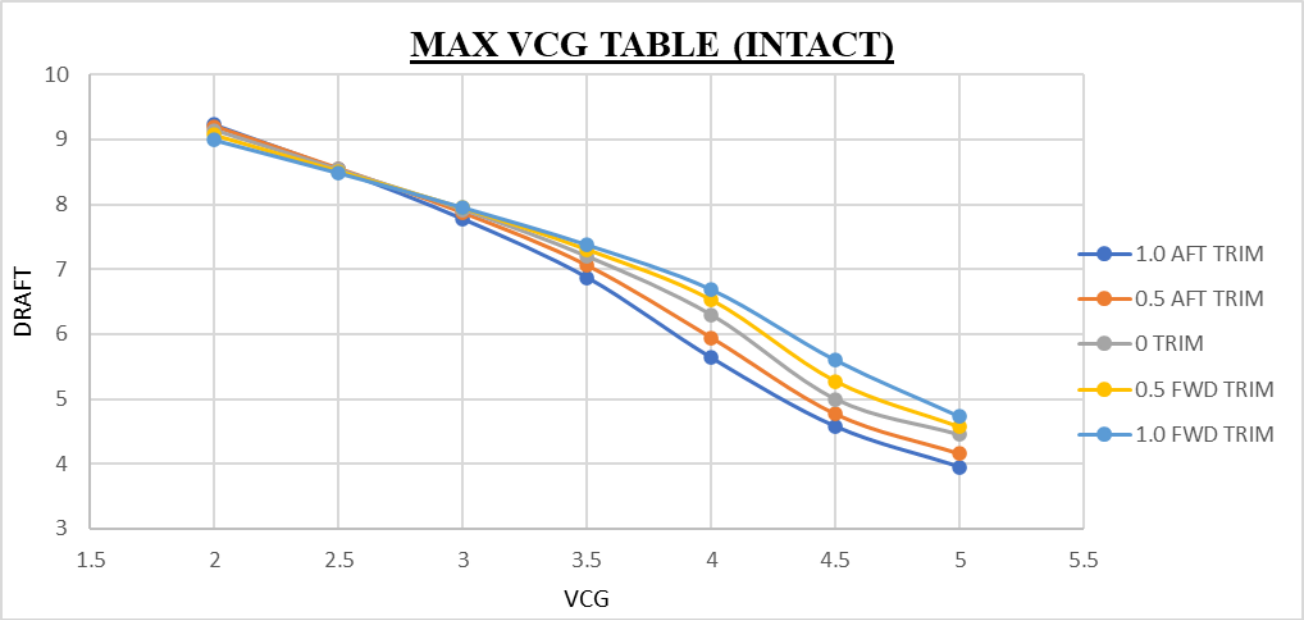
INTACT MAX VCG 0 TRIM	
DRAFT	**VCG**
2	9.142
2.5	8.537
3	7.917
3.5	7.2
4	6.298
4.5	5
5	4.453

INTACT MAX VCG 0.5 AFT TRIM	
DRAFT	**VCG**
2	9.198
2.5	8.549
3	7.866
3.5	7.055
4	5.94
4.5	4.764
5	4.152

INTACT MAX VCG 1.0 AFT TRIM	
DRAFT	**VCG**
2	9.233
2.5	8.541
3	7.778
3.5	6.874
4	5.632
4.5	4.571
5	3.947

INTACT MAX VCG 0.5 FWD TRIM	
DRAFT	**VCG**
2	9.071
2.5	8.513
3	7.944
3.5	7.305
4	6.53
4.5	5.272
5	4.567

INTACT MAX VCG 1.0 FWD TRIM	
DRAFT	**VCG**
2	8.989
2.5	8.482
3	7.952
3.5	7.376
4	6.687
4.5	5.6
5	4.73



LIMIT (1) AREA FROM 0 TO MAX AT 15 > 0.070
LIMIT (2) AREA FROM 0 TO MAX AT 30 > 0.055
LIMIT (3) AREA FROM 0 TO 40 OR FLD > 0.090
LIMIT (4) AREA FROM 30 TO 40 OR FLD > 0.030
LIMIT (5) RA AT 30 > 0.2
LIMIT (6) ABS ANGLE AT MAX RA > 15
LIMIT (7) GM UPRIGHT > 0.15

MAXIMUM VCG FOR SEVERE WIND AND ROLLING CRITERION

IMO SEVERE WIND & ROLLING

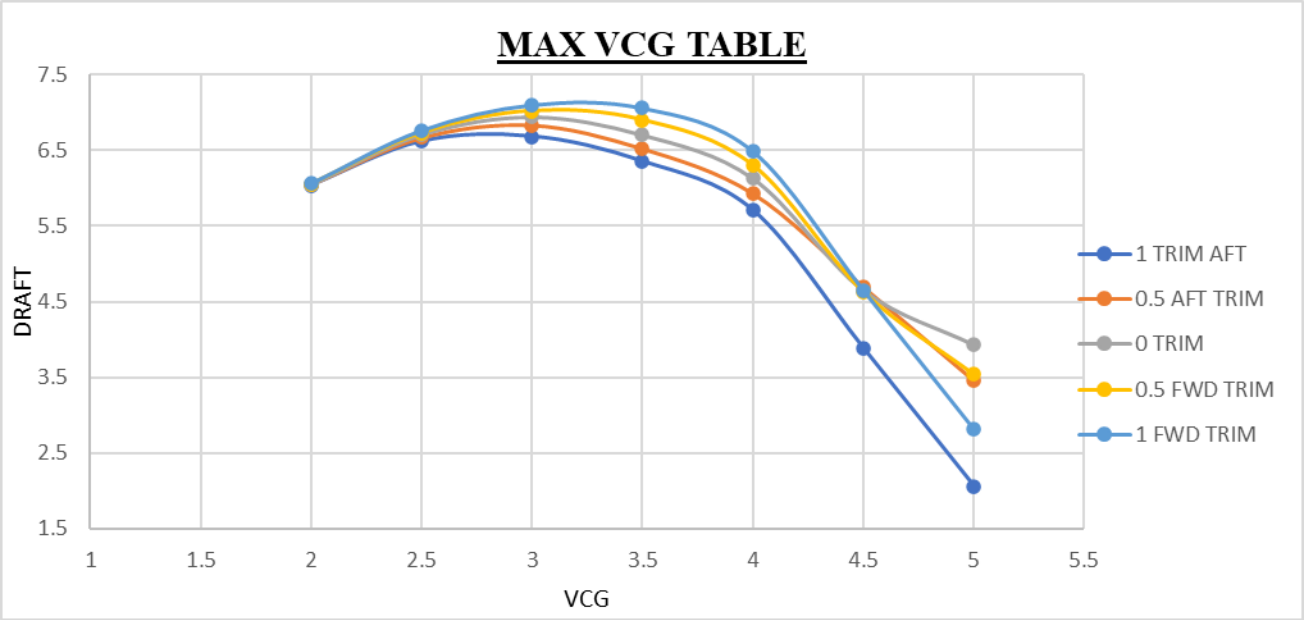
WIND MAX VCG 0 TRIM	
DRAFT	**VCG**
2	6.048
2.5	6.71
3	6.944
3.5	6.707
4	6.131
4.5	4.649
5	3.928

WIND MAX VCG 0.5 AFT TRIM	
DRAFT	**VCG**
2	6.047
2.5	6.667
3	6.833
3.5	6.523
4	5.93
4.5	4.693
5	3.456

WIND MAX VCG 1.0 AFT TRIM	
DRAFT	**VCG**
2	6.038
2.5	6.618
3	6.683
3.5	6.355
4	5.707
4.5	3.885
5	2.063

WIND MAX VCG 0.5 FWD TRIM	
DRAFT	**VCG**
2	6.055
2.5	6.732
3	7.022
3.5	6.902
4	6.302
4.5	4.63
5	3.54

WIND MAX VCG 1.0 FWD TRIM	
DRAFT	**VCG**
2	6.058
2.5	6.761
3	7.098
3.5	7.056
4	6.484
4.5	4.648
5	2.812



LIMIT (1) RESIDUAL RATIO FROM ROLL TO ABS 50 OR FLD>1
LIMIT(2) ANGLE FROM SSWEQ TO .8*DI OR ABS 16 > 0

MAXIMUM VCG FOR FIRE FIGHTING OPERATION

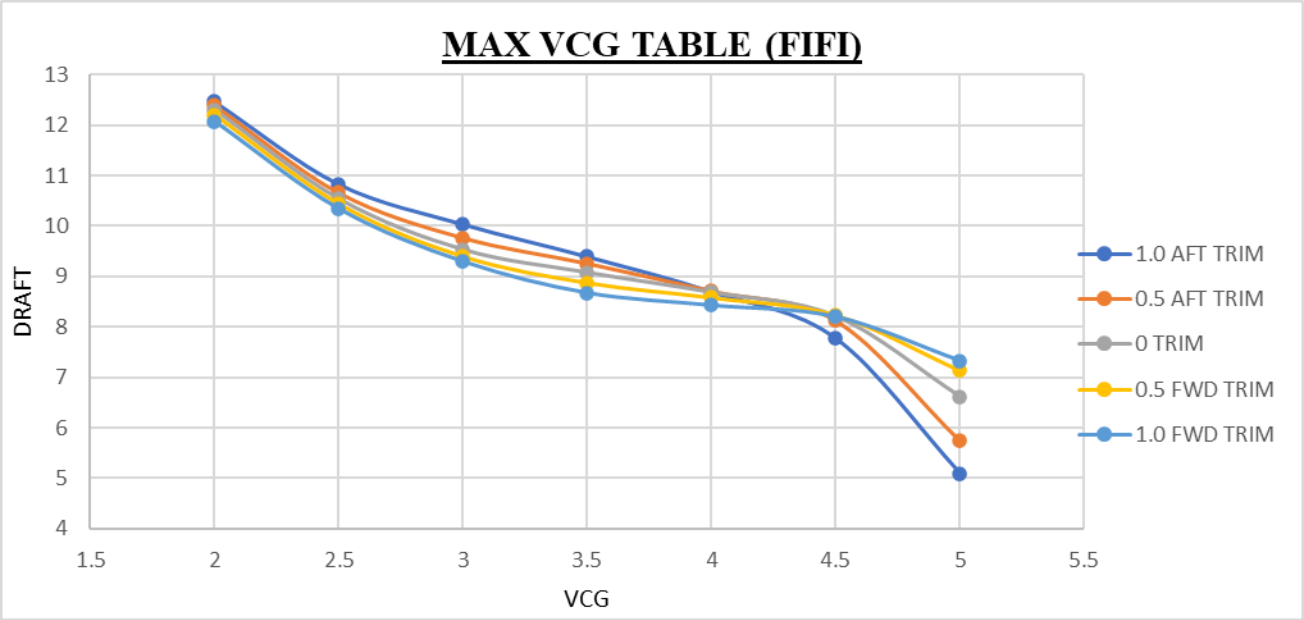
FIFI MAX VCG 0 TRIM	
DRAFT	**VCG**
2	12.301
2.5	10.538
3	9.531
3.5	9.078
4	8.685
4.5	8.213
5	6.621

FIFI MAX VCG 0.5 AFT TRIM	
DRAFT	**VCG**
2	12.388
2.5	10.659
3	9.766
3.5	9.261
4	8.712
4.5	8.126
5	5.742

FIFI MAX VCG 1.0 AFT TRIM	
DRAFT	**VCG**
2	12.468
2.5	10.828
3	10.033
3.5	9.391
4	8.692
4.5	7.769
5	5.09

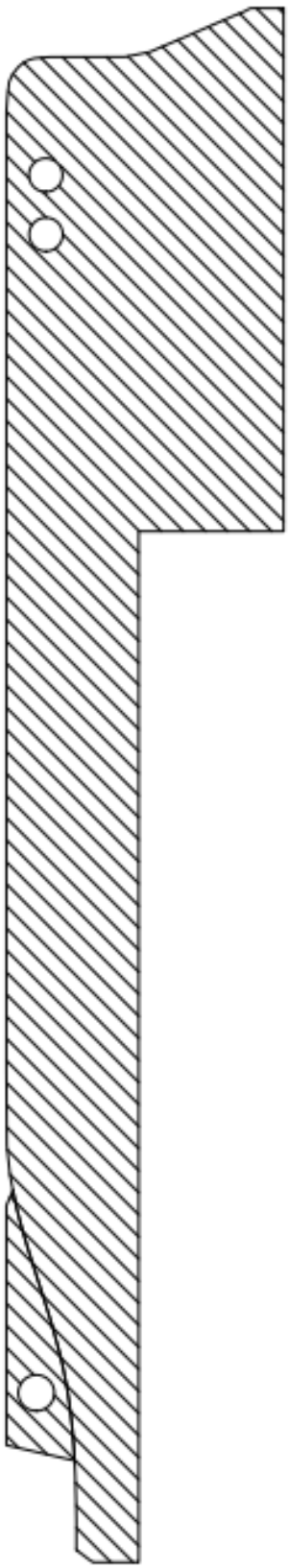
FIFI MAX VCG 0.5 FWD TRIM	
DRAFT	**VCG**
2	12.197
2.5	10.434
3	9.392
3.5	8.87
4	8.577
4.5	8.229
5	7.134

FIFI MAX VCG 1.0 FWD TRIM	
DRAFT	**VCG**
2	12.078
2.5	10.344
3	9.298
3.5	8.675
4	8.431
4.5	8.202
5	7.331

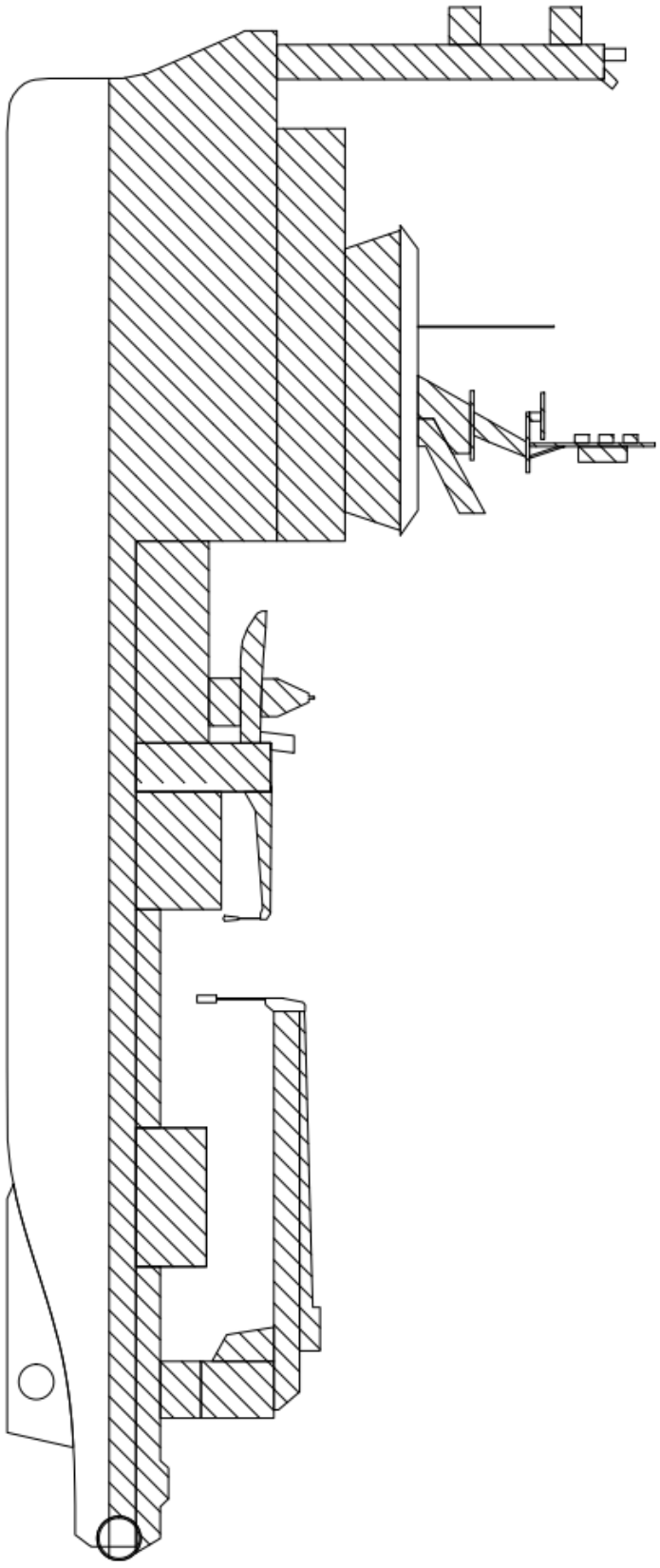


LIMIT (1) ANGLE AT EQUIL < 5

14.5 VOLUME USED FOR COMPUTATION OF KN CURVES



14.6 **LATERAL AREA FOR WEATHER CRITERIA**



15 APPENDIX

15.1 BLANK CALCULATION SHEET

List out the weights, LCG, VCG & FSM of each tank and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and the VCG of the vessel.

Fixed Weight Status

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
LIGHT SHIP	1911.974	0.592A	0.243p	5.792
01 PROVISION				
02 CREW AND EFFECTS				
Total Fixed:				

Tank Status

WATER BALLST

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	Volume (m^3)	Max FSM (MT-m)
DWSWBFP	1.025							
NO1_DWSWB.P	1.025							
NO1_DWSWB.S	1.025							
NO2_DWSWB.P	1.025							
NO2_DWSWB.S	1.025							
NO3_DWSWB.P	1.025							
NO3_DWSWB.S	1.025							
DWSWBAP.P	1.025							
DWSWBAP.S	1.025							
Subtotals:								

FRESH WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	Volume (m^3)	Max FSM (MT-m)
NO1_FW.P	1.000							
NO1_FW.S	1.000							
NO2_FW.P	1.000							
NO2_FW.S	1.000							
NO3_FW.P	1.000							
NO3_FW.S	1.000							

FUEL OIL

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	Volume (m3)	Max FSM (MT-m)
NO1_FO.P	0.850							
NO1_FO.S	0.850							
NO1_FO.U.P	0.850							
NO1_FO.U.S	0.850							
NO1_FO.U.C	0.850							
NO2_FO.U.P	0.850							
NO2_FO.U.S	0.850							
NO3_FO.P	0.850							
NO3_FO.S	0.850							
NO3_FO.U.P	0.850							
NO3_FO.U.S	0.850							
NO4_FO.U.P	0.850							
NO4_FO.U.S	0.850							
FODAY.P	0.850							
FODAY.S	0.850							

MISC

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	Volume (m^3)	Max FSM (MT-m)
SLOP	0.850							
DISPERSANT.P	1.163							
FOAM.S	1.145							
HYDOIL.P	0.924							
LUBOIL.S	0.924							
DIRTYOIL.P	0.900							
BILGEHOLD.S	1.000							
SEWAGE	1.000							
OVERFLOW	0.850							

SCALE INHIBITOR

Tank Name	**Spgr**	**Load (%)**	**Weight (MT)**	**LCG (m)**	**TCG (m)**	**VCG (m)**	**Volume (m^3)**	**Max FSM (MT-m)**
SCLEIHBTR.P	1.100							
SCLEIHBTR.S	1.100							
SCALE_DISLVR.S	1.100							

CORROSION INHIBITOR

Tank Name	**Spgr**	**Load (%)**	**Weight (MT)**	**LCG (m)**	**TCG (m)**	**VCG (m)**	**Volume (m^3)**	**Max FSM (MT-m)**
CORSNIHBTR.P	0.950							

TREATED INJECTION WATER

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	Volume (m^3)	Max FSM (MT-m)
TRTDINJW.P	1.000							
TRTDINJW.S	1.000							

BRINE

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	Volume (m^3)	Max FSM (MT-m)
NO1_BRINETK.C	1.550							
NO2_BRINETK.C	1.550							
NO3_BRINETK.C	1.550							
NO4_BRINETK.P	1.550							
NO4_BRINETK.S	1.550							

ACID

Tank Name	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	Volume (m^3)	Max FSM (MT-m)
NO1_ACIDTK.C	1.160							

All Tanks

	Spgr	Load (%)	Weight (MT)	LCG (m)	TCG (m)	VCG (m)	Volume (m^3)	Max FSM (MT-m)
Totals:								

FSM Notes – Max FSM taken into account for all tanks

All the weights onboard are to be listed along with their centers of gravity and Free surface moments. The individual Vertical, Longitudinal and Transverse Moments are found out by multiplying the individual weight by corresponding VCG, LCG and TCG. The individual Vertical, Longitudinal and Transverse Moments are added up. The resultant LCG, VCG and TCG are obtained by dividing these summations by the displacement. The VCG is now to be corrected for Free Surface Effect.

Loading Summary

Item	Weight (MT)	LCG (m)	TCG (m)	VCG (m)
Light Ship	1911.974	0.592A	0.243p	5.792
Deadweight				
Displacement				

After working out the displacement and the centers of gravity for a particular Loading condition, the Hydrostatic particulars of the vessel at that displacement is read out from the Hydrostatics Table given.

Displacer Status

Item	Status	Spgr	Displ (MT)	LCB (m)	TCB (m)	VCB (m)	Eff /Perm
HULL.C	Intact	1.025					1.000
SubTotals:							

From the hydrostatic particulars @ -------- m trim for ---------- T displacement the Hydrostatic particulars are obtained.

The mean draft corresponding to the displacement and the Metacentric Radius (KMt) are obtained. The Transverse metacentric height GMt of the vessel is thence calculated.
KGo is the corrected Vertical Centre of Gravity and is obtained as follows.

Free Surface Correction (Meters) = Sum of Free Surface Moments of all Tanks / Displacement of Vessel

=

KGo = KG + Free Surface correction

The trim and Metacentric Height are calculated using the Hydrostatic particulars and the values of the centers of gravity.

GMt = KMt - KGo

=

Trim (cm) = Displacement x (LCB - LCG) / (MCT x 100)

Floating Status

Draft FP		Heel		GM(Solid)	
Draft MS		Equil		F/S Corr.	
Draft AP		Wind		GM(Fluid)	
Trim		Wave		KMt	
LCG		VCG		TPcm	
Displacement		WaterSpgr			

The Cross curves or the KN curves are used to obtain the staitical stability curves or the GZ curves of the vessel. The Intact stability of the vessel is determined based on the statical stability curves based on the requirements of the IMO regulations in this regard.

Obtain the values of KN from the Displacement, corresponding to vessel's displacement and trim, for each angle of Heel. GZ is obtained using the relationship given in the formula below,

GZ = KN - KGo Sinθ

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the Maxima of GZ curve, and the area under the GZ curve as required by the IMO criteria.

GZ = KN - KGo Sinθ

Angle (Deg)	10	20	30	40	50	60
KN						
KG Sin θ (m)						
GZ (m)						

The area under the righting lever curve (GZ curve) shall not be less than 0.055 metre-radians up to φ= 30° angle of heel and not less than 0.09 metre-radians up toφ= 40° or the angle of down-floodingφf, if this angle is less than 40°. Additionally, the area under the righting lever curve (GZ curve) between the angles of heel of 30° and 40° or between 30° andφf, if this angle is less than 40°, shall not be less than 0.03 metre-radians.

The righting lever GZ shall be at least 0.2 m at an angle of heel equal to or greater than 30°.

The maximum righting lever shall occur at an angle of heel not less than 25°. If this is not practicable, alternative criteria, based on an equivalent level of safety, may be applied subject to the approval of the Administration.

The initial metacentric height GM0 shall not be less than 0.15 m.

GZ CURVE

Limit	Min/Max	Actual	Margin	Pass
(1) Area from 0.00 deg to MaxRA	>0.0550 m-R			
(2) Area from 0.00 deg to 40.00 or Flood	>0.0900 m-R			
(3) Area from 30.00 deg to 40.00 or Flood	>0.0300 m-R			
(4) Righting Arm at 30.00 deg	>0.200 m			
(5) Absolute Angle at MaxRA	>15.00 deg			
(6) GM Upright	>0.150 m			

Heeling Moment Derivation

Part	LPA (m2)	HCP (m)	Arm (m)	Pressure (MT/m2)	Moment (m-MT)
HULL.C					

The wind heeling levers lw1 and lw2 referred to are constant values at all angles of inclination and are calculated as follows:

P =

A =

Z =

Disp =

lw1 = P A Z / Disp.

=

lw2 = 1.5 lw1

=

The angle of roll (θ_1) should be calculated as follows:

θ_1 = 109 k X1 X2 $\sqrt{}$(r s) (degrees)

Where:

X1 = factor as shown in table 1

=

X2 = factor as shown in table 2

=

k = factor as follows:

k = 1.0 for round-bilged ship having no bilge or bar keels;

k = 0.7 for ship having sharp bilges;

k = as shown in table 3 for a ship having bilge Keels, a bar keel or both

=

OG = distance between the center of gravity and the waterline (- if it is below).

=

d = mean moulded draft of the ship (m)

=

s = factor as shown in table 4.

=

r = 0.73 ± 0.6 OG/d

=

Therefore,

θ_1 =

Table 1 - Values of coefficient X_1

B/d	**X_1**
≤ 2.4	1.00
2.5	0.98
2.6	0.96
2.7	0.95
2.8	0.93
2.9	0.91
3.0	0.90
3.1	0.88
3.2	0.86
3.4	0.82
≥ 3.5	0.80

Table 2 - Values of coefficient X2

C_B	**X_2**
≤ 0.45	0.75
0.50	0.82
0.55	0.89
0.60	0.95
0.65	0.97
≥0.70	1.00

Table 3 - Values of coefficient k

$\frac{A_K \times 100}{L \times B}$	**k**
0.0	1.00
1.0	0.98
1.5	0.95
2.0	0.88
2.5	0.79
3.0	0.74
3.5	0.72
≥4.0	0.70

Table 4 - Values of factor s

T_R	**s**
≤ 6	0.100
7	0.098
8	0.093
12	0.065
14	0.053
16	0.044
18	0.038
≥20	0.035

Rolling Period T = 2.C.B/√GM) seconds where: C = 0.373 + 0.023 (B/d) - 0.043 (L/100)

The symbols in the above tables and formula for the rolling period are defined as follows:

L = waterline length of the ship (m)

B = moulded breadth of the ship (m)

d = moulded draft of the ship (m)

Cb = block coefficient

Ak = total overall area of bilge keels, or area of the lateral projection of the bar keel, or sum of these areas. (m^2)

GM = metacentric height corrected for free surface effect (m)

C =

T =

GZ CURVE

IMO SEVERE WIND & ROLLING

Limit	Min/Max	Actual	Margin	Pass
(1) Res. Ratio from Roll to Abs 50.00 deg or Flood	>1.000			
(2) Angle from SSWEQ to 80% of Dk. Imm. or Abs 16.00 deg	>0.00 deg			

15.2 TANK CALIBRATION TABLE

Tank Capacities for DWSWBFP containing WATER BALLAST (1.025)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.66	2.73	26.254f	0.000	0.396	3.51
10.00%	5.32	5.46	26.333f	0.000	0.638	5.95
20.00%	10.65	10.91	26.400f	0.000	1.028	10.10
30.00%	15.97	16.37	26.433f	0.000	1.359	13.90
40.00%	21.29	21.83	26.454f	0.000	1.659	17.47
50.00%	26.62	27.28	26.469f	0.000	1.936	20.90
60.00%	31.94	32.74	26.480f	0.000	2.198	24.26
70.00%	37.26	38.20	26.489f	0.000	2.447	27.73
80.00%	42.59	43.65	26.496f	0.000	2.685	31.40
90.00%	47.91	49.11	26.503f	0.000	2.914	35.51
95.00%	50.57	51.84	26.506f	0.000	3.026	37.82
98.00%	52.17	53.47	26.508f	0.000	3.092	39.29
100.00%	53.24	54.57	26.510f	0.000	3.135	

Tank Capacities for NO1_DWSWB.P containing WATER BALLAST (1.025)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.93	1.97	20.705f	2.772p	0.099	10.53
10.00%	3.85	3.95	20.794f	2.841p	0.182	13.73
20.00%	7.71	7.90	20.885f	2.937p	0.328	16.11
30.00%	11.56	11.85	20.844f	3.048p	0.475	17.37
40.00%	15.41	15.80	20.806f	3.141p	0.630	19.75
50.00%	19.26	19.74	20.800f	3.212p	0.786	23.60
60.00%	23.12	23.69	20.833f	3.260p	0.935	29.55
70.00%	26.97	27.64	20.901f	3.288p	1.074	36.22
80.00%	30.82	31.59	20.960f	3.311p	1.199	38.08
90.00%	34.67	35.54	21.009f	3.332p	1.316	39.85
95.00%	36.60	37.51	21.030f	3.343p	1.372	40.69
98.00%	37.76	38.70	21.041f	3.349p	1.405	41.18
100.00%	38.53	39.49	21.049f	3.353p	1.426	

Tank Capacities for NO1_DWSWB.S containing WATER BALLAST (1.025)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.93	1.97	20.705f	2.772s	0.099	10.53
10.00%	3.85	3.95	20.794f	2.841s	0.182	13.73
20.00%	7.71	7.90	20.885f	2.937s	0.328	16.11
30.00%	11.56	11.85	20.844f	3.048s	0.475	17.37
40.00%	15.41	15.80	20.806f	3.141s	0.630	19.75
50.00%	19.26	19.74	20.800f	3.212s	0.786	23.60
60.00%	23.12	23.69	20.833f	3.260s	0.935	29.55
70.00%	26.97	27.64	20.901f	3.288s	1.074	36.22
80.00%	30.82	31.59	20.960f	3.311s	1.199	38.08
90.00%	34.67	35.54	21.009f	3.332s	1.316	39.85
95.00%	36.60	37.51	21.030f	3.343s	1.372	40.69
98.00%	37.76	38.70	21.041f	3.349s	1.405	41.18
100.00%	38.53	39.49	21.049f	3.353s	1.426	

Tank Capacities for NO2_DWSWB.P containing WATER BALLAST (1.025)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.85	2.92	16.599f	3.709p	0.068	79.44
10.00%	5.70	5.84	16.592f	3.799p	0.133	87.63
20.00%	11.39	11.68	16.584f	3.915p	0.256	98.35
30.00%	17.09	17.52	16.579f	3.994p	0.374	106.01
40.00%	22.79	23.35	16.577f	4.056p	0.489	112.13
50.00%	28.48	29.19	16.538f	4.237p	0.605	41.23
60.00%	34.18	35.03	16.503f	4.405p	0.728	43.99
70.00%	39.87	40.87	16.479f	4.533p	0.853	46.28
80.00%	45.57	46.71	16.461f	4.633p	0.980	48.25
90.00%	51.27	52.55	16.448f	4.716p	1.107	49.92
95.00%	54.11	55.47	16.443f	4.751p	1.170	50.69
98.00%	55.82	57.22	16.440f	4.771p	1.208	51.11
100.00%	56.96	58.39	16.438f	4.784p	1.234	

Tank Capacities for NO2_DWSWB.S containing WATER BALLAST (1.025)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.85	2.92	16.599f	3.709s	0.068	79.44
10.00%	5.70	5.84	16.592f	3.799s	0.133	87.63
20.00%	11.39	11.68	16.584f	3.915s	0.256	98.35
30.00%	17.09	17.52	16.579f	3.994s	0.374	106.01
40.00%	22.79	23.35	16.577f	4.056s	0.489	112.13
50.00%	28.48	29.19	16.538f	4.237s	0.605	41.23
60.00%	34.18	35.03	16.503f	4.405s	0.728	43.99
70.00%	39.87	40.87	16.479f	4.533s	0.853	46.28
80.00%	45.57	46.71	16.461f	4.633s	0.980	48.25
90.00%	51.27	52.55	16.448f	4.716s	1.107	49.92
95.00%	54.11	55.47	16.443f	4.751s	1.170	50.69
98.00%	55.82	57.22	16.440f	4.771s	1.208	51.11
100.00%	56.96	58.39	16.438f	4.784s	1.234	

Tank Capacities for NO3_DWSWB.P containing WATER BALLAST (1.025)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.40	2.46	13.749a	6.267p	0.134	10.82
10.00%	4.80	4.92	13.984a	6.300p	0.227	13.71
20.00%	9.60	9.84	14.290a	6.370p	0.384	18.70
30.00%	14.40	14.76	14.467a	6.421p	0.519	20.97
40.00%	19.20	19.68	14.564a	6.454p	0.645	22.20
50.00%	24.00	24.60	14.626a	6.478p	0.767	22.90
60.00%	28.80	29.52	14.670a	6.497p	0.887	23.27
70.00%	33.60	34.44	14.702a	6.511p	1.004	23.45
80.00%	38.40	39.36	14.726a	6.522p	1.121	23.51
90.00%	43.20	44.28	14.746a	6.531p	1.236	23.53
95.00%	45.60	46.74	14.754a	6.534p	1.294	23.53
98.00%	47.04	48.21	14.758a	6.536p	1.328	23.53
100.00%	48.00	49.20	14.761a	6.538p	1.351	

Tank Capacities for NO3_DWSWB.S containing WATER BALLAST (1.025)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.40	2.46	13.749a	6.267s	0.134	10.82
10.00%	4.80	4.92	13.984a	6.300s	0.227	13.71
20.00%	9.60	9.84	14.290a	6.370s	0.384	18.70
30.00%	14.40	14.76	14.467a	6.421s	0.519	20.97
40.00%	19.20	19.68	14.564a	6.454s	0.645	22.20
50.00%	24.00	24.60	14.626a	6.478s	0.767	22.90
60.00%	28.80	29.52	14.670a	6.497s	0.887	23.27
70.00%	33.60	34.44	14.702a	6.511s	1.004	23.45
80.00%	38.40	39.36	14.726a	6.522s	1.121	23.51
90.00%	43.20	44.28	14.746a	6.531s	1.236	23.53
95.00%	45.60	46.74	14.754a	6.534s	1.294	23.53
98.00%	47.04	48.21	14.758a	6.536s	1.328	23.53
100.00%	48.00	49.20	14.761a	6.538s	1.351	

Tank Capacities for DWSWBAP.P containing WATER BALLAST (1.025)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.64	1.68	29.301a	5.492p	3.524	25.08
10.00%	3.28	3.36	29.362a	5.777p	3.595	29.30
20.00%	6.55	6.72	29.434a	5.968p	3.711	35.19
30.00%	9.83	10.07	29.488a	6.055p	3.814	39.07
40.00%	13.10	13.43	29.534a	6.102p	3.910	41.09
50.00%	16.38	16.79	29.568a	6.135p	4.002	41.69
60.00%	19.65	20.15	29.591a	6.157p	4.091	41.71
70.00%	22.93	23.50	29.607a	6.173p	4.179	41.71
80.00%	26.21	26.86	29.619a	6.185p	4.267	41.71
90.00%	29.48	30.22	29.629a	6.195p	4.353	41.71
95.00%	31.12	31.90	29.633a	6.199p	4.396	41.71
98.00%	32.10	32.91	29.635a	6.201p	4.422	41.71
100.00%	32.76	33.58	29.636a	6.202p	4.439	

Tank Capacities for DWSWBAP.S containing WATER BALLAST (1.025)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.64	1.68	29.301a	5.492s	3.524	25.08
10.00%	3.28	3.36	29.362a	5.777s	3.595	29.30
20.00%	6.55	6.72	29.434a	5.968s	3.711	35.19
30.00%	9.83	10.07	29.488a	6.055s	3.814	39.07
40.00%	13.10	13.43	29.534a	6.102s	3.910	41.09
50.00%	16.38	16.79	29.568a	6.135s	4.002	41.69
60.00%	19.65	20.15	29.591a	6.157s	4.091	41.71
70.00%	22.93	23.50	29.607a	6.173s	4.179	41.71
80.00%	26.21	26.86	29.619a	6.185s	4.267	41.71
90.00%	29.48	30.22	29.629a	6.195s	4.353	41.71
95.00%	31.12	31.90	29.633a	6.199s	4.396	41.71
98.00%	32.10	32.91	29.635a	6.201s	4.422	41.71
100.00%	32.76	33.58	29.636a	6.202s	4.439	

Tank Capacities for NO1_FW.P containing FRESH WATER (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	3.69	3.69	21.423f	3.563p	2.578	42.13
10.00%	7.38	7.38	21.429f	3.575p	2.655	43.57
20.00%	14.77	14.77	21.439f	3.599p	2.808	46.31
30.00%	22.15	22.15	21.448f	3.621p	2.957	48.82
40.00%	29.53	29.53	21.456f	3.642p	3.104	51.19
50.00%	36.92	36.92	21.464f	3.662p	3.249	53.43
60.00%	44.30	44.30	21.471f	3.680p	3.392	55.58
70.00%	51.68	51.68	21.478f	3.698p	3.533	57.71
80.00%	59.07	59.07	21.485f	3.715p	3.672	59.77
90.00%	66.45	66.45	21.491f	3.731p	3.810	61.82
95.00%	70.14	70.14	21.494f	3.739p	3.878	62.83
98.00%	72.35	72.35	21.496f	3.744p	3.919	63.42
100.00%	73.83	73.83	21.497f	3.747p	3.946	

Tank Capacities for NO1_FW.S containing FRESH WATER (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	3.69	3.69	21.423f	3.563s	2.578	42.13
10.00%	7.38	7.38	21.429f	3.575s	2.655	43.57
20.00%	14.77	14.77	21.439f	3.599s	2.808	46.31
30.00%	22.15	22.15	21.448f	3.621s	2.957	48.82
40.00%	29.53	29.53	21.456f	3.642s	3.104	51.19
50.00%	36.92	36.92	21.464f	3.662s	3.249	53.43
60.00%	44.30	44.30	21.471f	3.680s	3.392	55.58
70.00%	51.68	51.68	21.478f	3.698s	3.533	57.71
80.00%	59.07	59.07	21.485f	3.715s	3.672	59.77
90.00%	66.45	66.45	21.491f	3.731s	3.810	61.82
95.00%	70.14	70.14	21.494f	3.739s	3.878	62.83
98.00%	72.35	72.35	21.496f	3.744s	3.919	63.42
100.00%	73.83	73.83	21.497f	3.747s	3.946	

Tank Capacities for NO2_FW.P containing FRESH WATER (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	3.32	3.32	16.352f	5.423p	2.573	50.83
10.00%	6.64	6.64	16.354f	5.430p	2.645	51.54
20.00%	13.29	13.29	16.356f	5.442p	2.789	52.82
30.00%	19.93	19.93	16.359f	5.453p	2.932	53.90
40.00%	26.57	26.57	16.361f	5.464p	3.074	54.84
50.00%	33.22	33.22	16.364f	5.473p	3.216	55.68
60.00%	39.86	39.86	16.366f	5.481p	3.356	56.39
70.00%	46.51	46.51	16.368f	5.489p	3.497	57.03
80.00%	53.15	53.15	16.370f	5.496p	3.636	57.62
90.00%	59.79	59.79	16.372f	5.502p	3.775	58.16
95.00%	63.12	63.12	16.373f	5.505p	3.845	58.41
98.00%	65.11	65.11	16.374f	5.507p	3.886	58.58
100.00%	66.44	66.44	16.374f	5.508p	3.914	

Tank Capacities for NO2_FW.S containing FRESH WATER (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	3.32	3.32	16.352f	5.423s	2.573	50.83
10.00%	6.64	6.64	16.354f	5.430s	2.645	51.54
20.00%	13.29	13.29	16.356f	5.442s	2.789	52.82
30.00%	19.93	19.93	16.359f	5.453s	2.932	53.90
40.00%	26.57	26.57	16.361f	5.464s	3.074	54.84
50.00%	33.22	33.22	16.364f	5.473s	3.216	55.68
60.00%	39.86	39.86	16.366f	5.481s	3.356	56.39
70.00%	46.51	46.51	16.368f	5.489s	3.497	57.03
80.00%	53.15	53.15	16.370f	5.496s	3.636	57.62
90.00%	59.79	59.79	16.372f	5.502s	3.775	58.16
95.00%	63.12	63.12	16.373f	5.505s	3.845	58.41
98.00%	65.11	65.11	16.374f	5.507s	3.886	58.58
100.00%	66.44	66.44	16.374f	5.508s	3.914	

Tank Capacities for NO3_FW.P containing FRESH WATER (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.70	2.70	20.385a	6.323p	1.952	5.98
10.00%	5.39	5.39	20.667a	6.378p	2.176	9.32
20.00%	10.79	10.79	21.087a	6.430p	2.484	15.05
30.00%	16.18	16.18	21.448a	6.459p	2.709	18.74
40.00%	21.57	21.57	21.677a	6.487p	2.895	19.99
50.00%	26.97	26.97	21.820a	6.508p	3.064	20.48
60.00%	32.36	32.36	21.916a	6.524p	3.224	20.55
70.00%	37.76	37.76	21.985a	6.535p	3.379	20.55
80.00%	43.15	43.15	22.037a	6.543p	3.531	20.55
90.00%	48.54	48.54	22.078a	6.549p	3.680	20.55
95.00%	51.24	51.24	22.095a	6.552p	3.754	20.55
98.00%	52.86	52.86	22.104a	6.553p	3.798	20.55
100.00%	53.94	53.94	22.110a	6.554p	3.828	

Tank Capacities for NO3_FW.S containing FRESH WATER (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	3.56	3.56	19.284a	6.326s	1.548	6.71
10.00%	7.12	7.12	19.589a	6.401s	1.805	10.67
20.00%	14.24	14.24	20.039a	6.449s	2.164	16.29
30.00%	21.36	21.36	20.431a	6.471s	2.431	22.06
40.00%	28.47	28.47	20.752a	6.493s	2.650	24.09
50.00%	35.59	35.59	20.957a	6.512s	2.843	24.93
60.00%	42.71	42.71	21.097a	6.527s	3.023	25.13
70.00%	49.83	49.83	21.198a	6.537s	3.196	25.13
80.00%	56.95	56.95	21.273a	6.545s	3.363	25.13
90.00%	64.07	64.07	21.332a	6.551s	3.527	25.13
95.00%	67.62	67.62	21.356a	6.554s	3.608	25.13
98.00%	69.76	69.76	21.370a	6.555s	3.657	25.13
100.00%	71.18	71.18	21.378a	6.556s	3.689	

Tank Capacities for NO1_FO.P containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.91	1.62	9.400f	7.500p	2.570	3.13
10.00%	3.81	3.24	9.400f	7.500p	2.640	3.13
20.00%	7.63	6.48	9.400f	7.500p	2.780	3.13
30.00%	11.44	9.72	9.400f	7.500p	2.920	3.13
40.00%	15.25	12.96	9.400f	7.500p	3.060	3.13
50.00%	19.07	16.21	9.400f	7.500p	3.200	3.13
60.00%	22.88	19.45	9.400f	7.500p	3.340	3.13
70.00%	26.69	22.69	9.400f	7.500p	3.480	3.13
80.00%	30.50	25.93	9.400f	7.500p	3.620	3.13
90.00%	34.32	29.17	9.400f	7.500p	3.760	3.13
95.00%	36.22	30.79	9.400f	7.500p	3.830	3.13
98.00%	37.37	31.76	9.400f	7.500p	3.872	3.13
100.00%	38.13	32.41	9.400f	7.500p	3.900	

Tank Capacities for NO1_FO.S containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.91	1.62	9.400f	7.500s	2.570	3.13
10.00%	3.81	3.24	9.400f	7.500s	2.640	3.13
20.00%	7.63	6.48	9.400f	7.500s	2.780	3.13
30.00%	11.44	9.72	9.400f	7.500s	2.920	3.13
40.00%	15.25	12.96	9.400f	7.500s	3.060	3.13
50.00%	19.07	16.21	9.400f	7.500s	3.200	3.13
60.00%	22.88	19.45	9.400f	7.500s	3.340	3.13
70.00%	26.69	22.69	9.400f	7.500s	3.480	3.13
80.00%	30.50	25.93	9.400f	7.500s	3.620	3.13
90.00%	34.32	29.17	9.400f	7.500s	3.760	3.13
95.00%	36.22	30.79	9.400f	7.500s	3.830	3.13
98.00%	37.37	31.76	9.400f	7.500s	3.872	3.13
100.00%	38.13	32.41	9.400f	7.500s	3.900	

Tank Capacities for NO1_FO.U.P containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.59	2.20	9.357f	5.819p	0.048	28.87
10.00%	5.18	4.40	9.359f	5.874p	0.093	32.61
20.00%	10.35	8.80	9.363f	5.946p	0.181	37.53
30.00%	15.53	13.20	9.365f	5.994p	0.265	40.77
40.00%	20.70	17.60	9.367f	6.030p	0.348	42.97
50.00%	25.88	22.00	9.369f	6.058p	0.430	44.35
60.00%	31.05	26.40	9.371f	6.079p	0.510	45.02
70.00%	36.23	30.80	9.372f	6.270p	0.604	3.09
80.00%	41.41	35.19	9.374f	6.423p	0.722	3.12
90.00%	46.58	39.59	9.377f	6.543p	0.856	3.12
95.00%	49.17	41.79	9.378f	6.593p	0.928	3.13
98.00%	50.72	43.11	9.379f	6.621p	0.972	3.13
100.00%	51.76	43.99	9.379f	6.638p	1.002	

Tank Capacities for NO1_FO.U.S containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.59	2.20	9.357f	5.819s	0.048	28.87
10.00%	5.18	4.40	9.359f	5.874s	0.093	32.61
20.00%	10.35	8.80	9.363f	5.946s	0.181	37.53
30.00%	15.53	13.20	9.365f	5.994s	0.265	40.77
40.00%	20.70	17.60	9.367f	6.030s	0.348	42.97
50.00%	25.88	22.00	9.369f	6.058s	0.430	44.35
60.00%	31.05	26.40	9.371f	6.079s	0.510	45.02
70.00%	36.23	30.80	9.372f	6.270s	0.604	3.09
80.00%	41.41	35.19	9.374f	6.423s	0.722	3.12
90.00%	46.58	39.59	9.377f	6.543s	0.856	3.12
95.00%	49.17	41.79	9.378f	6.593s	0.928	3.13
98.00%	50.72	43.11	9.379f	6.621s	0.972	3.13
100.00%	51.76	43.99	9.379f	6.638s	1.002	

Tank Capacities for NO1_FO.U.C containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.57	2.19	9.400f	0.000	0.021	274.40
10.00%	5.14	4.37	9.400f	0.000	0.042	274.40
20.00%	10.29	8.75	9.400f	0.000	0.085	274.40
30.00%	15.43	13.12	9.400f	0.000	0.127	274.40
40.00%	20.58	17.49	9.400f	0.000	0.170	274.40
50.00%	25.72	21.87	9.400f	0.000	0.212	274.40
60.00%	30.87	26.24	9.400f	0.000	0.255	274.40
70.00%	36.01	30.61	9.400f	0.000	0.297	274.40
80.00%	41.16	34.99	9.400f	0.000	0.340	274.40
90.00%	46.30	39.36	9.400f	0.000	0.382	274.40
95.00%	48.88	41.55	9.400f	0.000	0.404	274.40
98.00%	50.42	42.86	9.400f	0.000	0.416	274.40
100.00%	51.45	43.73	9.400f	0.000	0.425	

Tank Capacities for NO2_FO.U.P containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.59	2.21	2.866f	3.718p	0.034	159.86
10.00%	5.19	4.41	2.871f	3.757p	0.067	167.52
20.00%	10.38	8.82	2.878f	3.810p	0.132	178.01
30.00%	15.57	13.23	2.882f	3.848p	0.197	185.13
40.00%	20.75	17.64	2.886f	3.877p	0.261	190.36
50.00%	25.94	22.05	2.889f	3.901p	0.324	194.10
60.00%	31.13	26.46	2.891f	3.920p	0.387	196.76
70.00%	36.32	30.87	2.893f	3.935p	0.450	198.07
80.00%	41.51	35.28	2.902f	3.997p	0.514	1.68
90.00%	46.70	39.69	2.957f	4.386p	0.616	1.68
95.00%	49.29	41.90	2.980f	4.550p	0.687	1.68
98.00%	50.85	43.22	2.993f	4.640p	0.735	1.68
100.00%	51.88	44.10	3.001f	4.698p	0.769	

Tank Capacities for NO2_FO.U.S containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.30	1.96	2.865f	3.712s	0.030	158.63
10.00%	4.60	3.91	2.870f	3.749s	0.059	165.91
20.00%	9.20	7.82	2.876f	3.800s	0.118	175.92
30.00%	13.81	11.73	2.881f	3.836s	0.175	182.91
40.00%	18.41	15.65	2.885f	3.865s	0.232	188.03
50.00%	23.01	19.56	2.887f	3.888s	0.288	192.10
60.00%	27.61	23.47	2.890f	3.907s	0.345	195.00
70.00%	32.21	27.38	2.892f	3.923s	0.401	197.02
80.00%	36.81	31.29	2.893f	3.937s	0.456	198.08
90.00%	41.42	35.20	2.901f	3.991s	0.512	1.68
95.00%	43.72	37.16	2.927f	4.175s	0.550	1.68
98.00%	45.10	38.33	2.941f	4.277s	0.579	1.68
100.00%	46.02	39.12	2.951f	4.342s	0.600	

Tank Capacities for NO3_FO.P containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.32	1.12	2.600a	7.500p	2.570	2.16
10.00%	2.64	2.24	2.600a	7.500p	2.640	2.16
20.00%	5.28	4.49	2.600a	7.500p	2.780	2.16
30.00%	7.92	6.73	2.600a	7.500p	2.920	2.16
40.00%	10.56	8.98	2.600a	7.500p	3.060	2.16
50.00%	13.20	11.22	2.600a	7.500p	3.200	2.16
60.00%	15.84	13.46	2.600a	7.500p	3.340	2.16
70.00%	18.48	15.71	2.600a	7.500p	3.480	2.16
80.00%	21.12	17.95	2.600a	7.500p	3.620	2.16
90.00%	23.76	20.20	2.600a	7.500p	3.760	2.16
95.00%	25.08	21.32	2.600a	7.500p	3.830	2.16
98.00%	25.87	21.99	2.600a	7.500p	3.872	2.16
100.00%	26.40	22.44	2.600a	7.500p	3.900	

Tank Capacities for NO3_FO.S containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.32	1.12	2.600a	7.500s	2.570	2.16
10.00%	2.64	2.24	2.600a	7.500s	2.640	2.16
20.00%	5.28	4.49	2.600a	7.500s	2.780	2.16
30.00%	7.92	6.73	2.600a	7.500s	2.920	2.16
40.00%	10.56	8.98	2.600a	7.500s	3.060	2.16
50.00%	13.20	11.22	2.600a	7.500s	3.200	2.16
60.00%	15.84	13.46	2.600a	7.500s	3.340	2.16
70.00%	18.48	15.71	2.600a	7.500s	3.480	2.16
80.00%	21.12	17.95	2.600a	7.500s	3.620	2.16
90.00%	23.76	20.20	2.600a	7.500s	3.760	2.16
95.00%	25.08	21.32	2.600a	7.500s	3.830	2.16
98.00%	25.87	21.99	2.600a	7.500s	3.872	2.16
100.00%	26.40	22.44	2.600a	7.500s	3.900	

Tank Capacities for NO3_FO.U.P containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.89	1.60	2.544a	5.613p	0.046	30.51
10.00%	3.77	3.21	2.546a	5.668p	0.089	33.72
20.00%	7.55	6.41	2.547a	5.739p	0.174	37.89
30.00%	11.32	9.62	2.548a	5.788p	0.256	40.62
40.00%	15.09	12.83	2.549a	5.825p	0.336	42.47
50.00%	18.86	16.03	2.550a	5.852p	0.415	43.59
60.00%	22.64	19.24	2.550a	5.873p	0.494	44.05
70.00%	26.41	22.45	2.556a	6.051p	0.581	2.16
80.00%	30.18	25.66	2.561a	6.232p	0.696	2.16
90.00%	33.96	28.86	2.566a	6.373p	0.830	2.16
95.00%	35.84	30.47	2.567a	6.433p	0.902	2.16
98.00%	36.97	31.43	2.568a	6.465p	0.947	2.16
100.00%	37.73	32.07	2.569a	6.486p	0.977	

Tank Capacities for NO3_FO.U.S containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.89	1.60	2.544a	5.613s	0.046	30.51
10.00%	3.77	3.21	2.546a	5.668s	0.089	33.72
20.00%	7.55	6.41	2.547a	5.739s	0.174	37.89
30.00%	11.32	9.62	2.548a	5.788s	0.256	40.62
40.00%	15.09	12.83	2.549a	5.825s	0.336	42.47
50.00%	18.86	16.03	2.550a	5.852s	0.415	43.59
60.00%	22.64	19.24	2.550a	5.873s	0.494	44.05
70.00%	26.41	22.45	2.556a	6.051s	0.581	2.16
80.00%	30.18	25.66	2.561a	6.232s	0.696	2.16
90.00%	33.96	28.86	2.566a	6.373s	0.830	2.16
95.00%	35.84	30.47	2.567a	6.433s	0.902	2.16
98.00%	36.97	31.43	2.568a	6.465s	0.947	2.16
100.00%	37.73	32.07	2.569a	6.486s	0.977	

Tank Capacities for NO4_FO.U.P containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.39	2.03	8.663a	5.516p	0.043	40.18
10.00%	4.78	4.06	8.661a	5.568p	0.085	44.38
20.00%	9.55	8.12	8.659a	5.636p	0.166	49.91
30.00%	14.33	12.18	8.658a	5.684p	0.245	53.59
40.00%	19.11	16.24	8.657a	5.719p	0.322	56.16
50.00%	23.88	20.30	8.657a	5.746p	0.398	57.83
60.00%	28.66	24.36	8.656a	5.768p	0.473	58.62
70.00%	33.43	28.42	8.650a	5.961p	0.559	2.68
80.00%	38.21	32.48	8.644a	6.153p	0.673	2.68
90.00%	42.99	36.54	8.639a	6.303p	0.808	2.68
95.00%	45.37	38.57	8.637a	6.366p	0.881	2.68
98.00%	46.81	39.79	8.636a	6.401p	0.926	2.68
100.00%	47.76	40.60	8.635a	6.423p	0.957	

Tank Capacities for NO4_FO.U.S containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.39	2.03	8.663a	5.516s	0.043	40.18
10.00%	4.78	4.06	8.661a	5.568s	0.085	44.38
20.00%	9.55	8.12	8.659a	5.636s	0.166	49.91
30.00%	14.33	12.18	8.658a	5.684s	0.245	53.59
40.00%	19.11	16.24	8.657a	5.719s	0.322	56.16
50.00%	23.88	20.30	8.657a	5.746s	0.398	57.83
60.00%	28.66	24.36	8.656a	5.768s	0.473	58.62
70.00%	33.43	28.42	8.650a	5.961s	0.559	2.68
80.00%	38.21	32.48	8.644a	6.153s	0.673	2.68
90.00%	42.99	36.54	8.639a	6.303s	0.808	2.68
95.00%	45.37	38.57	8.637a	6.366s	0.881	2.68
98.00%	46.81	39.79	8.636a	6.401s	0.926	2.68
100.00%	47.76	40.60	8.635a	6.423s	0.957	

Tank Capacities for FODAY.P containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.32	1.12	8.000a	7.500p	2.570	2.16
10.00%	2.64	2.24	8.000a	7.500p	2.640	2.16
20.00%	5.28	4.49	8.000a	7.500p	2.780	2.16
30.00%	7.92	6.73	8.000a	7.500p	2.920	2.16
40.00%	10.56	8.98	8.000a	7.500p	3.060	2.16
50.00%	13.20	11.22	8.000a	7.500p	3.200	2.16
60.00%	15.84	13.46	8.000a	7.500p	3.340	2.16
70.00%	18.48	15.71	8.000a	7.500p	3.480	2.16
80.00%	21.12	17.95	8.000a	7.500p	3.620	2.16
90.00%	23.76	20.20	8.000a	7.500p	3.760	2.16
95.00%	25.08	21.32	8.000a	7.500p	3.830	2.16
98.00%	25.87	21.99	8.000a	7.500p	3.872	2.16
100.00%	26.40	22.44	8.000a	7.500p	3.900	

Tank Capacities for FODAY.S containing FUEL OIL (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.61	1.37	8.600a	7.500s	2.570	2.64
10.00%	3.23	2.74	8.600a	7.500s	2.640	2.64
20.00%	6.45	5.49	8.600a	7.500s	2.780	2.64
30.00%	9.68	8.23	8.600a	7.500s	2.920	2.64
40.00%	12.91	10.97	8.600a	7.500s	3.060	2.64
50.00%	16.13	13.71	8.600a	7.500s	3.200	2.64
60.00%	19.36	16.46	8.600a	7.500s	3.340	2.64
70.00%	22.59	19.20	8.600a	7.500s	3.480	2.64
80.00%	25.81	21.94	8.600a	7.500s	3.620	2.64
90.00%	29.04	24.68	8.600a	7.500s	3.760	2.64
95.00%	30.65	26.05	8.600a	7.500s	3.830	2.64
98.00%	31.62	26.88	8.600a	7.500s	3.872	2.64
100.00%	32.27	27.43	8.600a	7.500s	3.900	

Tank Capacities for SLOP containing MISC (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.95	0.81	10.373a	0.000	0.025	56.50
10.00%	1.90	1.62	10.373a	0.000	0.050	56.50
20.00%	3.80	3.23	10.373a	0.000	0.100	56.50
30.00%	5.70	4.85	10.373a	0.000	0.150	56.50
40.00%	7.60	6.46	10.373a	0.000	0.200	56.50
50.00%	9.51	8.08	10.373a	0.000	0.250	56.50
60.00%	11.41	9.70	10.373a	0.000	0.300	56.50
70.00%	13.31	11.31	10.373a	0.000	0.350	56.50
80.00%	15.21	12.93	10.373a	0.000	0.400	56.50
90.00%	17.11	14.54	10.373a	0.000	0.450	56.50
95.00%	18.06	15.35	10.373a	0.000	0.475	56.50
98.00%	18.63	15.84	10.373a	0.000	0.490	56.50
100.00%	19.01	16.16	10.373a	0.000	0.500	

Tank Capacities for DISPERSANT.P containing MISC (1.163)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.74	0.86	4.000f	7.500p	2.570	1.66
10.00%	1.48	1.72	4.000f	7.500p	2.640	1.66
20.00%	2.96	3.45	4.000f	7.500p	2.780	1.66
30.00%	4.45	5.17	4.000f	7.500p	2.920	1.66
40.00%	5.93	6.89	4.000f	7.500p	3.060	1.66
50.00%	7.41	8.62	4.000f	7.500p	3.200	1.66
60.00%	8.89	10.34	4.000f	7.500p	3.340	1.66
70.00%	10.37	12.06	4.000f	7.500p	3.480	1.66
80.00%	11.85	13.79	4.000f	7.500p	3.620	1.66
90.00%	13.34	15.51	4.000f	7.500p	3.760	1.66
95.00%	14.08	16.37	4.000f	7.500p	3.830	1.66
98.00%	14.52	16.89	4.000f	7.500p	3.872	1.66
100.00%	14.82	17.23	4.000f	7.500p	3.900	

Tank Capacities for FOAM.S containing MISC (1.145)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.59	0.68	3.700f	7.500s	2.570	1.31
10.00%	1.19	1.36	3.700f	7.500s	2.640	1.31
20.00%	2.37	2.71	3.700f	7.500s	2.780	1.31
30.00%	3.56	4.07	3.700f	7.500s	2.920	1.31
40.00%	4.74	5.43	3.700f	7.500s	3.060	1.31
50.00%	5.93	6.79	3.700f	7.500s	3.200	1.31
60.00%	7.11	8.14	3.700f	7.500s	3.340	1.31
70.00%	8.30	9.50	3.700f	7.500s	3.480	1.31
80.00%	9.48	10.86	3.700f	7.500s	3.620	1.31
90.00%	10.67	12.22	3.700f	7.500s	3.760	1.31
95.00%	11.26	12.89	3.700f	7.500s	3.830	1.31
98.00%	11.62	13.30	3.700f	7.500s	3.872	1.31
100.00%	11.85	13.57	3.700f	7.500s	3.900	

Tank Capacities for HYDOIL.P containing MISC (0.924)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.59	0.55	1.300f	7.500p	2.570	1.06
10.00%	1.19	1.10	1.300f	7.500p	2.640	1.06
20.00%	2.37	2.19	1.300f	7.500p	2.780	1.06
30.00%	3.56	3.29	1.300f	7.500p	2.920	1.06
40.00%	4.74	4.38	1.300f	7.500p	3.060	1.06
50.00%	5.93	5.48	1.300f	7.500p	3.200	1.06
60.00%	7.11	6.57	1.300f	7.500p	3.340	1.06
70.00%	8.30	7.67	1.300f	7.500p	3.480	1.06
80.00%	9.48	8.76	1.300f	7.500p	3.620	1.06
90.00%	10.67	9.86	1.300f	7.500p	3.760	1.06
95.00%	11.26	10.41	1.300f	7.500p	3.830	1.06
98.00%	11.62	10.73	1.300f	7.500p	3.872	1.06
100.00%	11.85	10.95	1.300f	7.500p	3.900	

Tank Capacities for LUBOIL.S containing MISC (0.924)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.44	0.41	1.000f	7.500s	2.570	0.79
10.00%	0.89	0.82	1.000f	7.500s	2.640	0.79
20.00%	1.78	1.64	1.000f	7.500s	2.780	0.79
30.00%	2.67	2.46	1.000f	7.500s	2.920	0.79
40.00%	3.56	3.29	1.000f	7.500s	3.060	0.79
50.00%	4.44	4.11	1.000f	7.500s	3.200	0.79
60.00%	5.33	4.93	1.000f	7.500s	3.340	0.79
70.00%	6.22	5.75	1.000f	7.500s	3.480	0.79
80.00%	7.11	6.57	1.000f	7.500s	3.620	0.79
90.00%	8.00	7.39	1.000f	7.500s	3.760	0.79
95.00%	8.45	7.80	1.000f	7.500s	3.830	0.79
98.00%	8.71	8.05	1.000f	7.500s	3.872	0.79
100.00%	8.89	8.21	1.000f	7.500s	3.900	

Tank Capacities for DIRTYOIL.P containing MISC (0.900)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.35	0.32	1.100a	1.500p	0.025	4.76
10.00%	0.71	0.64	1.100a	1.500p	0.050	4.76
20.00%	1.41	1.27	1.100a	1.500p	0.100	4.76
30.00%	2.12	1.91	1.100a	1.500p	0.150	4.76
40.00%	2.82	2.54	1.100a	1.500p	0.200	4.76
50.00%	3.53	3.18	1.100a	1.500p	0.250	4.76
60.00%	4.23	3.81	1.100a	1.500p	0.300	4.76
70.00%	4.94	4.45	1.100a	1.500p	0.350	4.76
80.00%	5.64	5.08	1.100a	1.500p	0.400	4.76
90.00%	6.35	5.72	1.100a	1.500p	0.450	4.76
95.00%	6.70	6.03	1.100a	1.500p	0.475	4.76
98.00%	6.91	6.22	1.100a	1.500p	0.490	4.76
100.00%	7.06	6.35	1.100a	1.500p	0.500	

Tank Capacities for BILGEHOLD.S containing MISC (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.35	0.35	1.100a	1.500s	0.025	5.29
10.00%	0.71	0.71	1.100a	1.500s	0.050	5.29
20.00%	1.41	1.41	1.100a	1.500s	0.100	5.29
30.00%	2.12	2.12	1.100a	1.500s	0.150	5.29
40.00%	2.82	2.82	1.100a	1.500s	0.200	5.29
50.00%	3.53	3.53	1.100a	1.500s	0.250	5.29
60.00%	4.23	4.23	1.100a	1.500s	0.300	5.29
70.00%	4.94	4.94	1.100a	1.500s	0.350	5.29
80.00%	5.64	5.64	1.100a	1.500s	0.400	5.29
90.00%	6.35	6.35	1.100a	1.500s	0.450	5.29
95.00%	6.70	6.70	1.100a	1.500s	0.475	5.29
98.00%	6.91	6.91	1.100a	1.500s	0.490	5.29
100.00%	7.06	7.06	1.100a	1.500s	0.500	

Tank Capacities for SEWAGE containing MISC (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.52	0.52	14.764f	0.000	0.025	30.30
10.00%	1.03	1.03	14.764f	0.000	0.050	30.30
20.00%	2.06	2.06	14.764f	0.000	0.100	30.30
30.00%	3.09	3.09	14.764f	0.000	0.150	30.30
40.00%	4.12	4.12	14.764f	0.000	0.200	30.30
50.00%	5.16	5.16	14.764f	0.000	0.250	30.30
60.00%	6.19	6.19	14.764f	0.000	0.300	30.30
70.00%	7.22	7.22	14.764f	0.000	0.350	30.30
80.00%	8.25	8.25	14.764f	0.000	0.400	30.30
90.00%	9.28	9.28	14.764f	0.000	0.450	30.30
95.00%	9.80	9.80	14.764f	0.000	0.475	30.30
98.00%	10.11	10.11	14.764f	0.000	0.490	30.30
100.00%	10.31	10.31	14.764f	0.000	0.500	

Tank Capacities for OVERFLOW containing MISC (0.850)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.24	1.90	5.618a	0.000	0.025	152.28
10.00%	4.47	3.80	5.618a	0.000	0.050	152.28
20.00%	8.94	7.60	5.618a	0.000	0.100	152.28
30.00%	13.41	11.40	5.618a	0.000	0.150	152.28
40.00%	17.88	15.20	5.618a	0.000	0.200	152.28
50.00%	22.35	19.00	5.618a	0.000	0.250	152.28
60.00%	26.83	22.80	5.618a	0.000	0.300	152.28
70.00%	31.30	26.60	5.618a	0.000	0.350	152.28
80.00%	35.77	30.40	5.618a	0.000	0.400	152.28
90.00%	40.24	34.20	5.618a	0.000	0.450	152.28
95.00%	42.47	36.10	5.618a	0.000	0.475	152.28
98.00%	43.82	37.24	5.618a	0.000	0.490	152.28
100.00%	44.71	38.00	5.617a	0.000	0.500	

Tank Capacities for SCLEIHBTR.P containing SCALE INHIBITOR (1.100)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.67	0.74	13.400a	7.048p	2.570	3.21
10.00%	1.34	1.48	13.400a	7.048p	2.640	3.21
20.00%	2.68	2.95	13.400a	7.048p	2.780	3.21
30.00%	4.03	4.43	13.400a	7.048p	2.920	3.21
40.00%	5.37	5.90	13.400a	7.048p	3.060	3.21
50.00%	6.71	7.38	13.400a	7.048p	3.200	3.21
60.00%	8.05	8.86	13.400a	7.048p	3.340	3.21
70.00%	9.39	10.33	13.400a	7.048p	3.480	3.21
80.00%	10.74	11.81	13.400a	7.048p	3.620	3.21
90.00%	12.08	13.28	13.400a	7.048p	3.760	3.21
95.00%	12.75	14.02	13.400a	7.048p	3.830	3.21
98.00%	13.15	14.47	13.400a	7.048p	3.872	3.21
100.00%	13.42	14.76	13.400a	7.048p	3.900	

Tank Capacities for SCLEIHBTR.S containing SCALE INHIBITOR (1.100)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	0.67	0.74	13.400a	7.048s	2.570	3.21
10.00%	1.34	1.48	13.400a	7.048s	2.640	3.21
20.00%	2.68	2.95	13.400a	7.048s	2.780	3.21
30.00%	4.03	4.43	13.400a	7.048s	2.920	3.21
40.00%	5.37	5.90	13.400a	7.048s	3.060	3.21
50.00%	6.71	7.38	13.400a	7.048s	3.200	3.21
60.00%	8.05	8.86	13.400a	7.048s	3.340	3.21
70.00%	9.39	10.33	13.400a	7.048s	3.480	3.21
80.00%	10.74	11.81	13.400a	7.048s	3.620	3.21
90.00%	12.08	13.28	13.400a	7.048s	3.760	3.21
95.00%	12.75	14.02	13.400a	7.048s	3.830	3.21
98.00%	13.15	14.47	13.400a	7.048s	3.872	3.21
100.00%	13.42	14.76	13.400a	7.048s	3.900	

Tank Capacities for CORSNIHBTR.P containing CORROSION INHIBITOR (0.950)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.34	1.27	16.100a	7.048p	2.570	5.54
10.00%	2.68	2.55	16.100a	7.048p	2.640	5.54
20.00%	5.37	5.10	16.100a	7.048p	2.780	5.54
30.00%	8.05	7.65	16.100a	7.048p	2.920	5.54
40.00%	10.74	10.20	16.100a	7.048p	3.060	5.54
50.00%	13.42	12.75	16.100a	7.048p	3.200	5.54
60.00%	16.10	15.30	16.100a	7.048p	3.340	5.54
70.00%	18.79	17.85	16.100a	7.048p	3.480	5.54
80.00%	21.47	20.40	16.100a	7.048p	3.620	5.54
90.00%	24.15	22.95	16.100a	7.048p	3.760	5.54
95.00%	25.50	24.22	16.100a	7.048p	3.830	5.54
98.00%	26.30	24.99	16.100a	7.048p	3.872	5.54
100.00%	26.84	25.50	16.100a	7.048p	3.900	

Tank Capacities for SCALE_DISLVR.S containing SCALE INHIBITOR (1.100)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.34	1.48	16.100a	7.048s	2.570	6.42
10.00%	2.68	2.95	16.100a	7.048s	2.640	6.42
20.00%	5.37	5.90	16.100a	7.048s	2.780	6.42
30.00%	8.05	8.86	16.100a	7.048s	2.920	6.42
40.00%	10.74	11.81	16.100a	7.048s	3.060	6.42
50.00%	13.42	14.76	16.100a	7.048s	3.200	6.42
60.00%	16.10	17.71	16.100a	7.048s	3.340	6.42
70.00%	18.79	20.66	16.100a	7.048s	3.480	6.42
80.00%	21.47	23.62	16.100a	7.048s	3.620	6.42
90.00%	24.15	26.57	16.100a	7.048s	3.760	6.42
95.00%	25.50	28.05	16.100a	7.048s	3.830	6.42
98.00%	26.30	28.93	16.100a	7.048s	3.872	6.42
100.00%	26.84	29.52	16.100a	7.048s	3.900	

Tank Capacities for TRTDINJW.P containing TREATED INJECTION WATER (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.55	1.55	11.800f	2.100p	1.387	8.13
10.00%	3.09	3.09	11.800f	2.100p	1.464	8.13
20.00%	6.18	6.18	11.800f	2.100p	1.618	8.13
30.00%	9.27	9.27	11.800f	2.100p	1.772	8.13
40.00%	12.36	12.36	11.800f	2.100p	1.926	8.13
50.00%	15.45	15.45	11.800f	2.100p	2.080	8.13
60.00%	18.54	18.54	11.800f	2.100p	2.234	8.13
70.00%	21.63	21.63	11.800f	2.100p	2.388	8.13
80.00%	24.72	24.72	11.800f	2.100p	2.542	8.13
90.00%	27.81	27.81	11.800f	2.100p	2.696	8.13
95.00%	29.36	29.36	11.800f	2.100p	2.773	8.13
98.00%	30.28	30.28	11.800f	2.100p	2.819	8.13
100.00%	30.90	30.90	11.800f	2.100p	2.850	

Tank Capacities for TRTDINJW.S containing TREATED INJECTION WATER (1.000)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.55	1.55	11.800f	2.100s	1.387	8.13
10.00%	3.09	3.09	11.800f	2.100s	1.464	8.13
20.00%	6.18	6.18	11.800f	2.100s	1.618	8.13
30.00%	9.27	9.27	11.800f	2.100s	1.772	8.13
40.00%	12.36	12.36	11.800f	2.100s	1.926	8.13
50.00%	15.45	15.45	11.800f	2.100s	2.080	8.13
60.00%	18.54	18.54	11.800f	2.100s	2.234	8.13
70.00%	21.63	21.63	11.800f	2.100s	2.388	8.13
80.00%	24.72	24.72	11.800f	2.100s	2.542	8.13
90.00%	27.81	27.81	11.800f	2.100s	2.696	8.13
95.00%	29.36	29.36	11.800f	2.100s	2.773	8.13
98.00%	30.28	30.28	11.800f	2.100s	2.819	8.13
100.00%	30.90	30.90	11.800f	2.100s	2.850	

Tank Capacities for NITROGENTK.P containing TK ON MAIN DECK (1.185)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.04	2.41	17.240a	1.050p	5.373	8.66
10.00%	4.07	4.82	17.240a	1.050p	5.445	8.66
20.00%	8.14	9.65	17.240a	1.050p	5.590	8.66
30.00%	12.21	14.47	17.240a	1.050p	5.735	8.66
40.00%	16.28	19.29	17.240a	1.050p	5.880	8.66
50.00%	20.35	24.12	17.240a	1.050p	6.025	8.66
60.00%	24.42	28.94	17.240a	1.050p	6.170	8.66
70.00%	28.49	33.76	17.240a	1.050p	6.315	8.66
80.00%	32.56	38.59	17.240a	1.050p	6.460	8.66
90.00%	36.63	43.41	17.240a	1.050p	6.605	8.66
95.00%	38.67	45.82	17.240a	1.050p	6.678	8.66
98.00%	39.89	47.27	17.240a	1.050p	6.721	8.66
100.00%	40.71	48.24	17.240a	1.050p	6.750	

Tank Capacities for NO1_BRINETK.C containing BRINE (1.550)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.70	2.63	22.011a	0.000	2.872	42.06
10.00%	3.39	5.26	22.222a	0.000	2.975	59.59
20.00%	6.79	10.52	22.455a	0.000	3.121	65.95
30.00%	10.18	15.78	22.537a	0.000	3.249	65.95
40.00%	13.57	21.04	22.577a	0.000	3.373	65.95
50.00%	16.97	26.30	22.602a	0.000	3.495	65.95
60.00%	20.36	31.56	22.618a	0.000	3.617	65.95
70.00%	23.75	36.82	22.630a	0.000	3.738	65.95
80.00%	27.15	42.08	22.639a	0.000	3.858	65.95
90.00%	30.54	47.34	22.646a	0.000	3.978	65.95
95.00%	32.24	49.97	22.648a	0.000	4.039	65.95
98.00%	33.26	51.55	22.650a	0.000	4.075	65.95
100.00%	33.93	52.60	22.651a	0.000	4.099	

Tank Capacities for NO2_BRINETK.C containing BRINE (1.550)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.64	2.54	18.362a	0.000	0.844	29.20
10.00%	3.28	5.08	18.540a	0.000	0.960	42.20
20.00%	6.56	10.16	18.785a	0.000	1.125	62.33
30.00%	9.84	15.25	18.977a	0.000	1.253	78.68
40.00%	13.11	20.33	19.141a	0.000	1.361	87.43
50.00%	16.39	25.41	19.341a	0.000	1.459	82.64
60.00%	19.67	30.49	19.578a	0.000	1.552	84.56
70.00%	22.95	35.57	19.836a	0.000	1.643	71.91
80.00%	26.23	40.66	20.105a	0.000	1.733	67.93
90.00%	29.51	45.74	20.377a	0.000	1.826	58.37
95.00%	31.15	48.28	20.514a	0.000	1.874	41.16
98.00%	32.13	49.80	20.598a	0.000	1.905	26.08
100.00%	32.79	50.82	20.658a	0.000	1.928	

Tank Capacities for NO3_BRINETK.C containing BRINE (1.550)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.87	2.90	13.544a	0.000	0.059	103.99
10.00%	3.75	5.81	13.736a	0.000	0.103	119.39
20.00%	7.50	11.62	13.992a	0.000	0.178	138.59
30.00%	11.24	17.43	14.165a	0.000	0.247	146.98
40.00%	14.99	23.24	14.320a	0.000	0.309	154.55
50.00%	18.74	29.05	14.429a	0.000	0.368	164.57
60.00%	22.49	34.86	14.508a	0.000	0.425	164.57
70.00%	26.24	40.67	14.564a	0.000	0.481	164.57
80.00%	29.99	46.48	14.606a	0.000	0.536	164.57
90.00%	33.73	52.29	14.781a	0.000	0.595	61.98
95.00%	35.61	55.19	14.894a	0.000	0.629	43.93
98.00%	36.73	56.94	14.966a	0.000	0.653	27.98
100.00%	37.48	58.10	15.017a	0.000	0.671	

Tank Capacities for NO4_BRINETK.P containing BRINE (1.550)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.55	2.39	7.600f	2.100p	1.387	12.60
10.00%	3.09	4.79	7.600f	2.100p	1.464	12.60
20.00%	6.18	9.58	7.600f	2.100p	1.618	12.60
30.00%	9.27	14.37	7.600f	2.100p	1.772	12.60
40.00%	12.36	19.16	7.600f	2.100p	1.926	12.60
50.00%	15.45	23.95	7.600f	2.100p	2.080	12.60
60.00%	18.54	28.74	7.600f	2.100p	2.234	12.60
70.00%	21.63	33.53	7.600f	2.100p	2.388	12.60
80.00%	24.72	38.32	7.600f	2.100p	2.542	12.60
90.00%	27.81	43.11	7.600f	2.100p	2.696	12.60
95.00%	29.36	45.50	7.600f	2.100p	2.773	12.60
98.00%	30.28	46.94	7.600f	2.100p	2.819	12.60
100.00%	30.90	47.90	7.600f	2.100p	2.850	

Tank Capacities for NO4_BRINETK.S containing BRINE (1.550)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	1.55	2.39	7.600f	2.100s	1.387	12.60
10.00%	3.09	4.79	7.600f	2.100s	1.464	12.60
20.00%	6.18	9.58	7.600f	2.100s	1.618	12.60
30.00%	9.27	14.37	7.600f	2.100s	1.772	12.60
40.00%	12.36	19.16	7.600f	2.100s	1.926	12.60
50.00%	15.45	23.95	7.600f	2.100s	2.080	12.60
60.00%	18.54	28.74	7.600f	2.100s	2.234	12.60
70.00%	21.63	33.53	7.600f	2.100s	2.388	12.60
80.00%	24.72	38.32	7.600f	2.100s	2.542	12.60
90.00%	27.81	43.11	7.600f	2.100s	2.696	12.60
95.00%	29.36	45.50	7.600f	2.100s	2.773	12.60
98.00%	30.28	46.94	7.600f	2.100s	2.819	12.60
100.00%	30.90	47.90	7.600f	2.100s	2.850	

Tank Capacities for CHEMADCTTK.P containing TK ON MAIN DECK (1.185)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	3.28	3.89	0.410f	2.050p	5.439	42.02
10.00%	6.56	7.77	0.410f	2.050p	5.578	42.02
20.00%	13.12	15.55	0.410f	2.050p	5.855	42.02
30.00%	19.68	23.32	0.410f	2.050p	6.133	42.02
40.00%	26.24	31.09	0.410f	2.050p	6.410	42.02
50.00%	32.80	38.87	0.410f	2.050p	6.688	42.02
60.00%	39.36	46.64	0.410f	2.050p	6.965	42.02
70.00%	45.92	54.42	0.410f	2.050p	7.243	42.02
80.00%	52.48	62.19	0.410f	2.050p	7.520	42.02
90.00%	59.04	69.96	0.410f	2.050p	7.798	42.02
95.00%	62.32	73.85	0.410f	2.050p	7.936	42.02
98.00%	64.29	76.18	0.410f	2.050p	8.020	42.02
100.00%	65.60	77.74	0.410f	2.050p	8.075	

Tank Capacities for SURGETK.S containing TK ON MAIN DECK (1.185)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	2.48	2.94	2.822a	4.215s	5.388	10.75
10.00%	4.97	5.88	2.822a	4.215s	5.475	10.75
20.00%	9.93	11.77	2.822a	4.215s	5.650	10.75
30.00%	14.90	17.65	2.822a	4.215s	5.825	10.75
40.00%	19.86	23.54	2.822a	4.215s	6.000	10.75
50.00%	24.83	29.42	2.822a	4.215s	6.175	10.75
60.00%	29.79	35.31	2.822a	4.215s	6.350	10.75
70.00%	34.76	41.19	2.822a	4.215s	6.525	10.75
80.00%	39.73	47.08	2.822a	4.215s	6.700	10.75
90.00%	44.69	52.96	2.822a	4.215s	6.875	10.75
95.00%	47.17	55.90	2.822a	4.215s	6.963	10.75
98.00%	48.66	57.67	2.822a	4.215s	7.015	10.75
100.00%	49.66	58.84	2.822a	4.215s	7.050	

Tank Capacities for NO1_ACIDTK.C containing ACID (1.160)

Load (%)	Volume (m^3)	Weight (MT)	Lcg (m)	Tcg (m)	Vcg (m)	FSMt (MT-m)
0.00%	0.00	0.00				
5.00%	4.25	4.92	14.138a	0.000	1.805	28.76
10.00%	8.49	9.85	14.432a	0.000	1.959	36.86
20.00%	16.98	19.70	14.899a	0.000	2.208	49.20
30.00%	25.47	29.55	15.279a	0.000	2.412	59.01
40.00%	33.96	39.40	15.599a	0.000	2.579	62.43
50.00%	42.46	49.25	15.817a	0.000	2.713	64.81
60.00%	50.95	59.10	15.963a	0.000	2.842	64.81
70.00%	59.44	68.95	16.068a	0.000	2.970	64.81
80.00%	67.93	78.80	16.146a	0.000	3.096	64.81
90.00%	76.42	88.65	16.207a	0.000	3.222	64.81
95.00%	80.67	93.57	16.233a	0.000	3.284	61.76
98.00%	83.21	96.53	16.247a	0.000	3.322	55.83
100.00%	84.91	98.50	16.255a	0.000	3.347	